

nRF24L01+

Single Chip 2.4GHz Transceiver

Product Specification v1.0

Key Features

- Worldwide 2.4GHz ISM band operation
- 250kbps, 1Mbps and 2Mbps on air data rates
- Ultra low power operation
- 11.3mA TX at 0dBm output power
- 13.5mA RX at 2Mbps air data rate
- 900nA in power down
- 26µA in standby-I
- On chip voltage regulator
- 1.9 to 3.6V supply range
- Enhanced ShockBurst™
- Automatic packet handling
- Auto packet transaction handling
- 6 data pipe MultiCeiver™
- Drop-in compatibility with nRF24L01
- On-air compatible in 250kbps and 1Mbps with nRF2401A, nRF2402, nRF24E1 and nRF24E2
- Low cost BOM
- ±60ppm 16MHz crystal
- 5V tolerant inputs
- Compact 20-pin 4x4mm QFN package

Applications

- Wireless PC Peripherals
- Mouse, keyboards and remotes
- 3-in-1 desktop bundles
- Advanced Media center remote controls
- VoIP headsets
- Game controllers
- Sports watches and sensors
- RF remote controls for consumer electronics
- Home and commercial automation
- Ultra low power sensor networks
- Active RFID
- Asset tracking systems
- Toys

nRF24L01+

单芯片 2.4GHz 收发器

产品规格 v1.0

主要特点

- 全球 2.4GHz ISM 频段作
- 250kbps、1Mbps 和 2Mbps 的空中数据速率
- 超低功耗运行
- 11.3mA TX (在 0dBm 输出功率)
- 13.5mA RX, 2Mbps 空中数据速率
- 掉电时为 900nA
- 待机 I 时 26μA
- 片上稳压器
- 1.9 至 3.6V 电源范围
- 增强的 ShockBurst™
- 自动数据包处理
- 自动数据包事务处理
- 6 个数据管道 MultiCeiver™
- 与 nRF24L01 的直接兼容性
- 与 nRF2401A、nRF2402、nRF24E1 和 nRF24E2 在 250kbps 和 1Mbps 的广播中兼容
- 低成本 BOM
- ±60ppm 16MHz 晶振
- 5V 容限输入
- 紧凑的 20 引脚 4x4mm QFN 封装

应用

- 无线 PC 外围设备
- 鼠标、键盘和遥控器
- 3 合 1 桌面捆绑包
- 高级 Media Center 遥控器
- VoIP 耳麦
- 游戏控制器
- 运动手表和传感器
- 消费电子产品的射频遥控器
- 家庭和商用自动化
- 超低功耗传感器网络
- 有源 RFID
- 资产跟踪系统
- 玩具

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Contact details

Visit www.nordicsemi.no for Nordic Semiconductor sales offices and distributors worldwide

Main office:

Otto Nielsens vei 12
7004 Trondheim
Phone: +47 72 89 89 00
Fax: +47 72 89 89 89
www.nordicsemi.no



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访问 www.nordicsemi.no 了解 Nordic Semiconductor 全球销售办事处和分销商

总公司：

Otto Nielsens vei 12 7004
特隆赫姆 电话：+47 72 89
89 00 传真：+47 72 89 89
89 www.nordicsemi.no



Writing Conventions

This product specification follows a set of typographic rules that makes the document consistent and easy to read. The following writing conventions are used:

- Commands, bit state conditions, and register names are written in `Courier`.
- Pin names and pin signal conditions are written in `Courier bold`.
- Cross references are [underlined and highlighted in blue](#).

Revision History

Date	Version	Description
September 2008	1.0	

Attention!

Observe precaution for handling
Electrostatic Sensitive Device.

HBM (Human Body Model) $\geq 1\text{Kv}$
MM (Machine Model) $\geq 200\text{V}$



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- 引脚名称和引脚信号条件以 Courier 粗体书写。
- 交叉引用带下划线，并以蓝色突出显示。

修订历史记录

日期版本	描述
2008 年 9 月1.0	

注意力！

请遵守处理静电敏感设备的注意事
项。

HBM（人体模型）> 1Kv MM（机器
模型）> 200V



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1 Introduction

The nRF24L01+ is a single chip 2.4GHz transceiver with an embedded baseband protocol engine (Enhanced ShockBurst™), suitable for ultra low power wireless applications. The nRF24L01+ is designed for operation in the world wide ISM frequency band at 2.400 - 2.4835GHz.

To design a radio system with the nRF24L01+, you simply need an MCU (microcontroller) and a few external passive components.

You can operate and configure the nRF24L01+ through a Serial Peripheral Interface (SPI). The register map, which is accessible through the SPI, contains all configuration registers in the nRF24L01+ and is accessible in all operation modes of the chip.

The embedded baseband protocol engine (Enhanced ShockBurst™) is based on packet communication and supports various modes from manual operation to advanced autonomous protocol operation. Internal FIFOs ensure a smooth data flow between the radio front end and the system's MCU. Enhanced ShockBurst™ reduces system cost by handling all the high speed link layer operations.

The radio front end uses GFSK modulation. It has user configurable parameters like frequency channel, output power and air data rate. nRF24L01+ supports an air data rate of 250 kbps, 1 Mbps and 2Mbps. The high air data rate combined with two power saving modes make the nRF24L01+ very suitable for ultra low power designs.

nRF24L01+ is drop-in compatible with nRF24L01 and on-air compatible with nRF2401A, nRF2402, nRF24E1 and nRF24E2. Intermodulation and wideband blocking values in nRF24L01+ are much improved in comparison to the nRF24L01 and the addition of internal filtering to nRF24L01+ has improved the margins for meeting RF regulatory standards.

Internal voltage regulators ensure a high Power Supply Rejection Ratio (PSRR) and a wide power supply range.

1 介绍

nRF24L01+ 是一款单芯片 2.4GHz 收发器，具有嵌入式基带协议引擎（增强型 ShockBurst™），适用于超低功耗无线应用。nRF24L01+ 设计用于在 2.400 - 2.4835GHz 的全球 ISM 频段内工作。

要使用 nRF24L01+ 设计无线电系统，您只需要一个 MCU（微控制器）和一些外部无源元件。

您可以通过串行外设接口（SPI）作和配置 nRF24L01+。寄存器映射可通过 SPI 访问，包含 nRF24L01+ 中的所有配置寄存器，并且可以在芯片的所有工作模式下访问。

嵌入式基带协议引擎（Enhanced ShockBurst™）基于分组通信，支持从手动作到高级自主协议作的各种模式。内部 FIFO 确保无线电前端和系统 MCU 之间的数据流顺畅。增强的 ShockBurst™ 通过处理所有高速链路层作来降低系统成本。

无线电前端使用 GFSK 调制。它具有用户可配置的参数，如频道、输出功率和空中数据速率。nRF24L01+ 支持 250 kbps、1 Mbps 和 2Mbps 的空中数据速率。高空气数据速率与两种省电模式相结合，使 nRF24L01+ 非常适合超低功耗设计。

nRF24L01+ 与 nRF24L01 直接兼容，并与 nRF2401A、nRF2402、nRF24E1 和 nRF24E2 广播兼容。与 nRF24L01 相比，nRF24L01+ 中的互调和宽带阻塞值大大提高，并且 nRF24L01+ 中增加的内部滤波提高了满足射频监管标准的裕量。

内部稳压器可确保高电源抑制比（PSRR）和宽电源范围。

1.1 Features

Features of the nRF24L01+ include:

- Radio
 - Worldwide 2.4GHz ISM band operation
 - 126 RF channels
 - Common RX and TX interface
 - GFSK modulation
 - 250kbps, 1 and 2Mbps air data rate
 - 1MHz non-overlapping channel spacing at 1Mbps
 - 2MHz non-overlapping channel spacing at 2Mbps
- Transmitter
 - Programmable output power: 0, -6, -12 or -18dBm
 - 11.3mA at 0dBm output power
- Receiver
 - Fast AGC for improved dynamic range
 - Integrated channel filters
 - 13.5mA at 2Mbps
 - -82dBm sensitivity at 2Mbps
 - -85dBm sensitivity at 1Mbps
 - -94dBm sensitivity at 250kbps
- RF Synthesizer
 - Fully integrated synthesizer
 - No external loop filter, VCO varactor diode or resonator
 - Accepts low cost ± 60 ppm 16MHz crystal
- Enhanced ShockBurst™
 - 1 to 32 bytes dynamic payload length
 - Automatic packet handling
 - Auto packet transaction handling
 - 6 data pipe MultiCeiver™ for 1:6 star networks
- Power Management
 - Integrated voltage regulator
 - 1.9 to 3.6V supply range
 - Idle modes with fast start-up times for advanced power management
 - 26 μ A Standby-I mode, 900nA power down mode
 - Max 1.5ms start-up from power down mode
 - Max 130 μ s start-up from standby-I mode
- Host Interface
 - 4-pin hardware SPI
 - Max 10Mbps
 - 3 separate 32 bytes TX and RX FIFOs
 - 5V tolerant inputs
- Compact 20-pin 4x4mm QFN package

1.1 特征

nRF24L01+ 的功能包括：

- 收音机
 - × 全球 2.4GHz ISM 频段作
 - × 126 个 RF 通道
 - × 通用 RX 和 TX 接口
 - × GFSK 调制
 - × 250kbps、1 和 2Mbps 空中数据速率
 - × 1Mbps 时 1MHz 非重叠通道间隔
 - × 2Mbps 时 2MHz 非重叠通道间隔
- 发射机
 - × 可编程输出功率：0、-6、-12 或 -18dBm
 - × 11.3mA（在 0dBm 输出功率）
- 接收器
 - × 快速 AGC 可改善动态范围
 - × 集成通道滤波器
 - × 2Mbps 时为 13.5mA
 - × 2Mbps 时灵敏度为 -82dBm
 - × 1Mbps 时灵敏度为 -85dBm
 - × -94dBm 灵敏度（250kbps 时）
- RF 合成器
 - × 完全集成的合成器
 - × 无需外部环路过滤波器、VCO 变容二极管或谐振器
 - × 接受低成本 $\pm 60\text{ppm}$ 16MHz 晶振
- 增强的 ShockBurst™
 - × 1 至 32 字节动态有效载荷长度
 - × 自动数据包处理
 - × 自动数据包事务处理
 - × 用于 1:6 星形网络的 6 数据管 MultiCeiver™
- 电源管理
 - × 集成稳压器
 - × 1.9 至 3.6V 电源范围
 - × 具有快速启动时间的空闲模式，可实现高级电源管理
 - × 26 μA 待机 I 模式，900nA 省电模式
 - × 从掉电模式启动最大 1.5ms
 - × 从待机 I 模式启动最大 130 μs
- 主机接口
 - × 4 引脚硬件 SPI
 - × 最大 10Mbps
 - × 3 个独立的 32 字节 TX 和 RX FIFO
 - × 5V 容限输入
- 紧凑的 20 引脚 4x4mm QFN 封装

1.2 Block diagram

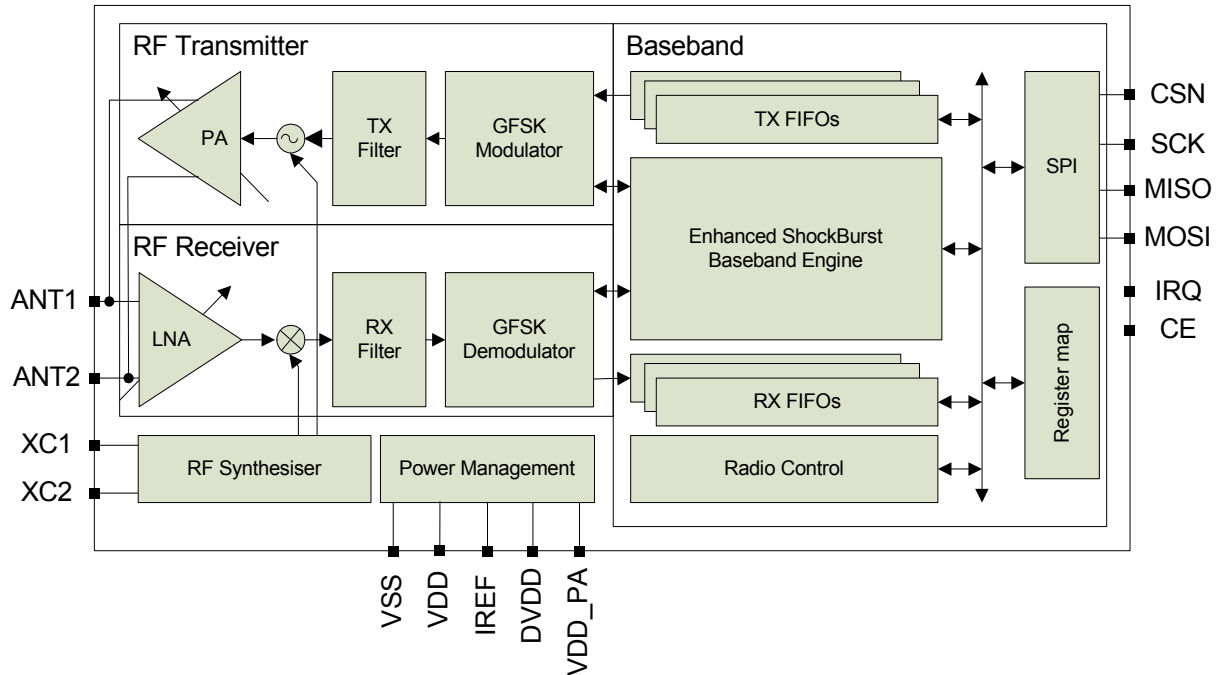


Figure 1. nRF24L01+ block diagram

1.2 方框图

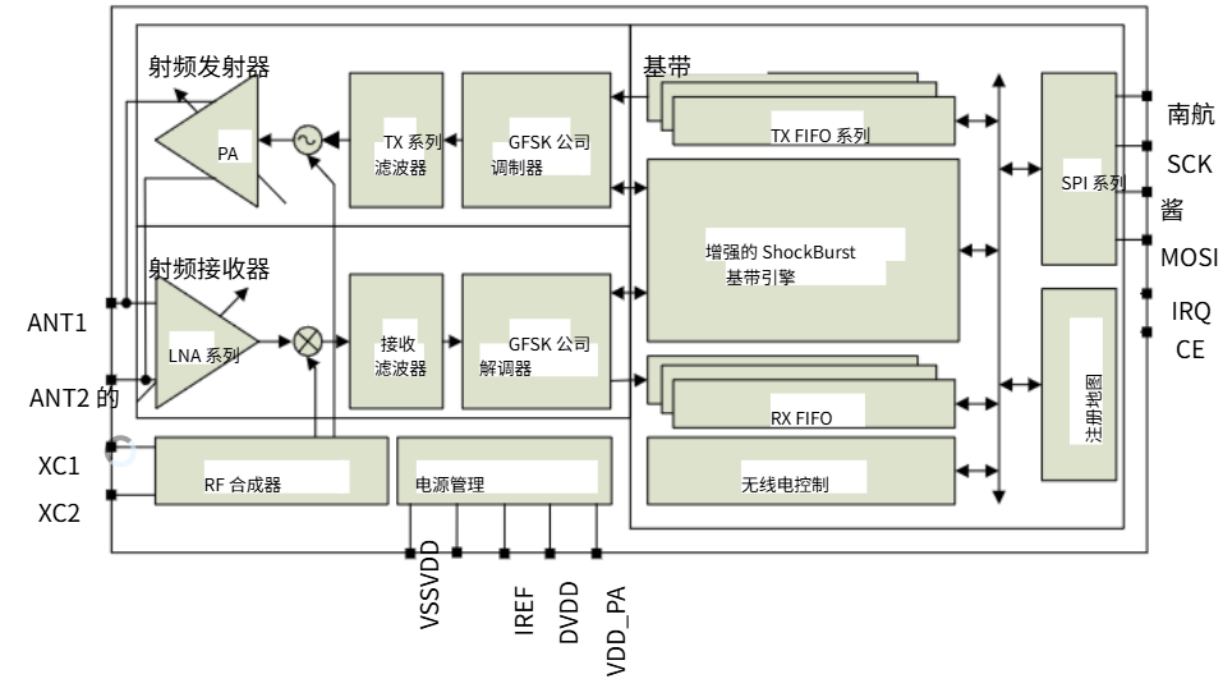


图 1.nRF24L01+ 框图

2 Pin Information

2.1 Pin assignment

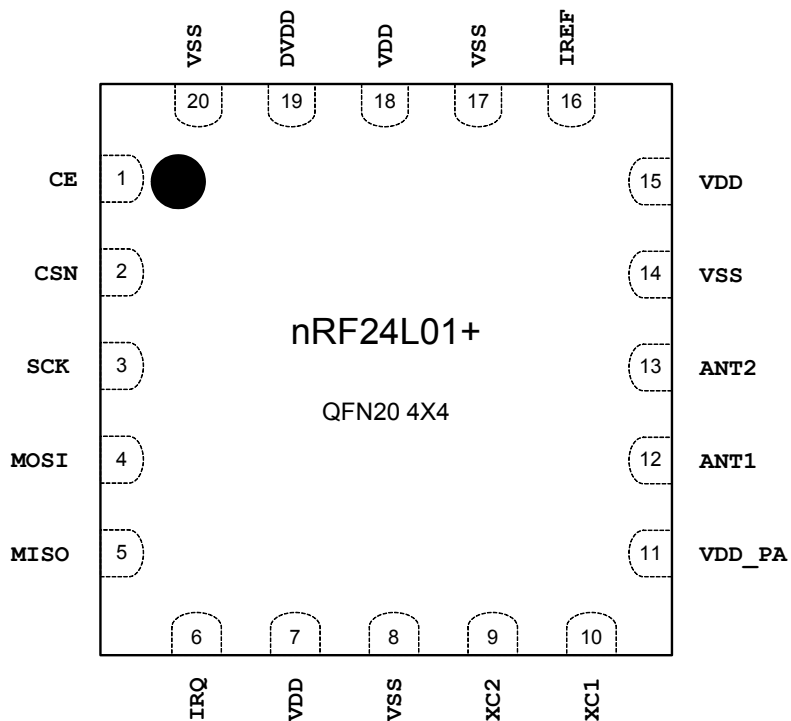


Figure 2. nRF24L01+ pin assignment (top view) for the QFN20 4x4 package

2 引脚信息

2.1 引脚分配

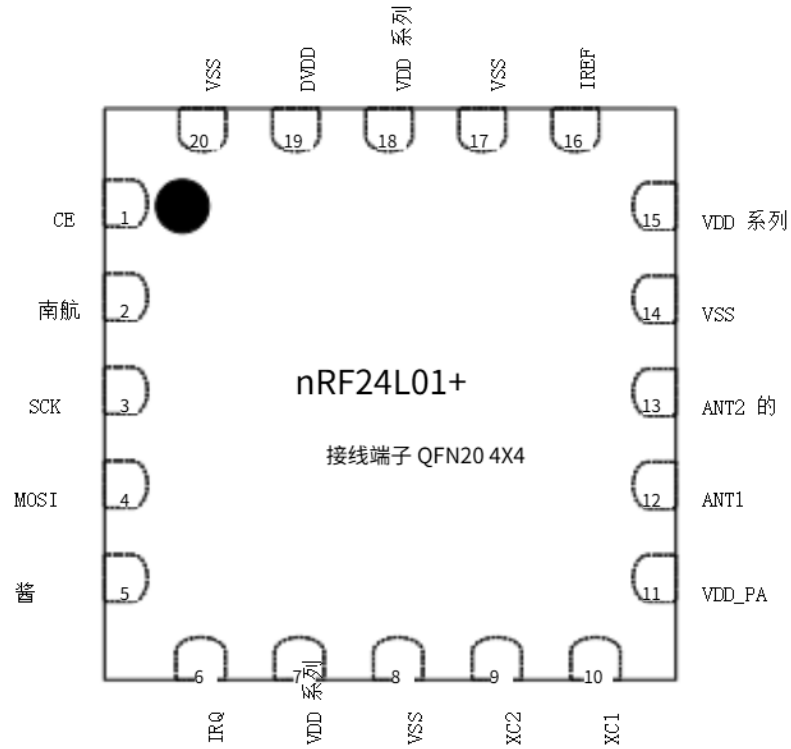


图 2.用于 QFN20 4x4 封装的 nRF24L01+ 引脚分配（顶视图）

2.2 Pin functions

Pin	Name	Pin function	Description
1	CE	Digital Input	Chip Enable Activates RX or TX mode
2	CSN	Digital Input	SPI Chip Select
3	SCK	Digital Input	SPI Clock
4	MOSI	Digital Input	SPI Slave Data Input
5	MISO	Digital Output	SPI Slave Data Output, with tri-state option
6	IRQ	Digital Output	Maskable interrupt pin. Active low
7	VDD	Power	Power Supply (+1.9V - +3.6V DC)
8	VSS	Power	Ground (0V)
9	XC2	Analog Output	Crystal Pin 2
10	XC1	Analog Input	Crystal Pin 1
11	VDD_PA	Power Output	Power Supply Output (+1.8V) for the internal nRF24L01+ Power Amplifier. Must be connected to ANT1 and ANT2 as shown in Figure 32 .
12	ANT1	RF	Antenna interface 1
13	ANT2	RF	Antenna interface 2
14	VSS	Power	Ground (0V)
15	VDD	Power	Power Supply (+1.9V - +3.6V DC)
16	IREF	Analog Input	Reference current. Connect a 22kΩ resistor to ground. See Figure 32 .
17	VSS	Power	Ground (0V)
18	VDD	Power	Power Supply (+1.9V - +3.6V DC)
19	DVDD	Power Output	Internal digital supply output for de-coupling purposes. See Figure 32 .
20	VSS	Power	Ground (0V)

Table 1. nRF24L01+ pin function

2.2 Pin 函数

Pin 名称	Pin 函数		描述
1	CE	数字输入	Chip Enable 激活 RX 或 TX 模式
2	南航	数字输入	SPI 片选
3	SCK	数字输入	SPI 时钟
4	MOSI	数字输入	SPI 从数据输入
5	酱	数字输出	SPI 从数据输出，带三态选项
6	IRQ	数字输出	可屏蔽中断引脚。低电平有效
7	VDD 系列	权力	电源 (+1.9V - +3.6V DC)
8	VSS	权力	接地 (0V)
9	XC2	模拟输出晶体引脚 2	
10	XC1	模拟输入	水晶针 1
11	VDD_PA	功率输出	内部 nRF24L01+ 电源的电源输出 (+1.8V) 放大器。必须连接到 ANT1 和 ANT2，如图 32 所示。
12	ANT1	射频	天线接口 1
13	ANT2 的	射频	天线接口 2
14	VSS	权力	接地 (0V)
15	VDD 系列	权力	电源 (+1.9V - +3.6V DC)
16	IREF	模拟输入	参考电流。将一个 22kΩ 电阻接地。参见图 32。
17	VSS	权力	接地 (0V)
18	VDD 系列	权力	电源 (+1.9V - +3.6V DC)
19	DVDD 电源输出 用于去耦目的的内部		数字电源输出。参见图 32。
20	VSS	权力	接地 (0V)

表 1.nRF24L01+ 引脚功能

3 Absolute maximum ratings

Note: Exceeding one or more of the limiting values may cause permanent damage to nRF24L01+.

Operating conditions	Minimum	Maximum	Units
Supply voltages			
VDD	-0.3	3.6	V
VSS		0	V
Input voltage			
V _I	-0.3	5.25	V
Output voltage			
V _O	VSS to VDD	VSS to VDD	
Total Power Dissipation			
P _D (T _A =85°C)		60	mW
Temperatures			
Operating Temperature	-40	+85	°C
Storage Temperature	-40	+125	°C

Table 2. Absolute maximum ratings

3 绝对最大额定值

请注意：超过一个或多个限制值可能会对 nRF24L01+ 造成永久性损坏。

工作条件	最低	最大	单位
电源电压			
VDD 系列	-0.3	3.6	V
VSS		0	V
输入电压			
V	-0.3	5.25	V
输出电压			
V	VSS 到 VDD	VSS 到 VDD	
总功耗			
P (T=85°C)		60	毫瓦
温度			
工作温度	-40	+85	°C
储存温度	-40	+125	°C

表 2.绝对最大额定值

4 Operating conditions

Symbol	Parameter (condition)	Notes	Min.	Typ.	Max.	Units
VDD	Supply voltage		1.9	3.0	3.6	V
VDD	Supply voltage if input signals >3.6V		2.7	3.0	3.3	V
TEMP	Operating Temperature		-40	+27	+85	°C

Table 3. Operating conditions

4 工作条件

象征	参数 (condition)	笔记	最小	典型值	最大	单位
VDD 系列	电源电压		1.9	3.0	3.6	V
VDD 系列	输入信号时的电源电压 >3.6V		2.7	3.0	3.3	V
临时	工作温度		-40	+27	+85	摄氏度

表 3.工作条件

5 Electrical specifications

Conditions: $V_{DD} = +3V$, $V_{SS} = 0V$, $T_A = -40^{\circ}C$ to $+85^{\circ}C$

5.1 Power consumption

Symbol	Parameter (condition)	Notes	Min.	Typ.	Max.	Units
	Idle modes					
I_{VDD_PD}	Supply current in power down			900		nA
I_{VDD_ST1}	Supply current in standby-I mode	a		26		μA
I_{VDD_ST2}	Supply current in standby-II mode			320		μA
I_{VDD_SU}	Average current during 1.5ms crystal oscillator startup			400		μA
	Transmit					
I_{VDD_TX0}	Supply current @ 0dBm output power	b		11.3		mA
I_{VDD_TX6}	Supply current @ -6dBm output power	b		9.0		mA
I_{VDD_TX12}	Supply current @ -12dBm output power	b		7.5		mA
I_{VDD_TX18}	Supply current @ -18dBm output power	b		7.0		mA
I_{VDD_AVG}	Average Supply current @ -6dBm output power, ShockBurst™	c		0.12		mA
I_{VDD_TXS}	Average current during TX settling	d		8.0		mA
	Receive					
I_{VDD_2M}	Supply current 2Mbps			13.5		mA
I_{VDD_1M}	Supply current 1Mbps			13.1		mA
I_{VDD_250}	Supply current 250kbps			12.6		mA
I_{VDD_RXS}	Average current during RX settling	e		8.9		mA

- This current is for a 12pF crystal. Current when using external clock is dependent on signal swing.
- Antenna load impedance = $15\Omega + j88\Omega$.
- Antenna load impedance = $15\Omega + j88\Omega$. Average data rate 10kbps and max. payload length packets.
- Average current consumption during TX startup (130 μs) and when changing mode from RX to TX (130 μs).
- Average current consumption during RX startup (130 μs) and when changing mode from TX to RX (130 μs).

Table 4. Power consumption

5 电气规格

条件: VDD = +3V, VSS = 0V, T = -40°C 至 +85°C

5.1 功耗

象征	参数 (condition)	笔记	最小	典型值	麦克斯。	单位
	空闲模式					
I	断电时电源电流			900		那
我 VDD_ST1	待机 I 模式下的电源电流	a		26		微安
I 待机 II 模式下的电源电流				320		微安
IVDD_SU	1.5ms 晶体振荡器启动期间的平均电流			400		微安
	传输					
我 VDD_TX0	电源电流 @ 0dBm 输出功率	b		11.3		马
我 VDD_TX6	电源电流 @ -6dBm 输出功率	b		9.0		马
我 VDD_TX12	电源电流 @ -12dBm 输出功率	b		7.5		马
I	电源电流 @ -18dBm 输出功率	b		7.0		马
IVDD_AVG	平均电源电流 @ -6dBm 输出功率, ShockBurst™	c		0.12		马
IVDD_TXS	TX 建立期间的平均电流	d		8.0		马
	收到					
IVDD_2M	电源电流 2Mbps			13.5		马
IVDD_1M	电源电流 1Mbps			13.1		马
我 VDD_250	电源电流 250kbps			12.6		马
IVDD_RXS	RX 建立期间的平均电流	e		8.9		马

一个。该电流适用于 12pF 晶体。使用外部 clock 时的电流取决于 signal swing。

b. 天线负载阻抗 = 15Ω+j88Ω..

c. 天线负载阻抗 = 15Ω+j88Ω。平均数据速率 10kbps 和最大有效载荷长度数据包。

d. TX 启动 (130μs) 和将模式从 RX 更改为 TX 时 (130μs) 的平均电流消耗。

e. RX 启动 (130μs) 和将模式从 TX 更改为 RX 时 (130μs) 的平均电流消耗。

表 4. 功耗

5.2 General RF conditions

Symbol	Parameter (condition)	Notes	Min.	Typ.	Max.	Units
f_{OP}	Operating frequency	a	2400		2525	MHz
PLL_{res}	PLL Programming resolution			1		MHz
f_{XTAL}	Crystal frequency			16		MHz
Δf_{250}	Frequency deviation @ 250kbps			± 160		kHz
Δf_{1M}	Frequency deviation @ 1Mbps			± 160		kHz
Δf_{2M}	Frequency deviation @ 2Mbps			± 320		kHz
R_{GFSK}	Air Data rate	b	250		2000	kbps
$F_{CHANNEL\ 1M}$	Non-overlapping channel spacing @ 250kbps/1Mbps	c		1		MHz
$F_{CHANNEL\ 2M}$	Non-overlapping channel spacing @ 2Mbps	c		2		MHz

a. Regulatory standards determine the band range you can use.

b. Data rate in each burst on-air

c. The minimum channel spacing is 1MHz

Table 5. General RF conditions

5.3 Transmitter operation

Symbol	Parameter (condition)	Notes	Min.	Typ.	Max.	Units
P_{RF}	Maximum Output Power	a		0	+4	dBm
P_{RFC}	RF Power Control Range		16	18	20	dB
P_{RFCR}	RF Power Accuracy				± 4	dB
P_{BW2}	20dB Bandwidth for Modulated Carrier (2Mbps)			1800	2000	kHz
P_{BW1}	20dB Bandwidth for Modulated Carrier (1Mbps)			900	1000	kHz
P_{BW250}	20dB Bandwidth for Modulated Carrier (250kbps)			700	800	kHz
$P_{RF1.2}$	1 st Adjacent Channel Transmit Power 2MHz (2Mbps)				-20	dBc
$P_{RF2.2}$	2 nd Adjacent Channel Transmit Power 4MHz (2Mbps)				-50	dBc
$P_{RF1.1}$	1 st Adjacent Channel Transmit Power 1MHz (1Mbps)				-20	dBc
$P_{RF2.1}$	2 nd Adjacent Channel Transmit Power 2MHz (1Mbps)				-45	dBc
$P_{RF1.250}$	1 st Adjacent Channel Transmit Power 1MHz (250kbps)				-30	dBc
$P_{RF2.250}$	2 nd Adjacent Channel Transmit Power 2MHz (250kbps)				-45	dBc

a. Antenna load impedance = $15\Omega + j88\Omega$

Table 6. Transmitter operation

5.2 一般 RF 条件

象征	参数 (condition)	笔记	最小	典型值	麦克斯。	单位
f	工作频率	a	2400		2525	兆赫
PLL PLL	编程分辨率			1		兆赫
f	晶体频率			16		兆赫
Δf	频率偏差 @ 250kbps			± 160		千赫
Δf	频率偏差 @ 1Mbps			± 160		千赫
Δf	频率偏差 @ 2Mbps			± 320		千赫
R	空中数据速率	b	250		2000	kbps 的
FCHANNEL 1M	非重叠信道间隔 @ 250kbps/ 1Mbps 的	c		1		兆赫
FCHANNEL 2M	非重叠信道间隔 @ 2Mbps	c		2		兆赫

一个。监管标准决定了您可以使用的频段范围。

b.每次广播突发的数据速率

c. 最小信道间隔为 1MHz

表 5.一般 RF 条件

5.3 发射器作

象征	参数 (condition)	笔记	最小	典型值	麦克斯。	单位
P	最大输出功率	a		0	+4	分贝米
P	RF 功率控制范围		-16	-18	-20	分贝
P	RF 功率精度				± 4	分贝
P	调制载波 20dB 带宽 (2Mbps)			-1800	-2000	千赫
P	调制载波 20dB 带宽 (1Mbps)			-900	-1000	千赫
P	调制载波 (250kbps) 的 20dB 带宽			-700	-800	千赫
P	1 个邻道发射功率 2MHz (2Mbps)				-20	分贝
P	2 邻道发射功率 4MHz (2Mbps)				-50	分贝
P	1 个邻道发射功率 1MHz (1Mbps)				-20	分贝
P	2 邻道发射功率 2MHz (1Mbps)				-45	分贝
P 射频 1.250	1 个邻道发射功率 1MHz (250kbps)				-30	分贝
P RF2.250	2 邻道发射功率 2MHz (250kbps)				-45	分贝

一个。天线负载阻抗 = $15\Omega + j88\Omega$

表 6.发射器作

5.4 Receiver operation

Datarate	Symbol	Parameter (condition)	Notes	Min.	Typ.	Max.	Units
	RX_{max}	Maximum received signal at <0.1% BER			0		dBm
2Mbps	RX_{SENS}	Sensitivity (0.1%BER) @2Mbps			-82		dBm
1Mbps	RX_{SENS}	Sensitivity (0.1%BER) @1Mbps			-85		dBm
250kbps	RX_{SENS}	Sensitivity (0.1%BER) @250kbps			-94		dBm

Table 7. RX Sensitivity

Datarate	Symbol	Parameter (condition)	Notes	Min.	Typ.	Max.	Units
2Mbps	C/I_{CO}	C/I Co-channel			7		dBc
	C/I_{1ST}	1 st ACS (Adjacent Channel Selectivity) C/I 2MHz			3		dBc
	C/I_{2ND}	2 nd ACS C/I 4MHz			-17		dBc
	C/I_{3RD}	3 rd ACS C/I 6MHz			-21		dBc
	C/I_{Nth}	N th ACS C/I, $f_i > 12\text{MHz}$			-40		dBc
	C/I_{Nth}	N th ACS C/I, $f_i > 36\text{MHz}$	a		-48		dBc
1Mbps	C/I_{CO}	C/I Co-channel			9		dBc
	C/I_{1ST}	1 st ACS C/I 1MHz			8		dBc
	C/I_{2ND}	2 nd ACS C/I 2MHz			-20		dBc
	C/I_{3RD}	3 rd ACS C/I 3MHz			-30		dBc
	C/I_{Nth}	N th ACS C/I, $f_i > 6\text{MHz}$			-40		dBc
	C/I_{Nth}	N th ACS C/I, $f_i > 25\text{MHz}$	a		-47		dBc
250kbps	C/I_{CO}	C/I Co-channel			12		dBc
	C/I_{1ST}	1 st ACS C/I 1MHz			-12		dBc
	C/I_{2ND}	2 nd ACS C/I 2MHz			-33		dBc
	C/I_{3RD}	3 rd ACS C/I 3MHz			-38		dBc
	C/I_{Nth}	N th ACS C/I, $f_i > 6\text{MHz}$			-50		dBc
	C/I_{Nth}	N th ACS C/I, $f_i > 25\text{MHz}$	a		-60		dBc

a. **Narrow Band (In Band) Blocking measurements:**

0 to $\pm 40\text{MHz}$; 1MHz step size

For Interferer frequency offsets $n \cdot 2 \cdot f_{xtal}$, blocking performance is degraded by approximately 5dB compared to adjacent figures.

Table 8. RX selectivity according to ETSI EN 300 440-1 V1.3.1 (2001-09) page 27

5.4 接收器作

数据速率	象征	参数 (condition)	注释	最小	典型值	最大单位	位数
	接收	最大接收信号为 <0.1% BER			0		分贝米
2Mbps RX	灵敏度 (0.1%BER) @2Mbps				-82		分贝米
1Mbps RX	灵敏度 (0.1%BER) @1Mbps				-85		分贝米
250kbps 的	接收	灵敏度 (0.1%BER) @250kbps			-94		分贝米

表 7.RX 灵敏度

Datarate Symbol	参数 (condition)	注释	最小	典型值	最大单位	
2Mbps 的 C/I	C/I 同频道			-7		分贝
	C/I 1ACS (邻道选择性) C/I 2MHz			-3		分贝
	C/I 2ACS C/I 4MHz			-17		分贝
	C/I 3ACS C/I 6MHz			-21		分贝
	C/I N ACS C/I, f> 12MHz			-40		分贝
	C/I N ACS C/I, f> 36MHz	a		-48		分贝
1Mbps 的 C/I	C/I 同频道			-9		分贝
	C/I 1ACS C/I 1MHz			-8		分贝
	C/I 2ACS C/I 2MHz			-20		分贝
	C/I 3ACS C/I 3MHz			-30		分贝
	C/I N ACS C/I, f> 6MHz			-40		分贝
	C/I N ACS C/I, f> 25MHz	a		-47		分贝
250kbps 的 C/I	C/I 同频道			-12		分贝
	C/I 1ACS C/I 1MHz			-12		分贝
	C/I 2ACS C/I 2MHz			-33		分贝
	C/I 3ACS C/I 3MHz			-38		分贝
	C/I N ACS C/I, f> 6MHz			-50		分贝
	C/I N ACS C/I, f> 25MHz	a		-60		分贝

一个。窄带（带内）阻塞测量：

0 至 $\pm 40\text{MHz}$; 1MHz 步长 对于干扰源频率偏移 $n \cdot 2 \cdot f_{\text{xtal}}$ ，与相邻数字相比，阻塞性能下降了大约 5dB。

表 8.RX 选择性符合 ETSI EN 300 440-1 V1.3.1 (2001-09) 第 27 页

Datarate	Symbol	Parameter (condition)	Notes	Min.	Typ.	Max.	Units
2Mbps	C/I _{CO}	C/I Co-channel (Modulated carrier)			11		dBc
	C/I _{1ST}	1 st ACS C/I 2MHz			4		dBc
	C/I _{2ND}	2 nd ACS C/I 4MHz			-18		dBc
	C/I _{3RD}	3 rd ACS C/I 6MHz			-24		dBc
	C/I _{Nth}	N th ACS C/I, $f_i > 12\text{MHz}$			-40		dBc
	C/I _{Nth}	N th ACS C/I, $f_i > 36\text{MHz}$	a		-48		dBc
1Mbps	C/I _{CO}	C/I Co-channel			12		dBc
	C/I _{1ST}	1 st ACS C/I 1MHz			8		dBc
	C/I _{2ND}	2 nd ACS C/I 2MHz			-21		dBc
	C/I _{3RD}	3 rd ACS C/I 3MHz			-30		dBc
	C/I _{Nth}	N th ACS C/I, $f_i > 6\text{MHz}$			-40		dBc
	C/I _{Nth}	N th ACS C/I, $f_i > 25\text{MHz}$	a		-50		dBc
250kbps	C/I _{CO}	C/I Co-channel			7		dBc
	C/I _{1ST}	1 st ACS C/I 1MHz			-12		dBc
	C/I _{2ND}	2 nd ACS C/I 2MHz			-34		dBc
	C/I _{3RD}	3 rd ACS C/I 3MHz			-39		dBc
	C/I _{Nth}	N th ACS C/I, $f_i > 6\text{MHz}$			-50		dBc
	C/I _{Nth}	N th ACS C/I, $f_i > 25\text{MHz}$	a		-60		dBc

a. **Narrow Band (In Band) Blocking measurements:**

0 to $\pm 40\text{MHz}$; 1MHz step size

Wide Band Blocking measurements:

30MHz to 2000MHz; 10MHz step size

2000MHz to 2399MHz; 3MHz step size

2484MHz to 3000MHz; 3MHz step size

3GHz to 12.75GHz; 25MHz step size

Wanted signal for wideband blocking measurements:

-67dBm in 1Mbps and 2Mbps mode

-77dBm in 250kbps mode

For Interferer frequency offsets $n \cdot 2 \cdot f_{xtal}$, blocking performance are degraded by approximately 5dB compared to adjacent figures.

If the wanted signal is 3dB or more above the sensitivity level then, the carrier/interferer ratio is independent of the wanted signal level for a given frequency offset.

Table 9. RX selectivity with nRF24L01+ equal modulation on interfering signal. Measured using $P_{in} = -67\text{dBm}$ for wanted signal.

Datarate	Symbol	参数 (condition)	注释	最小值	典型值	麦克斯。	单位
2Mbps 的	C/I	C/I 同通道 (调制载波)				-11	分贝
	C/I	1ACS C/I 2 兆赫				4	分贝
	C/I	2ACS C/I 4MHz				-18	分贝
	C/I	3ACS C/I 6MHz				-24	分贝
	C/I	N ACS C/I, f > 12MHz				-40	分贝
	C/I	N ACS C/I, f > 36MHz		a		-48	分贝
	C/I	C/I 同频道				-12	分贝
1Mbps 的	C/I	1ACS C/I 1 兆赫				8	分贝
	C/I	2ACS C/I 2MHz				-21	分贝
	C/I	3ACS C/I 3MHz				-30	分贝
	C/I	N ACS C/I, f > 6MHz				-40	分贝
	C/I	N ACS C/I, f > 25MHz		a		-50	分贝
	C/I	C/I 同频道				-7	分贝
	C/I	1ACS C/I 1 兆赫				-12	分贝
250kbps 的	C/I	2ACS C/I 2MHz				-34	分贝
	C/I	3ACS C/I 3MHz				-39	分贝
	C/I	N ACS C/I, f > 6MHz				-50	分贝
	C/I	N ACS C/I, f > 25MHz		a		-60	分贝
	C/I	C/I 同频道				-7	分贝
	C/I	1ACS C/I 1 兆赫				-12	分贝
	C/I	2ACS C/I 2MHz				-34	分贝

a. 窄带 (In Band) 阻塞测量:

0 至 $\pm 40\text{MHz}$; 1MHz 步长

宽带块测量:

30MHz 至 2000MHz; 10MHz 步长 2000MHz 至 2399MHz; 3MHz 步

长 2484MHz 至 3000MHz; 3MHz 步长 3GHz 至 12.75GHz; 25MHz

步长用于宽带阻塞测量的有用信号: 1Mbps 和 2Mbps 模式下为

-67dBm

-77dBm 在 250kbps 模式下对于干扰频率偏移 $n \cdot 2 \cdot f_{xtal}$, 与相邻数字相比, 阻塞性能下降了大约 5dB。如果有用信号比灵敏度电平高 3dB 或更高, 则载波/干扰比与给定频率偏移的有用信号电平无关。

表 9.nRF24L01+ 对干扰信号的 RX 选择性相等。测量条件: Pin = -67dBm 表示有用信号。

Datarate	Symbol	Parameter (condition)	Notes	Min.	Typ.	Max.	Units
2Mbps	P_IM(6)	Input power of IM interferers at 6 and 12MHz offset from wanted signal			-42		dBm
	P_IM(8)	Input power of IM interferers at 8 and 16MHz offset from wanted signal			-38		dBm
	P_IM(10)	Input power of IM interferers at 10 and 20MHz offset from wanted signal			-37		dBm
1Mbps	P_IM(3)	Input power of IM interferers at 3 and 6MHz offset from wanted signal			-36		dBm
	P_IM(4)	Input power of IM interferers at 4 and 8MHz offset from wanted signal			-36		dBm
	P_IM(5)	Input power of IM interferers at 5 and 10MHz offset from wanted signal			-36		dBm
250kbps	P_IM(3)	Input power of IM interferers at 3 and 6MHz offset from wanted signal			-36		dBm
	P_IM(4)	Input power of IM interferers at 4 and 8MHz offset from wanted signal			-36		dBm
	P_IM(5)	Input power of IM interferers at 5 and 10MHz offset from wanted signal			-36		dBm

Note: Wanted signal level at Pin = -64 dBm. Two interferers with equal input power are used. The interferer closest in frequency is unmodulated, the other interferer is modulated equal with the wanted signal. The input power of interferers where the sensitivity equals BER = 0.1% is presented.

Table 10. RX intermodulation test performed according to Bluetooth Specification version 2.0

Datarate	Symbol	参数 (condition)	备注	最小	典型值	最大	单位
2Mbps	P_IM (6)	IM 干扰源的输入功率在与有用信号相距 6 MHz 和 12MHz 时偏移			-42		分贝米
	P_IM (8)	IM 干扰源在 8 和 16MHz 偏移时的输入功率 来自通缉信号			-38		分贝米
	P_IM (10)	10 和 20MHz 偏移时 IM 干扰源的输入功率 来自通缉信号			-37		分贝米
1Mbps	P_IM (3)	3 MHz 和 6MHz 偏移时 IM 干扰源的输入功率 来自通缉信号			-36		分贝米
	P_IM (4)	4 MHz 和 8MHz 偏移时 IM 干扰源的输入功率 来自通缉信号			-36		分贝米
	P_IM (5)	5 MHz 和 10MHz 偏移时 IM 干扰源的输入功率 来自通缉信号			-36		分贝米
250kbps	P_IM (3)	IM 干扰源在 3 和 6MHz 偏移时的输入功率 来自通缉信号			-36		分贝米
	P_IM (4)	4 MHz 和 8MHz 偏移时 IM 干扰源的输入功率 来自通缉信号			-36		分贝米
	P_IM (5)	5 MHz 和 10MHz 偏移时 IM 干扰源的输入功率 来自通缉信号			-36		分贝米

注意：Pin = -64 dBm 时有用信号电平。使用两个输入功率相等的干扰源。这
频率最接近的干扰源未调制，另一个干扰源的调制与有用信号相等。给出了灵敏度等于 BER = 0.1% 的干
扰源的输入功率。

表 10.根据蓝牙规范 2.0 版执行 RX 互调测试

5.5 Crystal specifications

Symbol	Parameter (condition)	Notes	Min.	Typ.	Max.	Units
F _{xo}	Crystal Frequency			16		MHz
ΔF	Tolerance	a b			±60	ppm
C ₀	Equivalent parallel capacitance			1.5	7.0	pF
L _s	Equivalent serial inductance	c		30		mH
C _L	Load capacitance		8	12	16	pF
ESR	Equivalent Series Resistance				100	Ω

- a. Frequency accuracy including; tolerance at 25°C, temperature drift, aging and crystal loading.
b. Frequency regulations in certain regions set tighter requirements for frequency tolerance (For example, Japan and South Korea specify max. +/- 50ppm).
c. Startup time from power down to standby mode is dependant on the L_s parameter. See [Table 16. on page 24](#) for details.

Table 11. Crystal specifications

The crystal oscillator startup time is proportional to the crystal equivalent inductance. The trend in crystal design is to reduce the physical outline. An effect of a small outline is an increase in equivalent serial inductance L_s, which gives a longer startup time. The maximum crystal oscillator startup time, T_{pd2stby} = 1.5 ms, is set using a crystal with equivalent serial inductance of maximum 30mH.

An application specific worst case startup time can be calculated as :

T_{pd2stby} = L_s/30mH * 1.5ms if L_s exceeds 30mH.

Note: In some crystal datasheets L_s is called L1 or Lm and C_s is called C1 or Cm.

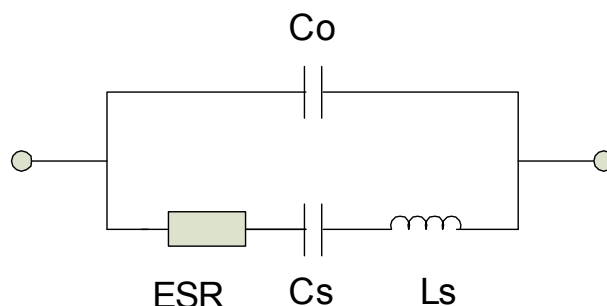


Figure 3. Equivalent crystal components

5.5 晶体规格

象征	参数 (condition)	笔记	最小	典型值	麦克斯。	单位
Fxo	晶体频率			16		兆赫
ΔF	宽容	一个 b			±60	ppm
C	等效并联电容			1.5	7.0	pF
Ls	等效串联电感	c		30		mH
C	负载电容		8	12	16	pF
红细胞沉降率	等效串联电阻				100	Ω

一个。频率精度包括;在 25 °C 下的耐受性、温度漂移、老化和晶体负载。b.某些地区的频率法规对频率容差提出了更严格的要求 (例如, 日本和韩国指定最大 +/- 50 ppm)。

c. 从断电到待机模式的启动时间取决于 Ls 参数。请参阅 表 16。有关详细信息, 请参阅第 24 页。

表 11.晶体规格

晶体振荡器启动时间与晶体等效电感成正比。晶体设计的趋势是减少物理轮廓。小轮廓的一个影响是等效串联电感 Ls 的增加, 从而提供更长的启动时间。晶体振荡器的最大启动时间 Tpd2stby = 1.5 ms, 使用等效串联电感最大为 30mH 的晶体进行设置。

应用程序特定的最坏情况启动时间可以计算为:

Tpd2stby= Ls/30mH *如果 Ls 超过 30mH, 则为 1.5ms。

注: 在一些晶体数据表中, Ls 称为 L1 或 Lm, Cs 称为 C1 或 Cm。

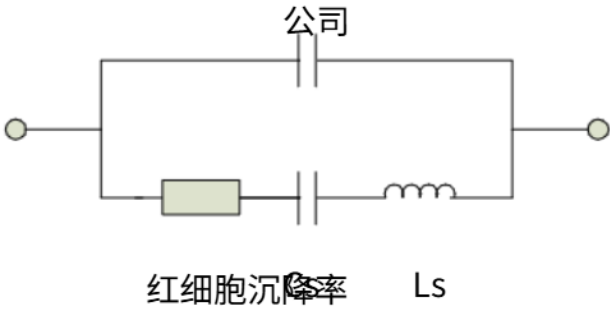


图 3.等效晶体组件

5.6 DC characteristics

Symbol	Parameter (condition)	Notes	Min.	Typ.	Max.	Units
V_{IH}	HIGH level input voltage		$0.7V_{DD}$		5.25^a	V
V_{IL}	LOW level input voltage		V_{SS}		$0.3V_{DD}$	V

a. If the input signal $>3.6V$, the V_{DD} of the nRF24L01+ must be between 2.7V and 3.3V ($3.0V \pm 10\%$)

Table 12. Digital input pin

Symbol	Parameter (condition)	Notes	Min.	Typ.	Max.	Units
V_{OH}	HIGH level output voltage ($I_{OH} = -0.25mA$)		$V_{DD} - 0.3$		V_{DD}	V
V_{OL}	LOW level output voltage ($I_{OL} = 0.25mA$)				0.3	V

Table 13. Digital output pin

5.7 Power on reset

Symbol	Parameter (condition)	Notes	Min.	Typ.	Max.	Units
T_{PUP}	Power ramp up time	a			100	ms
T_{POR}	Power on reset	b	1		100	ms

a. From 0V to 1.9V.

b. Measured from when the V_{DD} reaches 1.9V to when the reset finishes.

Table 14. Power on reset

5.6 直流特性

象征	参数 (condition)	笔记	最小	典型值	麦克斯。	单位
V	HIGH level 输入电压		0.7VDD		5.25	V
V	低电平输入电压		VSS		0.3VDD	V

一个。如果输入信号>3.6V，则 nRF24L01+ 的 VDD 必须在 2.7V 和 3.3V (3.0V±10%) 之间

表 12.数字输入引脚

象征	参数 (condition)	笔记	最小	典型值	麦克斯。	单位
V	高电平输出电压 (I =-0.25mA)		VDD -0.3		VDD 系列	V
V	低电平输出电压 (I =0.25mA)				0.3	V

表 13.数字输出引脚

5.7 开机重置

象征	参数 (condition)	笔记	最小	典型值	麦克斯。	单位
T	功率上升时间	a			100	女士
T	开机重置	b	1		100	女士

一个。从 0V 到 1.9V。

b.从 VDD 达到 1.9V 到 reset 完成测量。

表 14.开机重置

6 Radio Control

This chapter describes the nRF24L01+ radio transceiver's operating modes and the parameters used to control the radio.

The nRF24L01+ has a built-in state machine that controls the transitions between the chip's operating modes. The state machine takes input from user defined register values and internal signals.

6.1 Operational Modes

You can configure the nRF24L01+ in power down, standby, RX or TX mode. This section describes these modes in detail.

6.1.1 State diagram

The state diagram in [Figure 4](#), shows the operating modes and how they function. There are three types of distinct states highlighted in the state diagram:

- **Recommended operating mode:** is a recommended state used during normal operation.
- **Possible operating mode:** is a possible operating state, but is not used during normal operation.
- **Transition state:** is a time limited state used during start up of the oscillator and settling of the PLL.

When the V_{DD} reaches 1.9V or higher nRF24L01+ enters the Power on reset state where it remains in reset until entering the Power Down mode.

6 无线电控制

本章介绍 nRF24L01+ 无线电收发器的工作模式和用于控制无线电的参数。

nRF24L01+ 具有内置状态机，可控制芯片工作模式之间的转换。状态机从用户定义的寄存器值和内部信号中获取输入。

6.1 作模式

您可以将 nRF24L01+ 配置为掉电、待机、RX 或 TX 模式。本节将详细介绍这些模式。

6.1.1 状态图

图 4 中的状态图。显示作模式及其功能。状态图中突出显示了三种类型的不同状态：

- Recommended operating mode（推荐作模式）：是正常作期间使用的推荐状态。
- 可能的运行模式：是一种可能的运行状态，但在正常作期间不使用。
- Transition state：是振荡器启动和 PLL 建立期间使用的时间限制状态。

当 VDD 达到 1.9V 或更高时，nRF24L01+ 进入开机复位状态，保持复位状态，直到进入掉电模式。

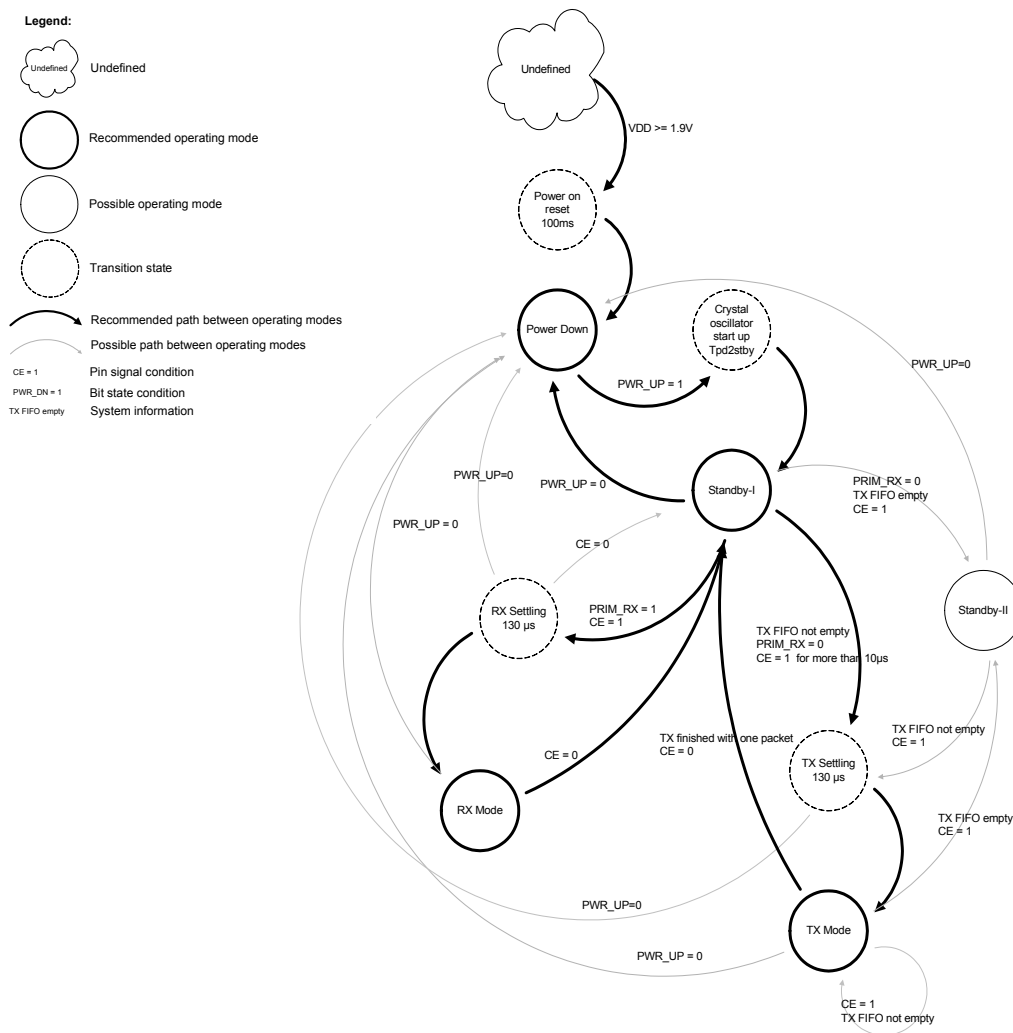


Figure 4. Radio control state diagram

6.1.2 Power Down Mode

In **power down mode** nRF24L01+ is disabled using minimal current consumption. All register values available are maintained and the SPI is kept active, enabling change of configuration and the uploading/downloading of data registers. For start up times see [Table 16. on page 24](#). Power down mode is entered by setting the `PWR_UP` bit in the `CONFIG` register low.

6.1.3 Standby Modes

6.1.3.1 Standby-I mode

By setting the `PWR_UP` bit in the `CONFIG` register to 1, the device enters standby-I mode. Standby-I mode is used to minimize average current consumption while maintaining short start up times. In this mode only part of the crystal oscillator is active. Change to active modes only happens if `CE` is set high and when `CE` is set low, the nRF24L01 returns to standby-I mode from both the TX and RX modes.

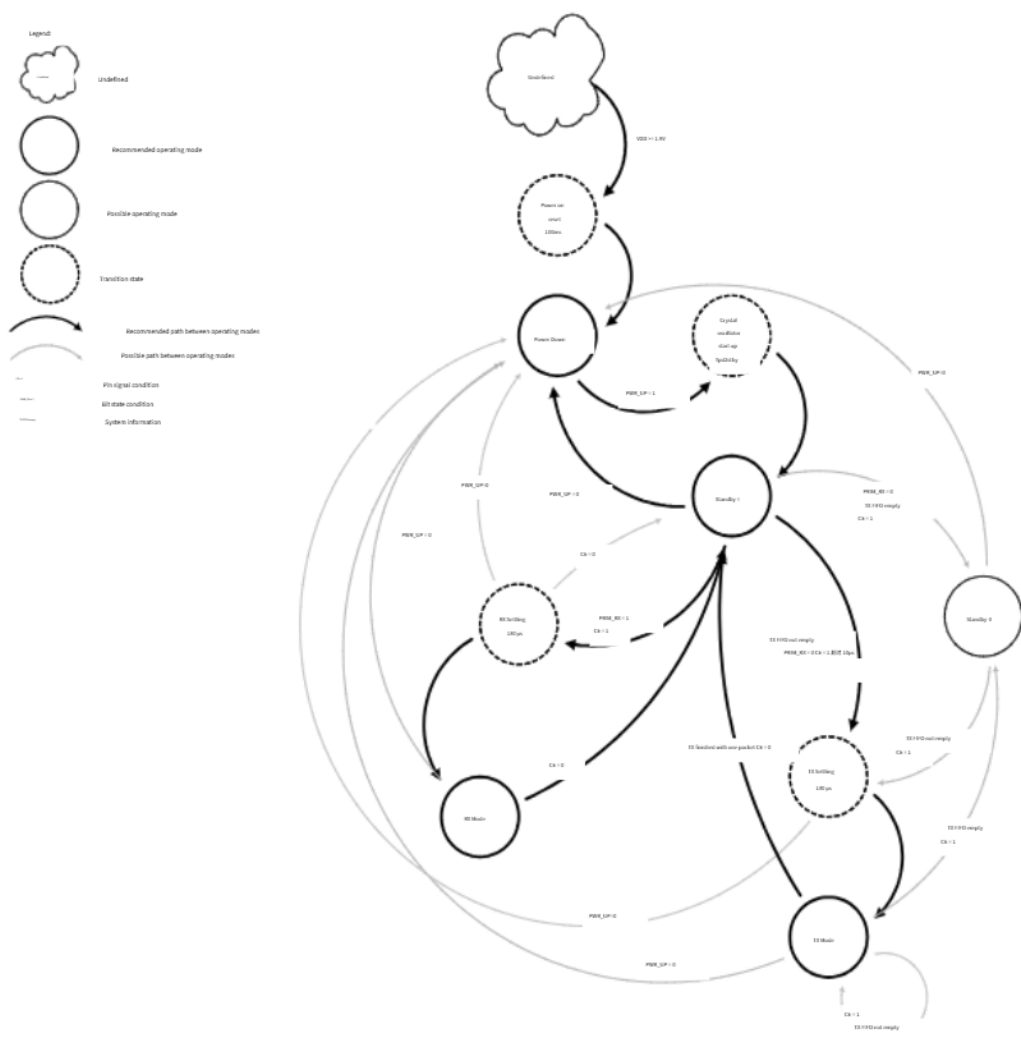


图 4.无线电控制状态图

6.1.2 关机模式

在掉电模式下，nRF24L01+ 以最小的电流消耗被禁用。所有可用的寄存器值都得到维护。SPI 保持活动状态，从而可以更改配置和上传/下载数据寄存器。有关启动时间，请参见表 16。在第 24 页。通过将 CONFIG 寄存器中的 PWR_UP 位设置为低电平来进入掉电模式。

6.1.3 待机模式

6.1.3.1 Standby-I 模式

通过将 CONFIG 寄存器中的 PWR_UP 位设置为 1，器件进入 standby-I 模式。Standby-I 模式用于最大限度地降低平均电流消耗，同时保持较短的启动时间。在此模式下，只有部分晶体振荡器处于活动状态。仅当 CE 设置为高电平时，才会更改为活动模式，当 CE 设置为低电平时，nRF24L01 会从 TX 和 RX 模式返回待机 I 模式。

6.1.3.2 Standby-II mode

In standby-II mode extra clock buffers are active and more current is used compared to standby-I mode. nRF24L01+ enters standby-II mode if **CE** is held high on a PTX device with an empty TX FIFO. If a new packet is uploaded to the TX FIFO, the PLL immediately starts and the packet is transmitted after the normal PLL settling delay (130µs).

Register values are maintained and the SPI can be activated during both standby modes. For start up times see [Table 16. on page 24](#).

6.1.4 RX mode

The RX mode is an active mode where the nRF24L01+ radio is used as a receiver. To enter this mode, the nRF24L01+ must have the **PWR_UP** bit, **PRIM_RX** bit and the **CE** pin set high.

In RX mode the receiver demodulates the signals from the RF channel, constantly presenting the demodulated data to the baseband protocol engine. The baseband protocol engine constantly searches for a valid packet. If a valid packet is found (by a matching address and a valid CRC) the payload of the packet is presented in a vacant slot in the RX FIFOs. If the RX FIFOs are full, the received packet is discarded.

The nRF24L01+ remains in RX mode until the MCU configures it to standby-I mode or power down mode. However, if the automatic protocol features (Enhanced ShockBurst™) in the baseband protocol engine are enabled, the nRF24L01+ can enter other modes in order to execute the protocol.

In RX mode a Received Power Detector (RPD) signal is available. The RPD is a signal that is set high when a RF signal higher than -64 dBm is detected inside the receiving frequency channel. The internal **RPD** signal is filtered before presented to the **RPD** register. The RF signal must be present for at least 40µs before the **RPD** is set high. How to use the RPD is described in [Section 6.4 on page 25](#).

6.1.5 TX mode

The TX mode is an active mode for transmitting packets. To enter this mode, the nRF24L01+ must have the **PWR_UP** bit set high, **PRIM_RX** bit set low, a payload in the TX FIFO and a high pulse on the **CE** for more than 10µs.

The nRF24L01+ stays in TX mode until it finishes transmitting a packet. If **CE** = 0, nRF24L01+ returns to standby-I mode. If **CE** = 1, the status of the TX FIFO determines the next action. If the TX FIFO is not empty the nRF24L01+ remains in TX mode and transmits the next packet. If the TX FIFO is empty the nRF24L01+ goes into standby-II mode. The nRF24L01+ transmitter PLL operates in open loop when in TX mode. It is important never to keep the nRF24L01+ in TX mode for more than 4ms at a time. If the Enhanced ShockBurst™ features are enabled, nRF24L01+ is never in TX mode longer than 4ms.

6.1.3.2 待机 II 模式

在 standby-II 模式下，额外的 clock buffers 处于活动状态，并且与 standby-I 模式相比，使用更多的电流。如果 CE 在具有空 TX FIFO 的 PTX 设备上保持高电平，nRF24L01+ 将进入待机 II 模式。如果将新数据包上传到 TX FIFO，则 PLL 立即启动，并在正常的 PLL 建立延迟（130μs）之后传输数据包。

寄存器值保持不变，并且 SPI 可以在两种待机模式下激活。有关启动时间，请参见表 16。在第 24 页。

6.1.4 RX 模式

RX 模式是一种主动模式，其中 nRF24L01+ 无线电用作接收器。要进入此模式，nRF24L01+ 必须将 PWR_UP 位、PRIM_RX 位和 CE 引脚设置为高电平。

在 RX 模式下，接收器解调来自 RF 通道的信号，不断将解调后的数据提供给基带协议引擎。基带协议引擎不断搜索有效的数据包。如果找到有效数据包（通过匹配地址和有效 CRC），则数据包的有效负载将显示在 RX FIFO 的空插槽中。如果 RX FIFO 已满，则接收到的数据包将被丢弃。

nRF24L01+ 保持 RX 模式，直到 MCU 将其配置为待机 I 模式或断电模式。但是，如果启用了基带协议引擎中的自动协议功能（增强型 ShockBurst™），则 nRF24L01+ 可以进入其他模式以执行协议。

在 RX 模式下，接收功率检测器（RPD）信号可用。RPD 是在接收频率信道内检测到高于 -64 dBm 的 RF 信号时设置的高信号。内部 RPD 信号在提供给 RPD 寄存器之前经过过滤。在 RPD 设置为高电平之前，RF 信号必须存在至少 40μs。第 25 页的第 6.4 节介绍了如何使用 RPD。

6.1.5 TX 模式

TX 模式是一种用于传输数据包的活动模式。要进入此模式，nRF24L01+ 必须将 PWR_UP 位设置为高位，PRIM_RX 位设置为低位，在 TX FIFO 中有一个有效载荷，并且在 CE 上有一个超过 10μs 的高脉冲。

nRF24L01+ 保持 TX 模式，直到完成数据包传输。如果 CE = 0，nRF24L01+ 返回待机 I 模式。如果 CE = 1，则 TX FIFO 的状态将决定下一个动作。如果 TX FIFO 不为空，则 nRF24L01+ 保持 TX 模式并发送下一个数据包。如果 TX FIFO 为空，nRF24L01+ 将进入待机 II 模式。nRF24L01+ 发射器 PLL 在 TX 模式下以开环方式运行。重要的是，切勿一次将 nRF24L01+ 保持在 TX 模式下超过 4 毫秒。如果启用了增强型 ShockBurst™ 功能，nRF24L01+ 处于 TX 模式的时间永远不会超过 4 毫秒。

6.1.6 Operational modes configuration

The following table ([Table 15.](#)) describes how to configure the operational modes.

Mode	PWR_UP register	PRIM_RX register	CE input pin	FIFO state
RX mode	1	1	1	-
TX mode	1	0	1	Data in TX FIFOs. Will empty all levels in TX FIFOs ^a .
TX mode	1	0	Minimum 10µs high pulse	Data in TX FIFOs. Will empty one level in TX FIFOs ^b .
Standby-II	1	0	1	TX FIFO empty.
Standby-I	1	-	0	No ongoing packet transmission.
Power Down	0	-	-	-

- If **CE** is held high all TX FIFOs are emptied and all necessary ACK and possible retransmits are carried out. The transmission continues as long as the TX FIFO is refilled. If the TX FIFO is empty when the **CE** is still high, nRF24L01+ enters standby-II mode. In this mode the transmission of a packet is started as soon as the **CSN** is set high after an upload (UL) of a packet to TX FIFO.
- This operating mode pulses the **CE** high for at least 10µs. This allows one packet to be transmitted. This is the normal operating mode. After the packet is transmitted, the nRF24L01+ enters standby-I mode.

Table 15. nRF24L01+ main modes

6.1.7 Timing Information

The timing information in this section relates to the transitions between modes and the timing for the **CE** pin. The transition from TX mode to RX mode or vice versa is the same as the transition from the standby modes to TX mode or RX mode (max. 130µs), as described in [Table 16.](#)

Name	nRF24L01+	Notes	Max.	Min.	Comments
Tpd2stby	Power Down → Standby mode	a	150µs		With external clock
			1.5ms		External crystal, Ls < 30mH
			3ms		External crystal, Ls = 60mH
			4.5ms		External crystal, Ls = 90mH
Tstby2a	Standby modes → TX/RX mode		130µs		
Thce	Minimum CE high			10µs	
Tpece2csn	Delay from CE positive edge to CSN low			4µs	

- See [Table 11. on page 19](#) for crystal specifications.

Table 16. Operational timing of nRF24L01+

For nRF24L01+ to go from power down mode to TX or RX mode it must first pass through stand-by mode. There must be a delay of Tpd2stby (see [Table 16.](#)) after the nRF24L01+ leaves power down mode before the **CE** is set high.

Note: If **VDD** is turned off the register value is lost and you must configure nRF24L01+ before entering the TX or RX modes.

6.2 Air data rate

The air data rate is the modulated signaling rate the nRF24L01+ uses when transmitting and receiving data. It can be 250kbps, 1Mbps or 2Mbps. Using lower air data rate gives better receiver sensitivity than higher air data rate. But, high air data rate gives lower average current consumption and reduced probability of on-air collisions.

The air data rate is set by the `RF_DR` bit in the `RF_SETUP` register. A transmitter and a receiver must be programmed with the same air data rate to communicate with each other.

nRF24L01+ is fully compatible with nRF24L01. For compatibility with nRF2401A, nRF2402, nRF24E1, and nRF24E2 the air data rate must be set to 250kbps or 1Mbps.

6.3 RF channel frequency

The RF channel frequency determines the center of the channel used by the nRF24L01+. The channel occupies a bandwidth of less than 1MHz at 250kbps and 1Mbps and a bandwidth of less than 2MHz at 2Mbps. nRF24L01+ can operate on frequencies from 2.400GHz to 2.525GHz. The programming resolution of the RF channel frequency setting is 1MHz.

At 2Mbps the channel occupies a bandwidth wider than the resolution of the RF channel frequency setting. To ensure non-overlapping channels in 2Mbps mode, the channel spacing must be 2MHz or more. At 1Mbps and 250kbps the channel bandwidth is the same or lower than the resolution of the RF frequency.

The RF channel frequency is set by the `RF_CH` register according to the following formula:

$$F_0 = 2400 + RF_CH [MHz]$$

You must program a transmitter and a receiver with the same RF channel frequency to communicate with each other.

6.4 Received Power Detector measurements

Received Power Detector (RPD), located in register 09, bit 0, triggers at received power levels above -64 dBm that are present in the RF channel you receive on. If the received power is less than -64 dBm, RDP = 0.

The RPD can be read out at any time while nRF24L01+ is in receive mode. This offers a snapshot of the current received power level in the channel. The RPD status is latched when a valid packet is received which then indicates signal strength from your own transmitter. If no packets are received the RPD is latched at the end of a receive period as a result of host MCU setting CE low or RX time out controlled by Enhanced ShockBurst™.

The status of RPD is correct when RX mode is enabled and after a wait time of $T_{stby2a} + T_{delay_AGC} = 130\mu s + 40\mu s$. The RX gain varies over temperature which means that the RPD threshold also varies over temperature. The RPD threshold value is reduced by - 5dB at $T = -40^{\circ}C$ and increased by + 5dB at $85^{\circ}C$.

6.2 空中数据速率

空中数据速率是 nRF24L01+ 在发送和接收数据时使用的调制信号速率。它可以是 250kbps、1Mbps 或 2Mbps。使用较低的空中数据速率比使用较高的空中数据速率提供更好的接收器灵敏度。但是，高空中数据速率可以降低平均电流消耗，并降低空中冲突的可能性。

空中数据速率由 RF_SETUP 寄存器中的 RF_DR 位设置。发射器和接收器必须使用相同的空中数据速率进行编程，才能相互通信。

nRF24L01+ 与 nRF24L01 完全兼容。为了与 nRF2401A、nRF2402、nRF24E1 和 nRF24E2 兼容，空中数据速率必须设置为 250kbps 或 1Mbps。

6.3 RF 信道频率

射频通道频率决定了 nRF24L01+ 使用的通道中心。该通道在 1kbps 和 250Mbps 时占用小于 1MHz 的带宽，在 2Mbps 时占用小于 2MHz 的带宽。nRF24L01+ 可以在 2.400GHz 至 2.525GHz 的频率上运行。RF 通道频率设置的编程分辨率为 1MHz。

在 2Mbps 时，通道占用的带宽比 RF 通道频率设置的分辨率更宽。为确保在 2Mbps 模式下不重叠的信道，信道间隔必须为 2MHz 或更大。在 1Mbps 和 250kbps 时，信道带宽等于或低于 RF 频率的分辨率。

RF 通道频率由 RF_CH 寄存器根据以下公式设置：

$$F = 2400 + RF_CH \text{ [MHz]}$$

您必须对具有相同 RF 信道频率的发射器和接收器进行编程，以便相互通信。

6.4 接收功率检测器测量

接收功率检测器（RPD）位于寄存器 09 位 0 中，在接收功率电平高于 -64 dBm 时触发，该电平存在于您接收的 RF 信道中。如果接收功率小于 -64 dBm，则 RPD = 0。

当 nRF24L01+ 处于接收模式时，可以随时读出 RPD。这提供了通道中当前接收功率电平的快照。当收到有效数据包时，RPD 状态将被锁定，然后指示来自您自己的发射器的信号强度。如果未收到数据包，则由于主机 MCU 设置、CE 低电平或 RX 超时由增强型 ShockBurst™ 控制，RPD 将在接收周期结束时被锁定。

当启用 RX 模式时，并且在等待时间为 $T_{stby2a} + T_{delay_AGC} = 130\mu s + 40\mu s$ 后，RPD 的状态是正确的。RX 增益随温度变化，这意味着 RPD 阈值也随温度变化。RPD 阈值在 $T = -40^{\circ}\text{C}$ 时降低 -5dB，在 85°C 时增加 +5dB。

6.5 PA control

The PA (Power Amplifier) control is used to set the output power from the nRF24L01+ power amplifier. In TX mode PA control has four programmable steps, see [Table 17](#).

The PA control is set by the `RF_PWR` bits in the `RF_SETUP` register.

SPI RF-SETUP (RF_PWR)	RF output power	DC current consumption
11	0dBm	11.3mA
10	-6dBm	9.0mA
01	-12dBm	7.5mA
00	-18dBm	7.0mA

Conditions: $V_{DD} = 3.0V$, $V_{SS} = 0V$, $T_A = 27^{\circ}C$, Load impedance = $15\Omega + j88\Omega$.

Table 17. RF output power setting for the nRF24L01+

6.6 RX/TX control

The RX/TX control is set by `PRIM_RX` bit in the `CONFIG` register and sets the nRF24L01+ in transmit/receive mode.

6.5 PA 控制

PA（功率放大器）控制用于设置 nRF24L01+ 功率放大器的输出功率。在 TX 模式下，PA 控制有四个可编程步骤，见表 17。

PA 控制由 RF_SETUP 寄存器中的 RF_PWR 位设置。

SPI 射频设置 (RF_PWR)	射频输出功率	直流电流 消费
11	0 分贝	11.3 安培
10	-6 分贝	9.0 安培
01	-12 分贝	7.5 安培
00	-18 分贝	7.0 安培

条件：VDD = 3.0V，VSS = 0V，T = 27°C，负载阻抗 = 15Ω+j88Ω。

表 17.nRF24L01+ 的射频输出功率设置

6.6 RX/TX 控制

RX/TX 控制由 CONFIG 寄存器中的 PRIM_RX 位设置，并将 nRF24L01+ 设置为发射/接收模式。

7 Enhanced ShockBurst™

Enhanced ShockBurst™ is a packet based data link layer that features automatic packet assembly and timing, automatic acknowledgement and retransmissions of packets. Enhanced ShockBurst™ enables the implementation of ultra low power and high performance communication with low cost host microcontrollers. The Enhanced ShockBurst™ features enable significant improvements of power efficiency for bi-directional and uni-directional systems, without adding complexity on the host controller side.

7.1 Features

The main features of Enhanced ShockBurst™ are:

- 1 to 32 bytes dynamic payload length
- Automatic packet handling
- Automatic packet transaction handling
 - Auto Acknowledgement with payload
 - Auto retransmit
- 6 data pipe MultiCeiver™ for 1:6 star networks

7.2 Enhanced ShockBurst™ overview

Enhanced ShockBurst™ uses ShockBurst™ for automatic packet handling and timing. During transmit, ShockBurst™ assembles the packet and clocks the bits in the data packet for transmission. During receive, ShockBurst™ constantly searches for a valid address in the demodulated signal. When ShockBurst™ finds a valid address, it processes the rest of the packet and validates it by CRC. If the packet is valid the payload is moved into a vacant slot in the RX FIFOs. All high speed bit handling and timing is controlled by ShockBurst™.

Enhanced ShockBurst™ features automatic packet transaction handling for the easy implementation of a reliable bi-directional data link. An Enhanced ShockBurst™ packet transaction is a packet exchange between two transceivers, with one transceiver acting as the Primary Receiver (PRX) and the other transceiver acting as the Primary Transmitter (PTX). An Enhanced ShockBurst™ packet transaction is always initiated by a packet transmission from the PTX, the transaction is complete when the PTX has received an acknowledgment packet (ACK packet) from the PRX. The PRX can attach user data to the ACK packet enabling a bi-directional data link.

The automatic packet transaction handling works as follows:

1. You begin the transaction by transmitting a data packet from the PTX to the PRX. Enhanced ShockBurst™ automatically sets the PTX in receive mode to wait for the ACK packet.
2. If the packet is received by the PRX, Enhanced ShockBurst™ automatically assembles and transmits an acknowledgment packet (ACK packet) to the PTX before returning to receive mode.
3. If the PTX does not receive the ACK packet immediately, Enhanced ShockBurst™ automatically retransmits the original data packet after a programmable delay and sets the PTX in receive mode to wait for the ACK packet.

In Enhanced ShockBurst™ it is possible to configure parameters such as the maximum number of retransmits and the delay from one transmission to the next retransmission. All automatic handling is done without the involvement of the MCU.

7 增强的 ShockBurst™

增强的 ShockBurst™ 是一个基于数据包的数据链路层，具有数据包的自动组装和定时、数据包的自动确认和重新传输功能。增强型 ShockBurst™ 支持使用低成本主机微控制器实现超低功耗和高性能通信。增强的 ShockBurst™ 功能可显著提高双向和单向系统的电源效率，而不会增加主机控制器端的复杂性。

7.1 特征

增强型 ShockBurst™ 的主要特点是：

- 1 至 32 字节动态有效载荷长度
- 自动数据包处理
- 自动数据包事务处理
 - ^x 使用有效负载自动确认
 - ^x 自动重传
- 用于 1: 6 星形网络的 6 数据管 MultiCeiver™

7.2 增强的 ShockBurst™ 概述

增强的 ShockBurst™ 使用 ShockBurst™ 进行自动数据包处理和计时。在传输过程中，ShockBurst™ 会组装数据包并对数据包中的位进行计时以进行传输。在接收过程中，ShockBurst™ 不断在解调信号中搜索有效地址。当 ShockBurst™ 找到有效地址时，它会处理数据包的其余部分并通过 CRC 对其进行验证。如果数据包有效，则有效负载将移动到 RX FIFO 中的空插槽中。所有高速位处理和定时均由 ShockBurst™ 控制。

增强的 ShockBurst™ 具有自动数据包事务处理功能，可实现可靠的双向数据链路。增强型 ShockBurst™ 数据包交易是两个收发器之间的数据包交换，一个收发器充当主接收器（PRX），另一个收发器充当主发射器（PTX）。增强型 ShockBurst™ 数据包事务始终由 PTX 的数据包传输发起，当 PTX 收到来自 PRX 的确认数据包（ACK 数据包）时，事务完成。PRX 可以将用户数据附加到 ACK 数据包，从而实现双向数据链路。

自动数据包事务处理的工作原理如下：

1. 您可以通过将数据包从 PTX 传输到 PRX 来开始事务。增强的 ShockBurst™ 会自动将 PTX 设置为接收模式以等待 ACK 数据包。
2. 如果 PRX 收到数据包，增强型 ShockBurst™ 会自动组装确认数据包（ACK 数据包）并将其传输到 PTX，然后再返回接收模式。
3. 如果 PTX 没有立即收到 ACK 数据包，Enhanced ShockBurst™ 会在可编程延迟后自动重新传输原始数据包，并将 PTX 设置为接收模式以等待 ACK 数据包。

在 Enhanced ShockBurst™ 中，可以配置参数，例如最大重传次数和从一次传输到下一次重传的延迟。所有自动处理都是在没有 MCU 参与的情况下完成的。

7.3 Enhanced Shockburst™ packet format

The format of the Enhanced ShockBurst™ packet is described in this section. The Enhanced ShockBurst™ packet contains a preamble, address, packet control, payload and CRC field. [Figure 5.](#) shows the packet format with MSB to the left.



Figure 5. An Enhanced ShockBurst™ packet with payload (0-32 bytes)

7.3.1 Preamble

The preamble is a bit sequence used to synchronize the receivers demodulator to the incoming bit stream. The preamble is one byte long and is either 01010101 or 10101010. If the first bit in the address is 1 the preamble is automatically set to 10101010 and if the first bit is 0 the preamble is automatically set to 01010101. This is done to ensure there are enough transitions in the preamble to stabilize the receiver.

7.3.2 Address

This is the address for the receiver. An address ensures that the packet is detected and received by the correct receiver, preventing accidental cross talk between multiple nRF24L01+ systems. You can configure the address field width in the `AW` register to be 3, 4 or 5 bytes, see [Table 28. on page 63](#).

Note: Addresses where the level shifts only one time (that is, 000FFFFFFFF) can often be detected in noise and can give a false detection, which may give a raised Packet Error Rate. Addresses as a continuation of the preamble (hi-low toggling) also raises the Packet Error Rate.

7.3.3 Packet control field

[Figure 6.](#) shows the format of the 9 bit packet control field, MSB to the left.

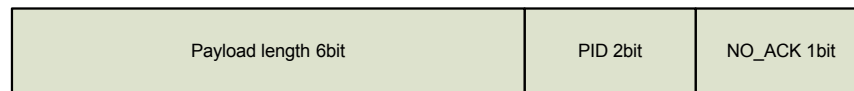


Figure 6. Packet control field

The packet control field contains a 6 bit payload length field, a 2 bit PID (Packet Identity) field and a 1 bit `NO_ACK` flag.

7.3 增强的 Shockburst™ 数据包格式

本节介绍了增强型 ShockBurst™ 数据包的格式。增强型 ShockBurst™ 数据包包含前导码、地址、数据包控制、有效载荷和 CRC 字段。图 5.显示 MSB 位于左侧的数据包格式。

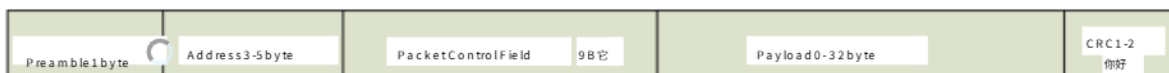


图 5.具有有效负载（0-32 字节）的增强型 ShockBurst™ 数据包

7.3.1 序言

preamble 是一个位序列，用于将接收器解调器与输入的 bit stream 同步。前导码长为一个字节，可以是 01010101 或 10101010。如果地址中的第一个位为 1，则前导码自动设置为 10101010 如果第一个位为 0，则前导码自动设置为 01010101。这样做是为了确保前导码中有足够的过渡来稳定接收器。

7.3.2 地址

这是接收方的地址。地址可确保正确的接收器检测和接收数据包，从而防止多个 nRF24L01+ 系统之间的意外串扰。您可以将 AW 寄存器中的地址字段宽度配置为 3、4 或 5 字节，参见表 28。在第 63 页。

注意：通常可以在 noise 的 Bug 的 Bug Programming Discount and can give a false detection, which may give a packet error rate.地址作为前导码的延续（高低切换）也会提高数据包错误率。

7.3.3 数据包控制字段

图 6.显示 9 位数据包控制字段的格式，MSB 在左侧。



图 6.数据包控制字段

数据包控制字段包含一个 6 位负载长度字段、一个 2 位 PID（数据包标识）字段和一个 1 位 NO_ACK 标志。

7.3.3.1 Payload length

This 6 bit field specifies the length of the payload in bytes. The length of the payload can be from 0 to 32 bytes.

Coding: 000000 = 0 byte (only used in empty ACK packets.) 100000 = 32 byte, 100001 = Don't care.

This field is only used if the Dynamic Payload Length function is enabled.

7.3.3.2 PID (Packet identification)

The 2 bit PID field is used to detect if the received packet is new or retransmitted. PID prevents the PRX device from presenting the same payload more than once to the receiving host MCU. The PID field is incremented at the TX side for each new packet received through the SPI. The PID and CRC fields (see [section 7.3.5 on page 30](#)) are used by the PRX device to determine if a packet is retransmitted or new. When several data packets are lost on the link, the PID fields may become equal to the last received PID. If a packet has the same PID as the previous packet, nRF24L01+ compares the CRC sums from both packets. If the CRC sums are also equal, the last received packet is considered a copy of the previously received packet and discarded.

7.3.3.3 No Acknowledgment flag (NO_ACK)

The Selective Auto Acknowledgement feature controls the NO_ACK flag.

This flag is only used when the auto acknowledgement feature is used. Setting the flag high tells the receiver that the packet is not to be auto acknowledged.

On the PTX you can set the NO_ACK flag bit in the Packet Control Field with this command:

```
W_TX_PAYLOAD_NOACK
```

However, the function must first be enabled in the FEATURE register by setting the EN_DYN_ACK bit. When you use this option the PTX goes directly to standby-I mode after transmitting the packet. The PRX does not transmit an ACK packet when it receives the packet.

7.3.4 Payload

The payload is the user defined content of the packet. It can be 0 to 32 bytes wide and is transmitted on-air when it is uploaded to nRF24L01+.

Enhanced ShockBurst™ provides two alternatives for handling payload lengths; static and dynamic.

The default is static payload length. With static payload length all packets between a transmitter and a receiver have the same length. Static payload length is set by the RX_PW_Px registers on the receiver side. The payload length on the transmitter side is set by the number of bytes clocked into the TX_FIFO and must equal the value in the RX_PW_Px register on the receiver side.

Dynamic Payload Length (DPL) is an alternative to static payload length. DPL enables the transmitter to send packets with variable payload length to the receiver. This means that for a system with different payload lengths it is not necessary to scale the packet length to the longest payload.

7.3.3.1 有效载荷长度

此 6 位字段指定负载的长度（以字节为单位）。负载的长度可以是 0 到 32 字节。

编码：000000 = 0 字节（仅用于空 ACK 数据包。100000 = 32 字节，100001 = 不在乎。

仅当启用了 Dynamic Payload Length 功能时，才使用此字段。

7.3.3.2 PID（数据包识别）

2 位 PID 字段用于检测接收到的数据包是新的还是重传的。PID 可防止 PRX 设备多次向接收主机 MCU 提供相同的有效负载。对于通过 SPI 接收的每个新数据包，PID 字段在 TX 端递增。PRX 设备使用 PID 和 CRC 字段（参见第 7.3.5 页第 30 节）来确定数据包是重新传输还是新数据包。

当链路上丢失多个数据包时，PID 字段可能会等于最后接收的 PID。如果一个数据包与前一个数据包具有相同的 PID，则 nRF24L01+ 会比较两个数据包的 CRC 和。如果 CRC 和也相等，则最后接收的数据包被视为先前接收数据包的副本并丢弃。

7.3.3.3 无鸣谢标志（NO_ACK）

选择性自动确认功能控制 NO_ACK 标志。

此标志仅在使用自动确认功能时使用。将标志设置为高电平会告诉接收方不会自动确认数据包。

在 PTX 上，您可以使用以下命令在 Packet Control 字段中设置 NO_ACK 标志位：

```
W_TX_PAYLOAD_NOACK
```

但是，必须首先在 FEATURE 寄存器中通过设置 EN_DYN_ACK 位来启用该功能。使用此选项时，PTX 在传输数据包后直接进入 standby-I 模式。PRX 收到 ACK 报文时，不会发送 ACK 报文。

7.3.4 有效载荷

payload 是数据包的用户定义内容。它可以是 0 到 32 字节宽，并在上传到 nRF24L01+ 时进行广播传输。

增强的 ShockBurst™ 提供了两种处理有效负载长度的替代方案；static 和 dynamic。

默认值为 static payload length。使用静态有效载荷长度时，发射器和接收器之间的所有数据包都具有相同的长度。静态有效载荷长度由接收器端的 RX_PW_Px 寄存器设置。发送器侧的有效载荷长度由 TX_FIFO 的字节数设置，并且必须等于接收器侧 RX_PW_Px 寄存器中的值。

动态负载长度（DPL）是静态负载长度的替代方法。DPL 使发送器能够向接收机发送具有可变有效载荷长度的数据包。这意味着，对于具有不同有效载荷长度的系统，无需将数据包长度缩放到最长的有效载荷。

With the DPL feature the nRF24L01+ can decode the payload length of the received packet automatically instead of using the `RX_PW_Px` registers. The MCU can read the length of the received payload by using the `R_RX_PL_WID` command.

Note: Always check if the packet width reported is 32 bytes or shorter when using the `R_RX_PL_WID` command. If its width is longer than 32 bytes then the packet contains errors and must be discarded. Discard the packet by using the `Flush_RX` command.

In order to enable DPL the `EN_DPL` bit in the `FEATURE` register must be enabled. In RX mode the `DYNPD` register must be set. A PTX that transmits to a PRX with DPL enabled must have the `DPL_P0` bit in `DYNPD` set.

7.3.5 CRC (Cyclic Redundancy Check)

The CRC is the mandatory error detection mechanism in the packet. It is either 1 or 2 bytes and is calculated over the address, Packet Control Field and Payload.

The polynomial for 1 byte CRC is $X^8 + X^2 + X + 1$. Initial value 0xFF.

The polynomial for 2 byte CRC is $X^{16} + X^{12} + X^5 + 1$. Initial value 0xFFFF.

The number of bytes in the CRC is set by the `CRCO` bit in the `CONFIG` register. No packet is accepted by Enhanced ShockBurst™ if the CRC fails.

借助 DPL 功能，nRF24L01+ 可以自动解码接收到的数据包的有效载荷长度，而不是使用 RX_PW_Px 寄存器。MCU 可以通过 R_RX_PL_WID 命令读取接收到的负载长度。

注意：使用

R_RX_PL_WID 命令。如果其宽度超过 32 字节，则数据包包含错误，必须丢弃。使用 Flush_RX 命令丢弃数据包。

为了使能 DPL，必须使能 FEATURE 寄存器中的 EN_DPL 位。在 RX 模式下，必须设置 DYNPD 寄存器。传输到启用了 DPL 的 PRX 的 PTX 必须在 DYNPD 中设置 DPL_P0 位。

7.3.5 CRC（循环冗余校验）

CRC 是数据包中的强制错误检测机制。它是 1 或 2 个字节，根据地址、数据包控制字段和负载计算。

1 字节 CRC 的多项式为 $X + X + X + 1$ 。初始值 0xFF。

2 字节 CRC 的多项式为 $X + X + X + 1$ 。初始值 0xFFFF。

CRC 中的字节数由 CONFIG 寄存器中的 CRCO 位设置。如果 CRC 失败，Enhanced ShockBurst™ 不接受任何数据包。

7.3.6 Automatic packet assembly

The automatic packet assembly assembles the preamble, address, packet control field, payload and CRC to make a complete packet before it is transmitted.

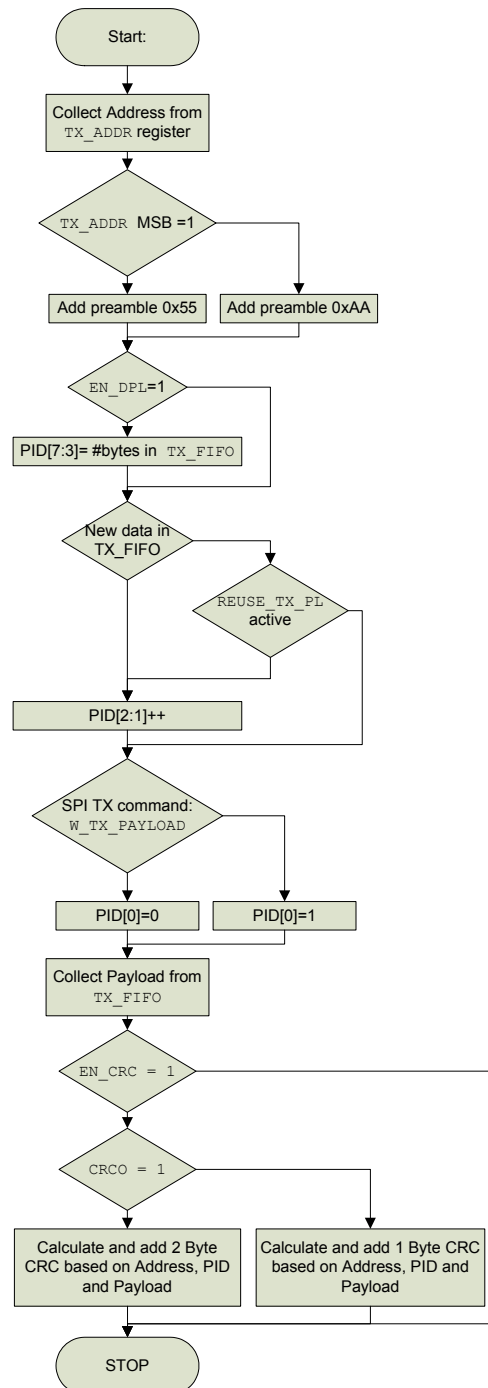


Figure 7. Automatic packet assembly

7.3.6 自动数据包组装

自动数据包组合将前导码、地址、数据包控制字段、有效载荷和 CRC 组合在一起，以便在传输之前制作一个完整的数据包。

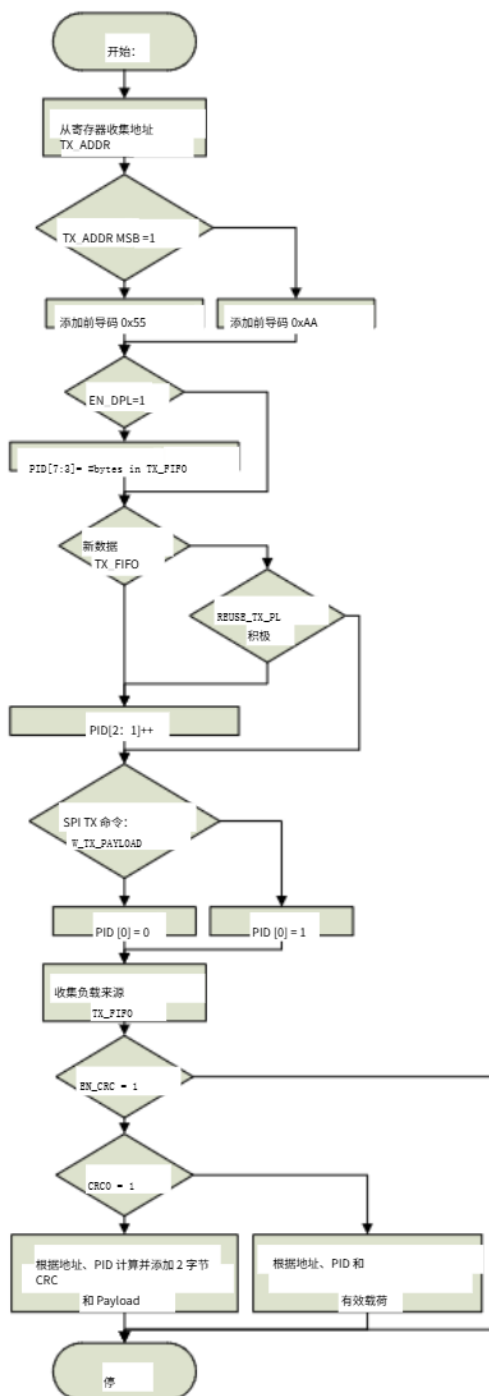


图 7.自动数据包组装

7.3.7 Automatic packet disassembly

After the packet is validated, Enhanced ShockBurst™ disassembles the packet and loads the payload into the RX FIFO, and asserts the `RX_DR` IRQ.

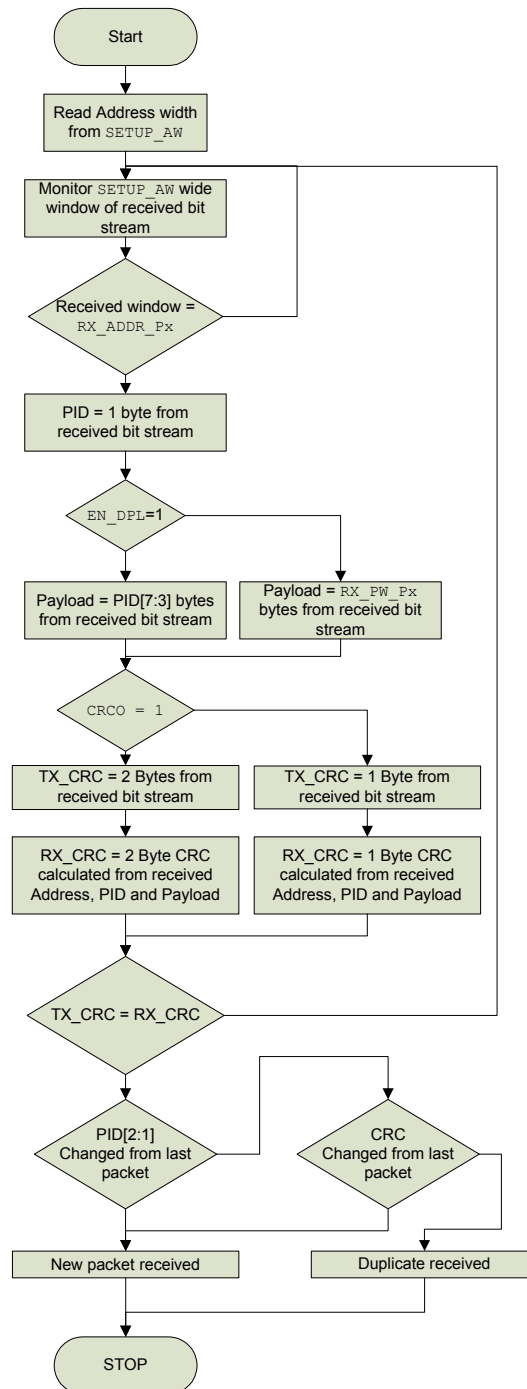


Figure 8. Automatic packet disassembly

7.3.7 自动拆包

数据包验证后, Enhanced ShockBurst™ 反汇编数据包并将有效负载加载到 RX FIFO 中, 并置位 RX_DR IRQ。

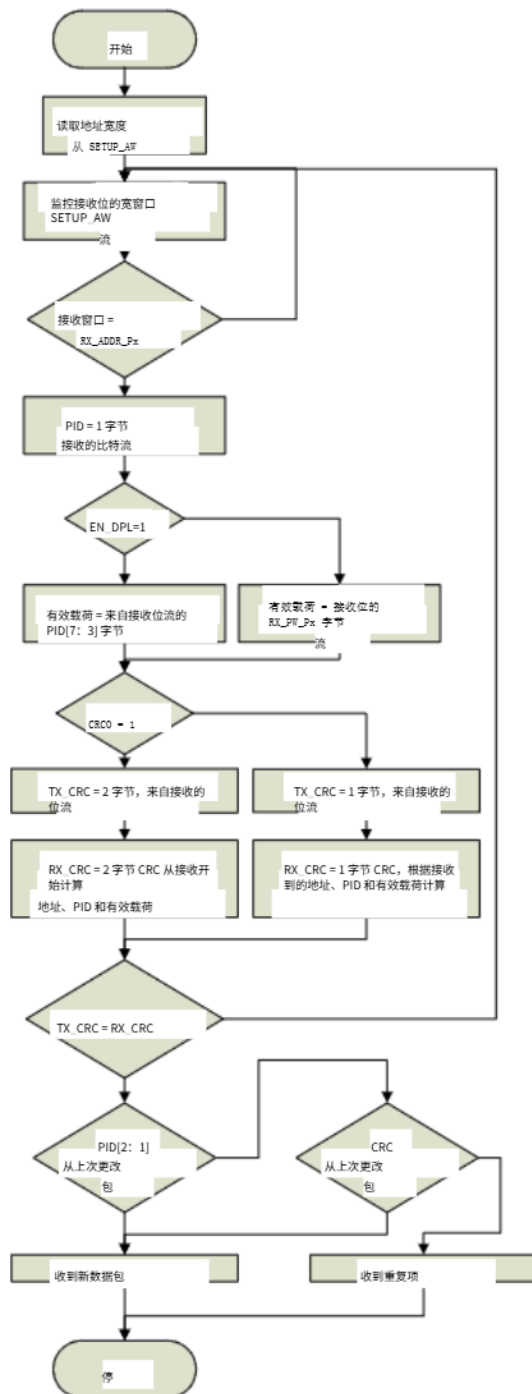


图 8.自动拆包

7.4 Automatic packet transaction handling

Enhanced ShockBurst™ has two functions for automatic packet transaction handling; auto acknowledgement and auto re-transmit.

7.4.1 Auto acknowledgement

Auto acknowledgement is a function that automatically transmits an ACK packet to the PTX after it has received and validated a packet. The auto acknowledgement function reduces the load of the system MCU and can remove the need for dedicated SPI hardware. This also reduces cost and average current consumption. The Auto Acknowledgement feature is enabled by setting the `EN_AA` register.

Note: If the received packet has the `NO_ACK` flag set, auto acknowledgement is not executed.

An ACK packet can contain an optional payload from PRX to PTX. In order to use this feature, the Dynamic Payload Length (DPL) feature must be enabled. The MCU on the PRX side has to upload the payload by clocking it into the TX FIFO by using the `W_ACK_PAYLOAD` command. The payload is pending in the TX FIFO (PRX) until a new packet is received from the PTX. nRF24L01+ can have three ACK packet payloads pending in the TX FIFO (PRX) at the same time.

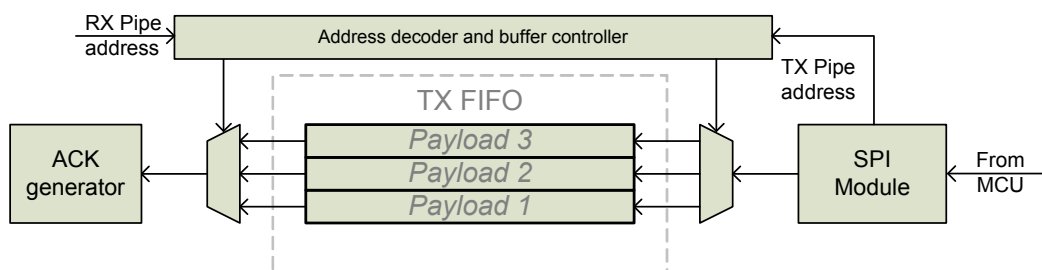


Figure 9. TX FIFO (PRX) with pending payloads

Figure 9. shows how the TX FIFO (PRX) is operated when handling pending ACK packet payloads. From the MCU the payload is clocked in with the `W_ACK_PAYLOAD` command. The address decoder and buffer controller ensure that the payload is stored in a vacant slot in the TX FIFO (PRX). When a packet is received, the address decoder and buffer controller are notified with the PTX address. This ensures that the right payload is presented to the ACK generator.

If the TX FIFO (PRX) contains more than one payload to a PTX, payloads are handled using the first in – first out principle. The TX FIFO (PRX) is blocked if all pending payloads are addressed to a PTX where the link is lost. In this case, the MCU can flush the TX FIFO (PRX) by using the `FLUSH_TX` command.

In order to enable Auto Acknowledgement with payload the `EN_ACK_PAY` bit in the `FEATURE` register must be set.

7.4.2 Auto Retransmission (ART)

The auto retransmission is a function that retransmits a packet if an ACK packet is not received. It is used in an auto acknowledgement system on the PTX. When a packet is not acknowledged, you can set the number of times it is allowed to retransmit by setting the ARC bits in the `SETUP_RETR` register. PTX enters RX mode and waits a short period for an ACK packet each time a packet is transmitted. The time period the PTX is in RX mode is based on the following conditions:

7.4 自动数据包事务处理

增强的 ShockBurst™ 有两个自动数据包事务处理功能;自动确认和自动重新传输。

7.4.1 自动确认

自动确认是在 PTX 接收并验证数据包后自动将 ACK 数据包传输到 PTX 的功能。自动确认功能减少了系统 MCU 的负载，并且无需专用的 SPI 硬件。这也降低了成本和平均电流消耗。Auto Acknowledgement 功能通过设置 EN_AA 寄存器来启用。

注意：如果收到的数据包设置了 NO_ACK 标志，则不会执行自动确认。

ACK 数据包可以包含从 PRX 到 PTX 的可选有效负载。要使用此功能，必须启用动态负载长度（DPL）功能。PRX 端的 MCU 必须使用 W_ACK_PAYLOAD 命令将 payload clocking（时钟）上传到 TX FIFO 中。有效负载在 TX FIFO（PRX）中处于挂起状态，直到从 PTX 收到新数据包。nRF24L01+ 可以同时从 TX FIFO（PRX）中等待三个 ACK 数据包有效负载。

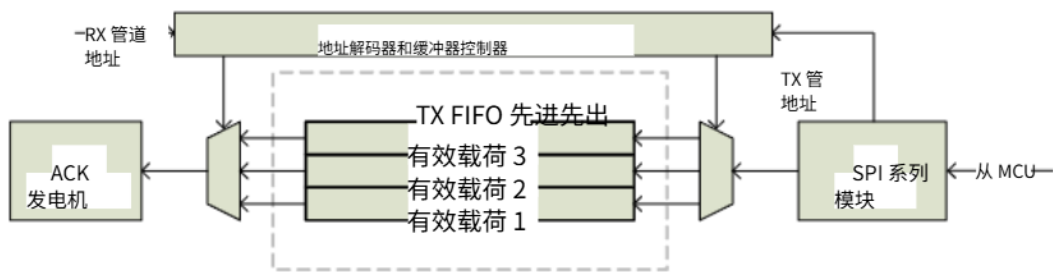


图 9.具有待处理负载的 TX FIFO（PRX）

图 9.显示了在处理待处理的 ACK 数据包负载时如何作 TX FIFO（PRX）。在 MCU 中，使用 W_ACK_PAYLOAD 命令对有效负载进行计时。地址解码器和缓冲器控制器确保有效负载存储在 TX FIFO（PRX）的空插槽中。收到数据包时，地址解码器和缓冲区控制器会收到 PTX 地址的通知。这可确保向 ACK 生成器提供正确的有效负载。

如果 TX FIFO（PRX）包含多个 PTX 负载，则使用先进先出原则处理负载。如果所有待处理的有效负载都发送到链路丢失的 PTX，则 TX FIFO（PRX）将被阻止。在这种情况下，MCU 可以使用 FLUSH_TX 命令刷新 TX FIFO（PRX）。

为了启用有效负载的 Auto Acknowledgement，必须设置 FEATURE 寄存器中的 EN_ACK_PAY 位。

7.4.2 自动重传（ART）

自动重传是在未收到 ACK 数据包时重传数据包的功能。它用于 PTX 上的自动确认系统。当数据包未被确认时，您可以通过在 SETUP_RETR 寄存器中设置 ARC 位来设置允许重传的次数。PTX 进入 RX 模式，并在每次传输数据包时等待一小段时间等待 ACK 数据包。PTX 处于 RX 模式的时间段基于以下条件：

- Auto Retransmit Delay (ARD) has elapsed.
- No address match within 250µs (or 500µs in 250kbps mode).
- After received packet (CRC correct or not).

nRF24L01+ asserts the `TX_DS` IRQ when the ACK packet is received.

nRF24L01+ enters standby-I mode if there is no more untransmitted data in the TX FIFO and the `CE` pin is low. If the ACK packet is not received, nRF24L01+ goes back to TX mode after a delay defined by ARD and retransmits the data. This continues until acknowledgment is received, or the maximum number of retransmits is reached.

Two packet loss counters are incremented each time a packet is lost, `ARC_CNT` and `PLOS_CNT` in the `OBSERVE_TX` register. The `ARC_CNT` counts the number of retransmissions for the current transaction. You reset `ARC_CNT` by initiating a new transaction. The `PLOS_CNT` counts the total number of retransmissions since the last channel change. You reset `PLOS_CNT` by writing to the `RF_CH` register. It is possible to use the information in the `OBSERVE_TX` register to make an overall assessment of the channel quality.

The ARD defines the time from the end of a transmitted packet to when a retransmit starts on the PTX. ARD is set in `SETUP_RETR` register in steps of 250µs. A retransmit is made if no ACK packet is received by the PTX.

There is a restriction on the length of ARD when using ACK packets with payload. The ARD time must never be shorter than the sum of the startup time and the time on-air for the ACK packet:

- For 2Mbps data rate and 5 byte address; 15 byte is maximum ACK packet payload length for ARD=250µs (reset value).
- For 1Mbps data rate and 5 byte address; 5 byte is maximum ACK packet payload length for ARD=250µs (reset value).

ARD=500µs is long enough for any ACK payload length in 1 or 2Mbps mode.

- For 250kbps data rate and 5byte address the following values apply:

ARD	ACK packet size (in bytes)
1500µs	All ACK payload sizes
1250µs	≤ 24
1000µs	≤ 16
750µs	≤ 8
500µs	Empty ACK with no payload

Table 18. Maximum ACK payload length for different retransmit delays at 250kbps

As an alternative to Auto Retransmit it is possible to manually set the nRF24L01+ to retransmit a packet a number of times. This is done by the `REUSE_TX_PL` command. The MCU must initiate each transmission of the packet with a pulse on the `CE` pin when this command is used.

- 自动重传延迟（ARD）已过。
- 在 250μs 内没有地址匹配（或 250kbps 模式下的 500μs）。
- 收到数据包后（CRC 正确与否）。

nRF24L01+ 在收到 ACK 数据包时置位 TX_DS IRQ。

如果 TX FIFO 中没有更多未传输的数据并且 CE 引脚为低电平，nRF24L01+ 将进入待机 I 模式。如果未收到 ACK 数据包，nRF24L01+ 将在 ARD 定义的延迟后返回 TX 模式并重新传输数据。这种情况一直持续到收到确认或达到最大重传次数。

每次数据包丢失时，两个数据包丢失计数器都会递增，ARC_CNT 和 PLOS_CNT 在 OBSERVE_TX 寄存器中。ARC_CNT 计算当前事务的重新传输次数。您可以通过启动新事务来重置 ARC_CNT。PLOS_CNT 会计算自上次频道更改以来的重新传输总数。您可以通过写入 RF_CH register 来重置 PLOS_CNT。可以使用 OBSERVE_TX 寄存器中的信息对频道质量进行整体评估。

ARD 定义从传输的数据包结束到 PTX 上开始重新传输的时间。ARD 以 250μs 的步长设置 SETUP_RETR 寄存器中。如果 PTX 未收到 ACK 数据包，则进行重新传输。

使用带有 payload 的 ACK 数据包时，ARD 的长度有限制。ARD 时间不得短于 ACK 数据包的启动时间和广播时间之和：

- 对于 2Mbps 数据速率和 5 字节地址；15 字节是 ARD=250μs（复位值）时的最大 ACK 数据包有效载荷长度。
- 对于 1Mbps 数据速率和 5 字节地址；5 字节是 ARD=250μs（复位值）时的最大 ACK 数据包有效载荷长度。

ARD=500μs 对于 1 或 2Mbps 模式下的任何 ACK 有效载荷长度来说都足够长。

- 对于 250kbps 数据速率和 5 字节地址，以下值适用：

ARD	ACK 数据包大小（以字节为单位）
1500 微秒	所有 ACK 负载大小
1250 微秒	< 24
1000 微秒	< 16
750 微秒	< 8
500 微秒	空 ACK，无负载

表 18.250kbps 时不同重传延迟的最大 ACK 有效载荷长度

作为自动重传的替代方案，可以手动设置 nRF24L01+ 以多次重传数据包。这是通过 REUSE_TX_PL 命令完成的。使用此命令时，MCU 必须在 CE 引脚上通过脉冲启动数据包的每次传输。

This section contains flowcharts outlining PTX and PRX operation in Enhanced ShockBurst™.

The flowchart in [Figure 10.](#) outlines how a nRF24L01+ configured as a PTX behaves after entering standby-I mode.

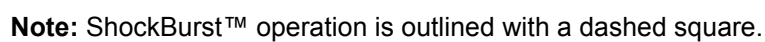


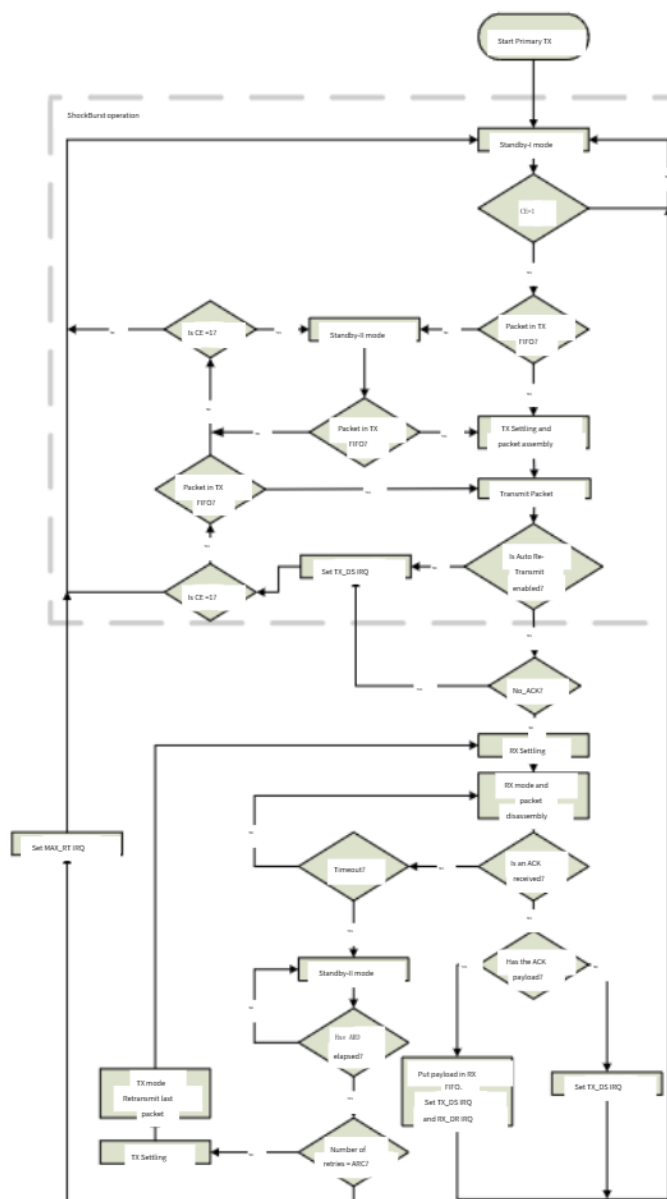
Figure 10. PTX operations in Enhanced ShockBurst™

7.5 增强的 ShockBurst 流程图

本节包含概述增强型 ShockBurst™ 中的 PTX 和 PRX 作的流程图。

7.5.1 PTX 作

图 10 中的流程图。概述了配置为 PTX 的 nRF24L01+ 在进入待机 I 模式后的行为方式。



注：ShockBurst™作用虚线方块勾勒出来。

图 10.增强型 ShockBurst 中的 PTX™作

Activate PTX mode by setting the `CE` pin high. If there is a packet present in the TX FIFO the nRF24L01+ enters TX mode and transmits the packet. If Auto Retransmit is enabled, the state machine checks if the `NO_ACK` flag is set. If it is not set, the nRF24L01+ enters RX mode to receive an ACK packet. If the received ACK packet is empty, only the `TX_DS` IRQ is asserted. If the ACK packet contains a payload, both `TX_DS` IRQ and `RX_DR` IRQ are asserted simultaneously before nRF24L01+ returns to standby-I mode.

If the ACK packet is not received before timeout occurs, the nRF24L01+ returns to standby-II mode. It stays in standby-II mode until the ARD has elapsed. If the number of retransmits has not reached the ARC, the nRF24L01+ enters TX mode and transmits the last packet once more.

While executing the Auto Retransmit feature, the number of retransmits can reach the maximum number defined in `ARC`. If this happens, the nRF24L01+ asserts the `MAX_RT` IRQ and returns to standby-I mode.

If the `CE` is high and the TX FIFO is empty, the nRF24L01+ enters Standby-II mode.

通过将 CE 引脚设置为高电平来激活 PTX 模式。如果 TX FIFO 中存在数据包，则 nRF24L01+ 进入 TX 模式并传输数据包。如果启用了 Auto Retransmit，则状态机将检查是否设置了 NO_ACK 标志。如果未设置，则 nRF24L01+ 进入 RX 模式以接收 ACK 数据包。如果接收到的 ACK 报文为空，则只置位 TX_DS IRQ。如果 ACK 数据包包含有效载荷，则在 nRF24L01+ 返回待机 I 模式之前，TX_DS IRQ 和 RX_DR IRQ 会同时置位。

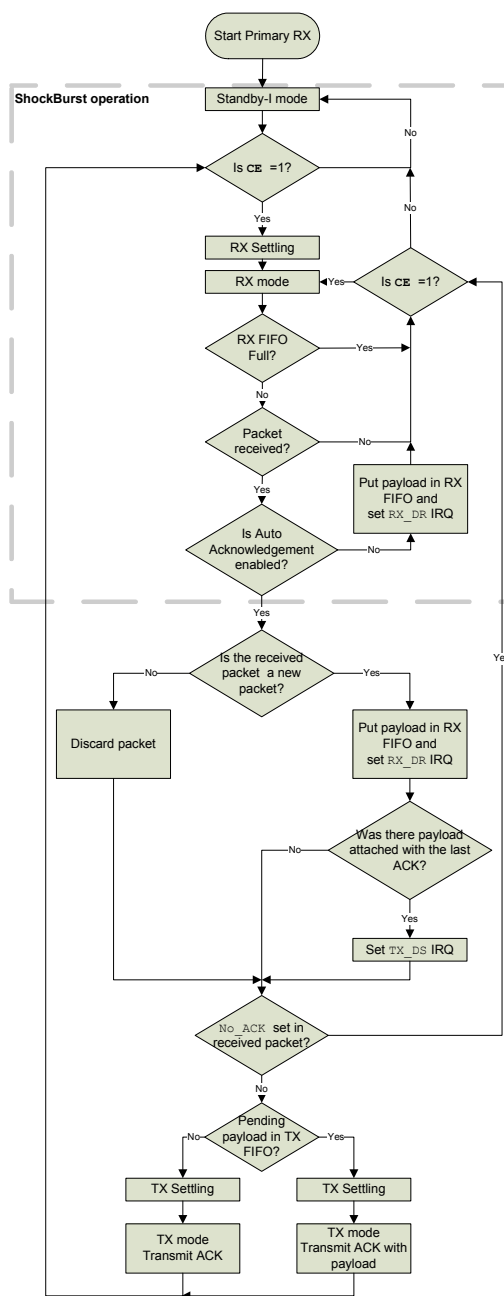
如果在超时之前没有收到 ACK 数据包，nRF24L01+ 将返回 standby-II 模式。它将保持待机 II 模式，直到 ARD 结束。如果重传次数未达到 ARC，nRF24L01+ 进入 TX 模式并再次发送最后一个数据包。

在执行 Auto Retransmit 功能时，重传次数可以达到 ARC 中定义的最大数量。如果发生这种情况，nRF24L01+ 将置位 MAX_RT IRQ 并返回待机 I 模式。

如果 CE 为高电平且 TX FIFO 为空，则 nRF24L01+ 进入 Standby-II 模式。

7.5.2 PRX operation

The flowchart in [Figure 11](#) outlines how a nRF24L01+ configured as a PRX behaves after entering standby-I mode.



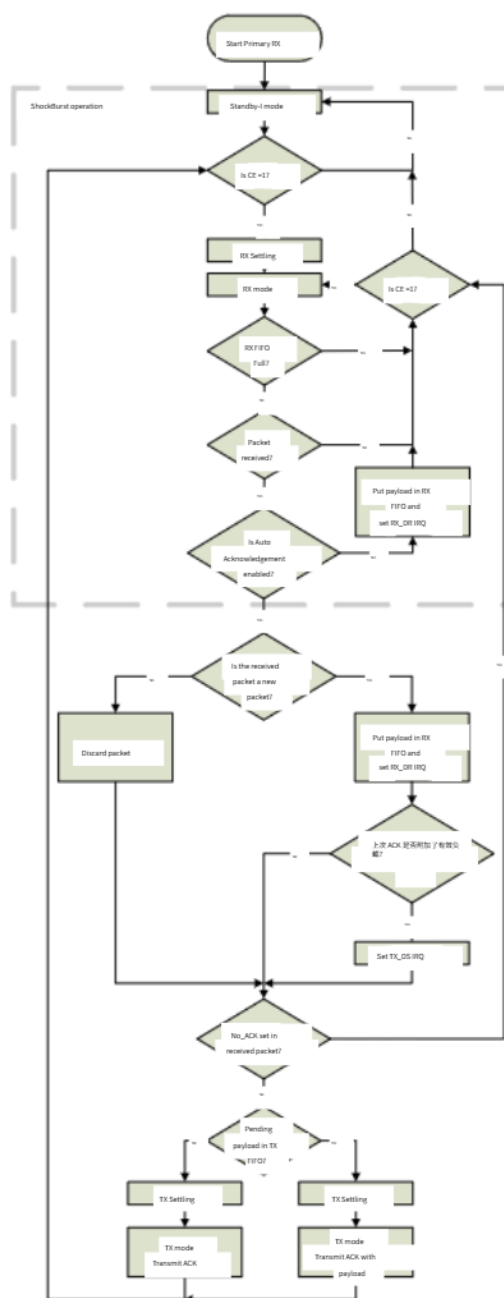
Note: ShockBurst™ operation is outlined with a dashed square.

Figure 11. PRX operations in Enhanced ShockBurst™

Activate PRX mode by setting the **CE** pin high. The nRF24L01+ enters RX mode and starts searching for packets. If a packet is received and Auto Acknowledgement is enabled, nRF24L01+ decides if the packet is new or a copy of a previously received packet. If the packet is new the payload is made available in the

7.5.2 PRX 作

图 11 中的流程图。概述了配置为 PRX 的 nRF24L01+ 在进入待机 I 模式后的行为方式。



注： ShockBurst™作用虚线方块勾勒出来。

图 11.增强型 ShockBurst™ 中的 PRX 作

通过将 CE 引脚设置为高电平来激活 PRX 模式。nRF24L01+ 进入 RX 模式并开始搜索数据包。如果收到数据包并启用了自动确认，nRF24L01+ 会决定该数据包是新的还是以前收到的数据包的副本。如果数据包是新的，则有效负载将在

RX FIFO and the `RX_DR` IRQ is asserted. If the last received packet from the transmitter is acknowledged with an ACK packet with payload, the `TX_DS` IRQ indicates that the PTX received the ACK packet with payload. If the `NO_ACK` flag is not set in the received packet, the PRX enters TX mode. If there is a pending payload in the TX FIFO it is attached to the ACK packet. After the ACK packet is transmitted, the nRF24L01+ returns to RX mode.

A copy of a previously received packet might be received if the ACK packet is lost. In this case, the PRX discards the received packet and transmits an ACK packet before it returns to RX mode.

RX FIFO 和 RX_DR IRQ 被置位。如果从发射器接收的最后一个数据包与带有有效负载的 ACK 数据包确认，则 TX_DS IRQ 表示 PTX 收到了带有有效负载的 ACK 数据包。如果收到的数据包中未设置 No_ACK 标志，则 PRX 将进入 TX 模式。如果 TX FIFO 中有待处理的 payload，则它将被附加到 ACK 数据包。发送 ACK 数据包后，nRF24L01+ 返回 RX 模式。

如果 ACK 数据包丢失，则可能会收到以前收到的数据包的副本。在这种情况下，PRX 会丢弃收到的数据包，并在返回 RX 模式之前传输 ACK 数据包。

7.6 MultiCeiver™

MultiCeiver™ is a feature used in RX mode that contains a set of six parallel data pipes with unique addresses. A data pipe is a logical channel in the physical RF channel. Each data pipe has its own physical address (data pipe address) decoding in the nRF24L01+.

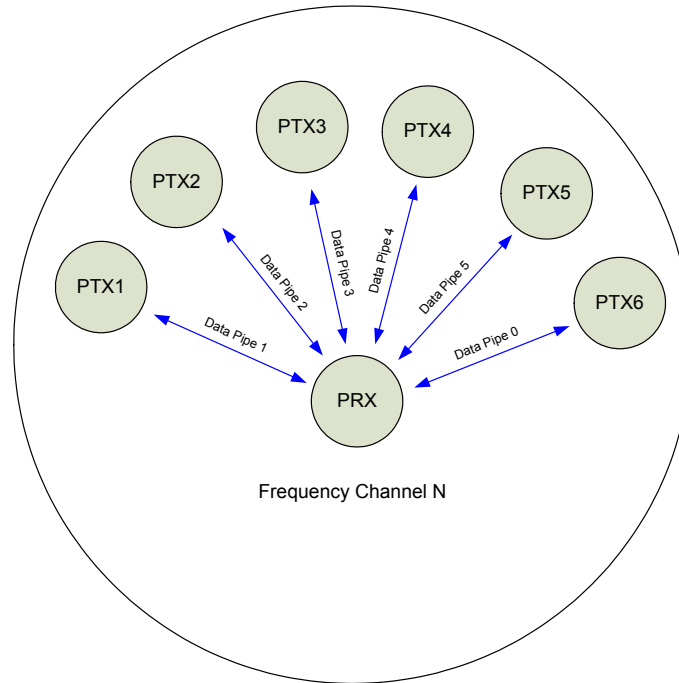


Figure 12. PRX using MultiCeiver™

nRF24L01+ configured as PRX (primary receiver) can receive data addressed to six different data pipes in one frequency channel as shown in [Figure 12](#). Each data pipe has its own unique address and can be configured for individual behavior.

Up to six nRF24L01+s configured as PTX can communicate with one nRF24L01+ configured as a PRX. All data pipe addresses are searched for simultaneously. Only one data pipe can receive a packet at a time. All data pipes can perform Enhanced ShockBurst™ functionality.

The following settings are common to all data pipes:

- CRC enabled/disabled (CRC always enabled when Enhanced ShockBurst™ is enabled)
- CRC encoding scheme
- RX address width
- Frequency channel
- Air data rate
- LNA gain

The data pipes are enabled with the bits in the `EN_RXADDR` register. By default only data pipe 0 and 1 are enabled. Each data pipe address is configured in the `RX_ADDR_PX` registers.

Note: Always ensure that none of the data pipes have the same address.

7.6 MultiCeiver™ 系列

MultiCeiver™ 是 RX 模式中使用的一项功能，它包含一组六个具有唯一地址的并行数据管道。数据管道是物理 RF 通道中的逻辑通道。每个数据管道在 nRF24L01+ 中都有自己的物理地址（数据管道地址）解码。

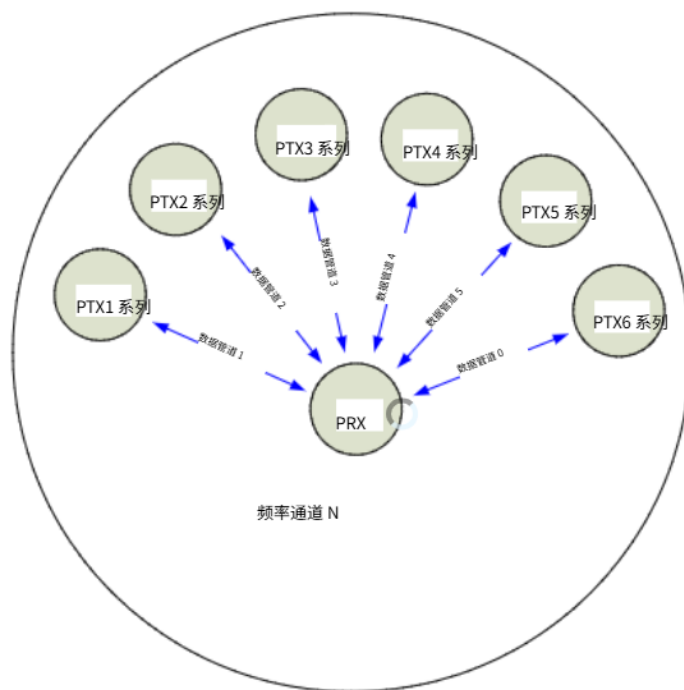


图 12.使用 MultiCeiver™ 的 PRX

配置为 PRX（主接收器）的 nRF24L01+ 可以在一个频道中接收寻址到六个不同数据管道的数据，如图 12 所示。每个数据管道都有自己唯一的地址，并且可以针对个人行为进行配置。

最多六个配置为 PTX 的 nRF24L01+ 可以与一个配置为 PRX 的 nRF24L01+ 通信。同时搜索所有数据管道地址。一次只能有一个数据管道接收数据包。

所有数据管道都可以执行增强的 ShockBurst™ 功能。

以下设置对于所有数据管道都是通用的：

- CRC 启用/禁用（启用 Enhanced ShockBurst™ 时 CRC 始终启用）
- CRC 编码方案
- RX 地址宽度
- 频率通道
- 空中数据速率
- LNA 增益

数据管道通过 EN_RXADDR register 中的 bits 使能。默认情况下，仅启用数据管道 0 和 1。每个 data pipe 地址都在 RX_ADDR_PX registers 中配置。

注意：始终确保没有任何数据管道具有相同的地址。

Each pipe can have up to a 5 byte configurable address. Data pipe 0 has a unique 5 byte address. Data pipes 1-5 share the four most significant address bytes. The LSByte must be unique for all six pipes. [Figure 13](#) is an example of how data pipes 0-5 are addressed.

	Byte 4	Byte 3	Byte 2	Byte 1	Byte 0
Data pipe 0 (RX_ADDR_P0)	0xE7	0xD3	0xF0	0x35	0x77
Data pipe 1 (RX_ADDR_P1)	0xC2	0xC2	0xC2	0xC2	0xC2
	↓	↓	↓	↓	
Data pipe 2 (RX_ADDR_P2)	0xC2	0xC2	0xC2	0xC2	0xC3
	↓	↓	↓	↓	
Data pipe 3 (RX_ADDR_P3)	0xC2	0xC2	0xC2	0xC2	0xC4
	↓	↓	↓	↓	
Data pipe 4 (RX_ADDR_P4)	0xC2	0xC2	0xC2	0xC2	0xC5
	↓	↓	↓	↓	
Data pipe 5 (RX_ADDR_P5)	0xC2	0xC2	0xC2	0xC2	0xC6

Figure 13. Addressing data pipes 0-5

每个管道最多可以有一个 5 字节的可配置地址。数据管道 0 具有唯一的 5 字节地址。数据管道 1-5 共享四个最重要的地址字节。LSByte 对于所有 6 个管道必须是唯一的。图 13.是如何对数据管道 0-5 进行寻址的示例。

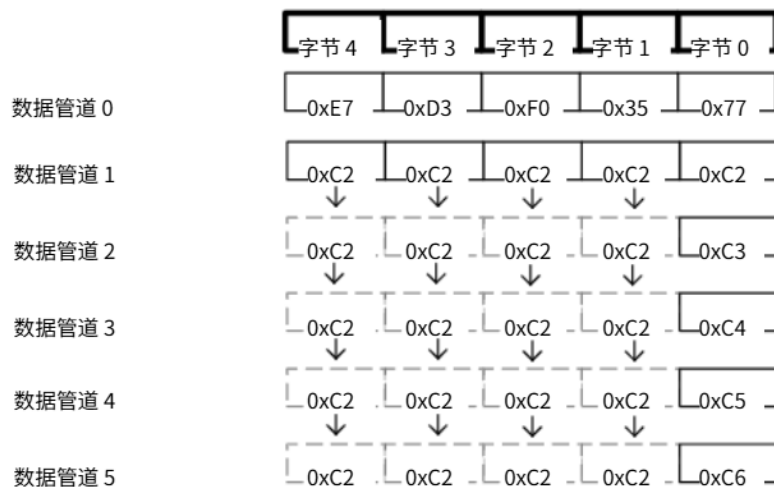


图 13.寻址数据管道 0-5

The PRX, using MultiCeiver™ and Enhanced ShockBurst™, receives packets from more than one PTX. To ensure that the ACK packet from the PRX is transmitted to the correct PTX, the PRX takes the data pipe address where it received the packet and uses it as the TX address when transmitting the ACK packet. [Figure 14](#) is an example of an address configuration for the PRX and PTX. On the PRX the RX_ADDR_Pn, defined as the pipe address, must be unique. On the PTX the TX_ADDR must be the same as the RX_ADDR_P0 and as the pipe address for the designated pipe.

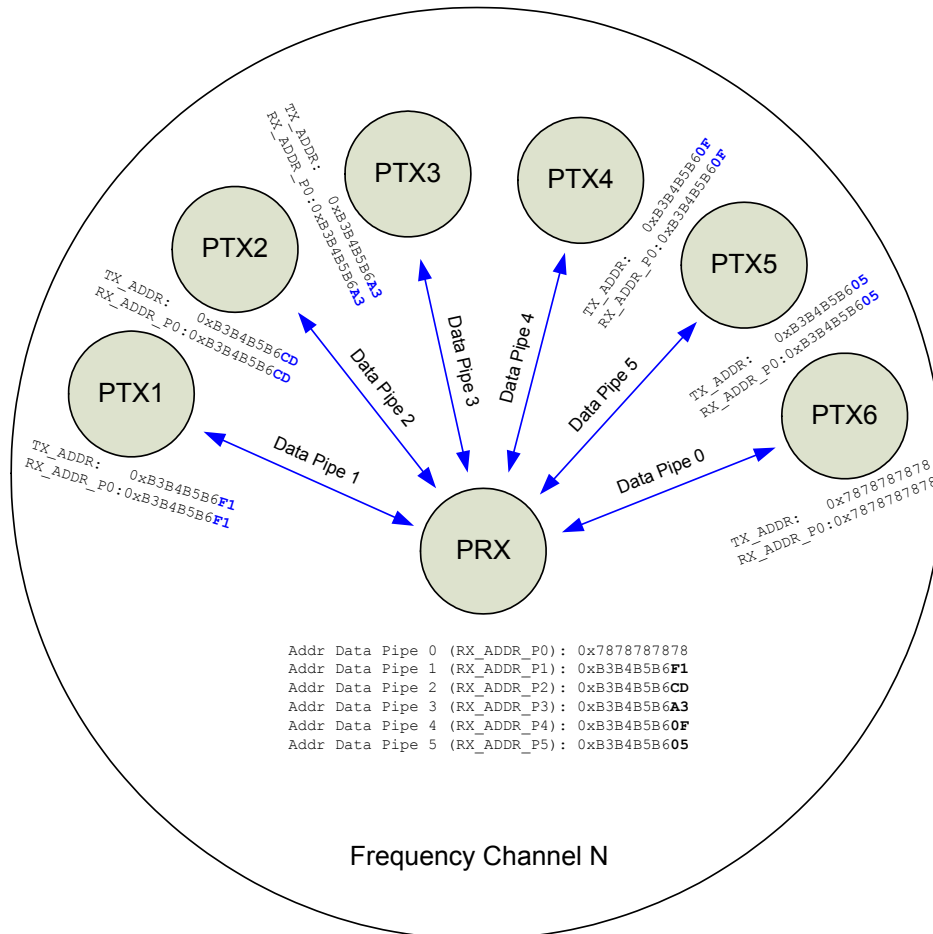


Figure 14. Example of data pipe addressing in MultiCeiver™

Only when a data pipe receives a complete packet can other data pipes begin to receive data. When multiple PTXs are transmitting to a PRX, the ARD can be used to skew the auto retransmission so that they only block each other once.

PRX 使用 MultiCeiver™ 和增强型 ShockBurst™ 从多个 PTX 接收数据包。为确保来自 PRX 的 ACK 数据包传输到正确的 PTX，PRX 会采用接收数据包的数据管道地址，并在传输 ACK 数据包时将其用作 TX 地址。图 14.是 PRX 和 PTX 的地址配置示例。在 PRX 上，定义为管道地址的 RX_ADDR_Pn 必须是唯一的。在 PTX 上，TX_ADDR 必须与 RX_ADDR_P0 相同，并且与指定管道的管道地址相同。

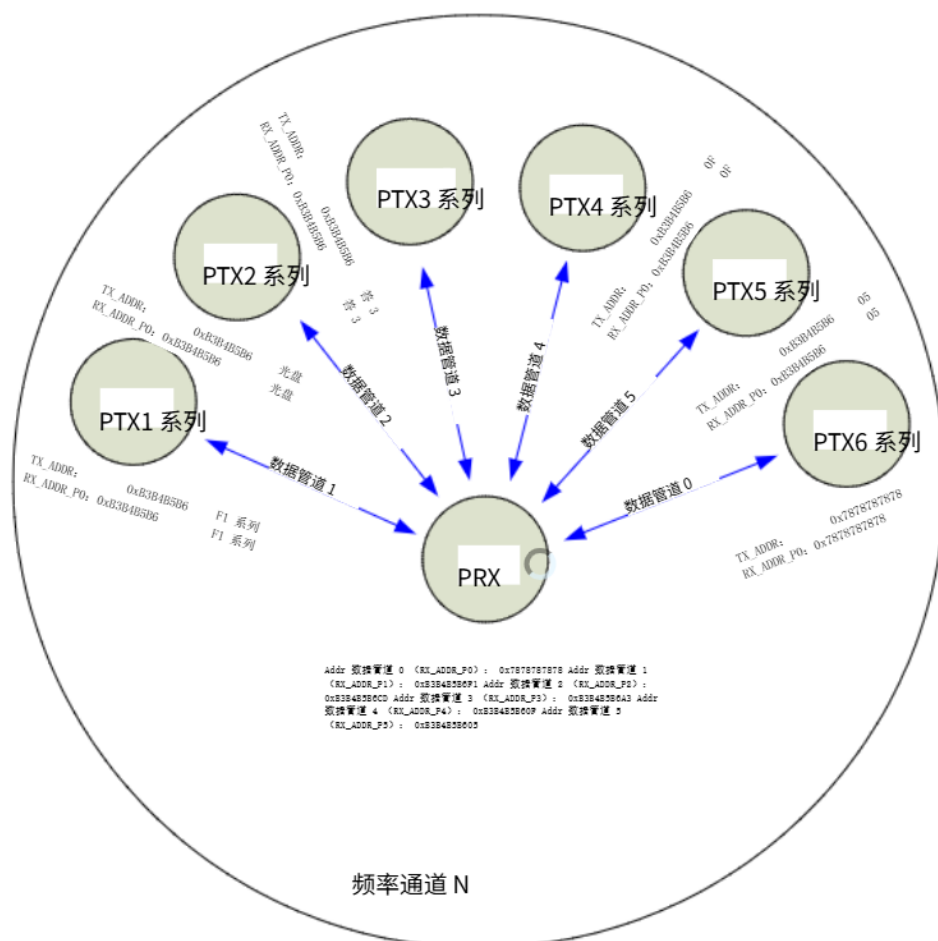


图 14. MultiCeiver™ 中的数据管道寻址示例

只有当数据管道接收到完整的数据包时，其他数据管道才能开始接收数据。当多个 PTX 传输到 PRX 时，ARD 可用于扭曲自动重新传输，以便它们仅相互阻塞一次。

7.7 Enhanced ShockBurst™ timing

This section describes the timing sequence of Enhanced ShockBurst™ and how all modes are initiated and operated. The Enhanced ShockBurst™ timing is controlled through the Data and Control interface. The nRF24L01+ can be set to static modes or autonomous modes where the internal state machine controls the events. Each autonomous mode/sequence ends with an interrupt at the **IRQ** pin. All the interrupts are indicated as IRQ events in the timing diagrams.

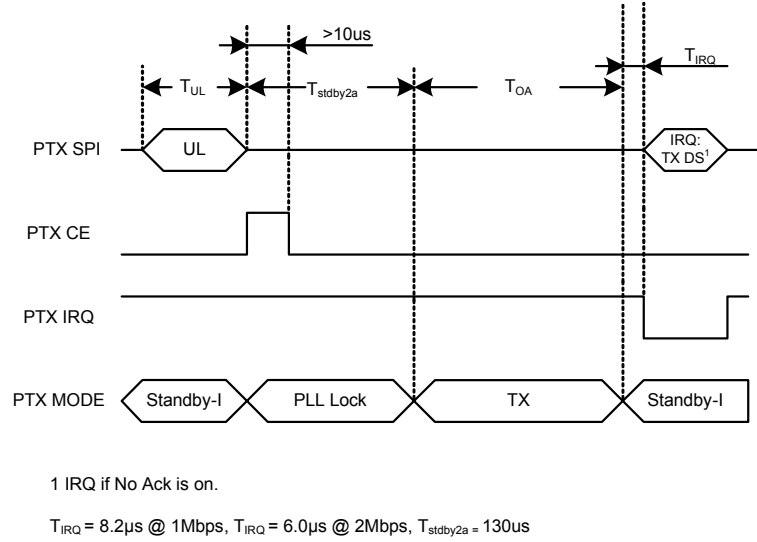


Figure 15. Transmitting one packet with NO_ACK on

The following equations calculate various timing measurements:

Symbol	Description	Equation
T_{OA}	Time on-air	$T_{\text{OA}} = \frac{\text{packet length}}{\text{air data rate}} = \frac{8 \left[\frac{\text{bit}}{\text{byte}} \right] \cdot \left(1 \left[\frac{\text{byte}}{\text{preamble}} \right] + 3, 4 \text{ or } 5 \left[\frac{\text{bytes}}{\text{address}} \right] + N \left[\frac{\text{bytes}}{\text{payload}} \right] + 1 \text{ or } 2 \left[\frac{\text{bytes}}{\text{CRC}} \right] \right) + 9 \left[\frac{\text{bit}}{\text{packet control field}} \right]}{\text{air data rate} \left[\frac{\text{bit}}{\text{s}} \right]}$
T_{ACK}	Time on-air Ack	$T_{\text{ACK}} = \frac{\text{packet length}}{\text{air data rate}} = \frac{8 \left[\frac{\text{bit}}{\text{byte}} \right] \cdot \left(1 \left[\frac{\text{byte}}{\text{preamble}} \right] + 3, 4 \text{ or } 5 \left[\frac{\text{bytes}}{\text{address}} \right] + N \left[\frac{\text{bytes}}{\text{payload}} \right] + 1 \text{ or } 2 \left[\frac{\text{bytes}}{\text{CRC}} \right] \right) + 9 \left[\frac{\text{bit}}{\text{packet control field}} \right]}{\text{air data rate} \left[\frac{\text{bit}}{\text{s}} \right]}$
T_{UL}	Time Upload	$T_{\text{UL}} = \frac{\text{payload length}}{\text{SPI data rate}} = \frac{8 \left[\frac{\text{bit}}{\text{byte}} \right] \cdot N \left[\frac{\text{bytes}}{\text{payload}} \right]}{\text{SPI data rate} \left[\frac{\text{bit}}{\text{s}} \right]}$
T_{ESB}	Time Enhanced ShockBurst™ cycle	$T_{\text{ESB}} = T_{\text{UL}} + 2 \cdot T_{\text{stby2a}} + T_{\text{OA}} + T_{\text{ACK}} + T_{\text{IRQ}}$

Table 19. Timing equations

7.7 增强的 ShockBurst™ 计时

本节介绍了 Enhanced ShockBurst™ 的时序，以及所有模式是如何启动和作的。增强型 ShockBurst™ 时序通过 Data and Control 接口进行控制。nRF24L01+ 可以设置为静态模式或自主模式，其中内部状态机控制事件。每个自主模式/序列都以 IRQ 引脚的中断结束。所有中断在 timing diagrams 中都表示为 IRQ 事件。

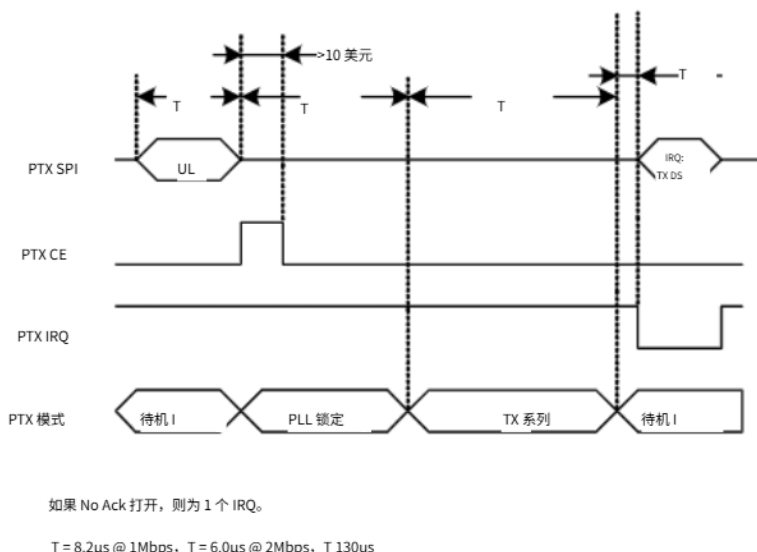


图 15.在 NO_ACK 开启的情况下传输一个数据包

以下 equation 计算各种 timing measurement:

符号描述	直播时间	方程
T	直播时间	$T_{\text{tx}} = \frac{\text{包长度}}{\text{空气数据率}} = \frac{8 \left\lceil \frac{\text{字节}}{8} \right\rceil \left(1 \left\lceil \frac{\text{字节}}{8} \right\rceil + 3, 4 \text{ 或 } 5 \left\lceil \frac{\text{字节}}{8} \right\rceil + \left\lceil \frac{N}{8} \right\rceil \right) + 1 \text{ 或 } 2 \text{ 字节}}{\text{空气数据率}} + \frac{9 \text{ 位}}{\text{空气数据率}}$
T	Time on-air Ack	$T_{\text{tx}} = \frac{\text{包长度}}{\text{空气数据率}} = \frac{8 \left\lceil \frac{\text{字节}}{8} \right\rceil \left(1 \left\lceil \frac{\text{字节}}{8} \right\rceil + 3, 4 \text{ 或 } 5 \left\lceil \frac{\text{字节}}{8} \right\rceil + \left\lceil \frac{N}{8} \right\rceil \right) + 1 \text{ 或 } 2 \text{ 字节}}{\text{空气数据率}} + \frac{9 \text{ 位}}{\text{空气数据率}}$
T	时间上传	$T_{\text{tx}} = \frac{\text{有效载荷长度}}{\text{SPI 数据率}} = \frac{8 \left\lceil \frac{\text{字节}}{8} \right\rceil \left\lceil \frac{N}{8} \right\rceil}{\text{SPI 数据率}}$
T	时间增强冲击- 突发™周期	$TT + 2 \cdot T_{\text{stby2a}} + T + T + T$

表 19.时序方程

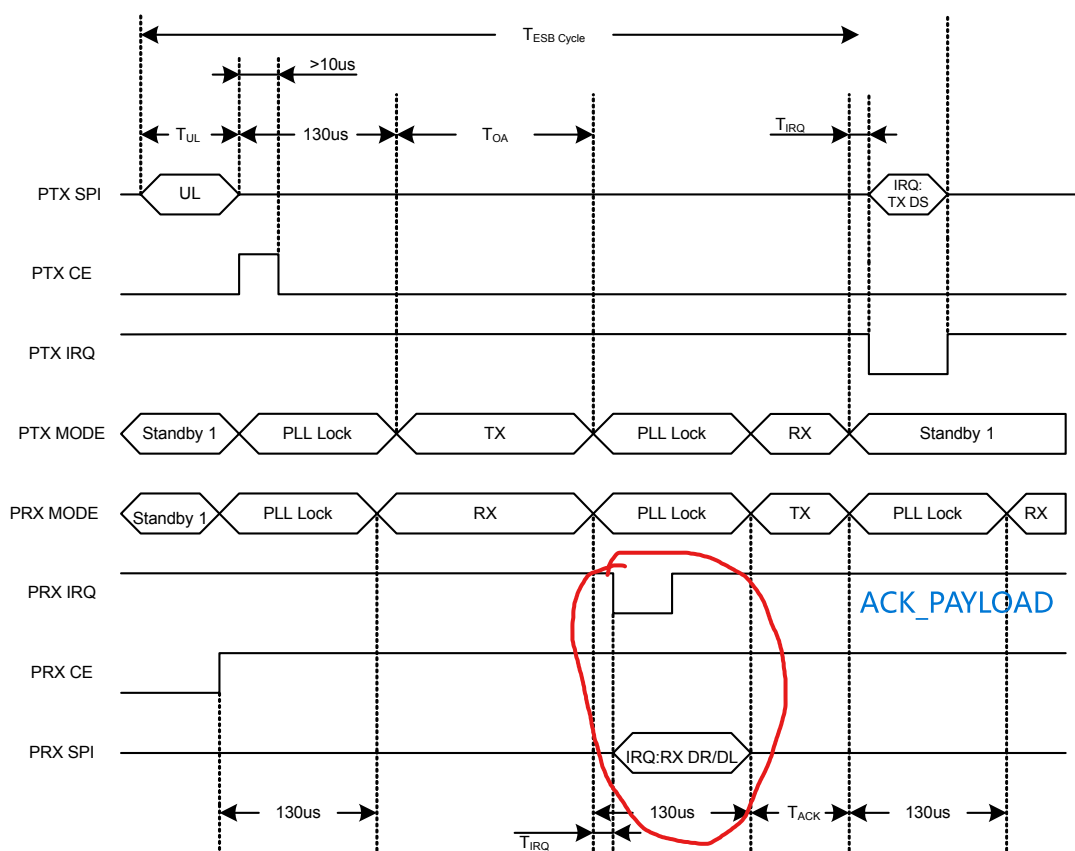


Figure 16. Timing of Enhanced ShockBurst™ for one packet upload (2Mbps)

In [Figure 16](#), the transmission and acknowledgement of a packet is shown. The PRX device activates RX mode ($CE=1$), and the PTX device is activated in TX mode ($CE=1$ for minimum 10µs). After 130µs the transmission starts and finishes after the elapse of T_{OA} .

When the transmission ends the PTX device automatically switches to RX mode to wait for the ACK packet from the PRX device. When the PRX device receives the packet it sets the interrupt for the host MCU and switches to TX mode to send an ACK. After the PTX device receives the ACK packet it sets the interrupt to the MCU and clears the packet from the TX FIFO.

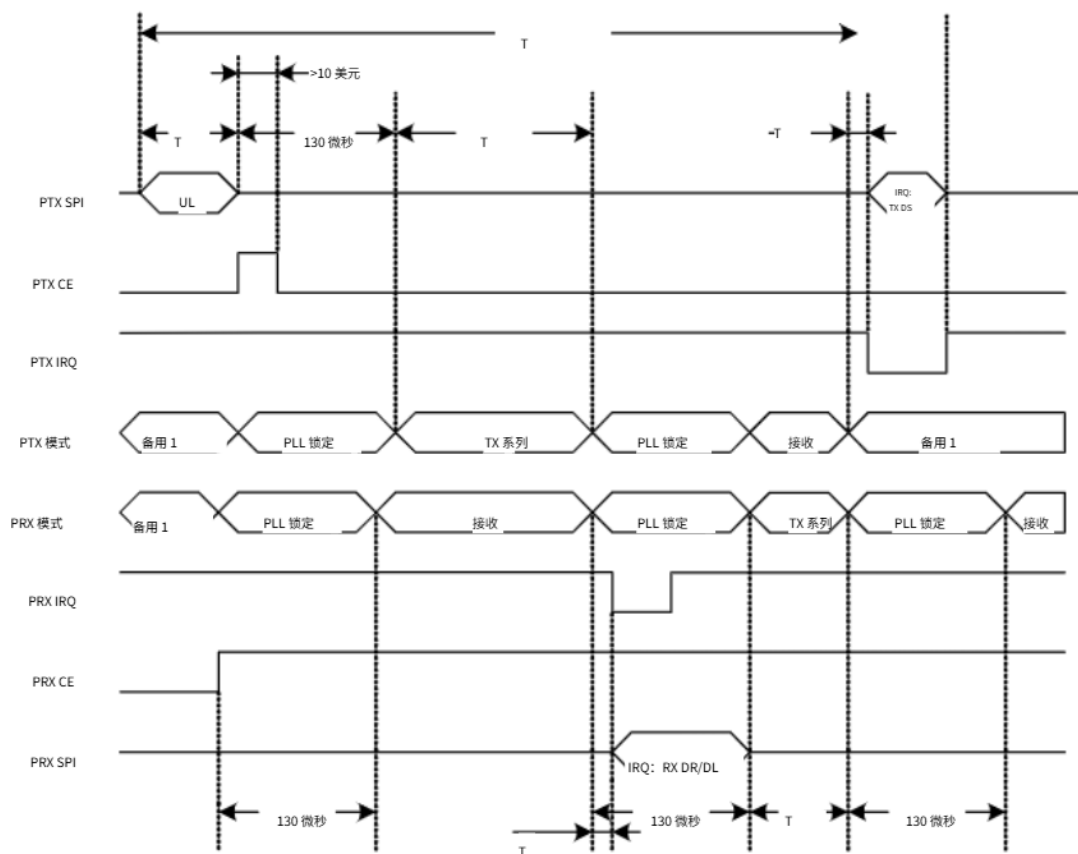


图 16.一个数据包上传的增强型 ShockBurst™ 时间 (2Mbps)

在图 16 中。将显示数据包的传输和确认。PRX 器件激活 RX 模式 (CE=1)，PTX 器件在 TX 模式下激活 (CE=1 持续至少 10μs)。130μs 后，传输开始并在 T 经过后结束。

传输结束时，PTX 设备会自动切换到 RX 模式，等待来自 PRX 设备的 ACK 数据包。当 PRX 设备收到数据包时，它会为主机 MCU 设置中断，并切换到 TX 模式以发送 ACK。PTX 设备收到 ACK 数据包后，将中断设置为 MCU 并从 TX FIFO 中清除数据包。

In [Figure 17](#), the PTX timing of a packet transmission is shown when the first ACK packet is lost. To see the complete transmission when the ACK packet fails see [Figure 20. on page 46](#).

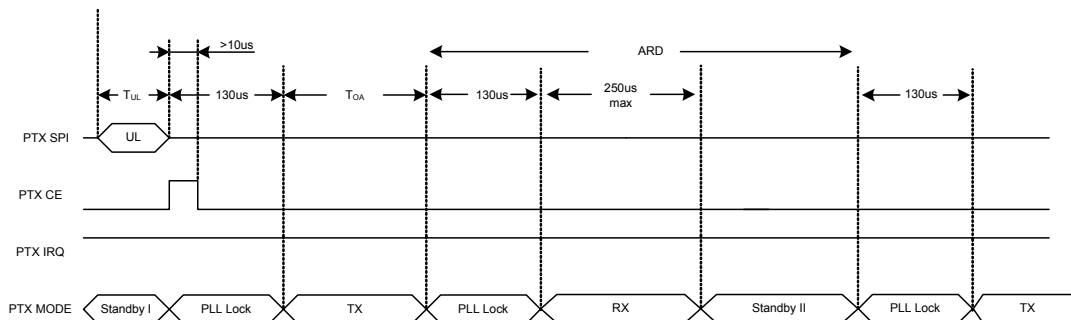


Figure 17. Timing of Enhanced ShockBurst™ when the first ACK packet is lost (2Mbps)

在图 17 中。当第一个 ACK 数据包丢失时，会显示数据包传输的 PTX 计时。要查看 ACK 数据包失败时的完整传输，请参见图 20。在第 46 页。

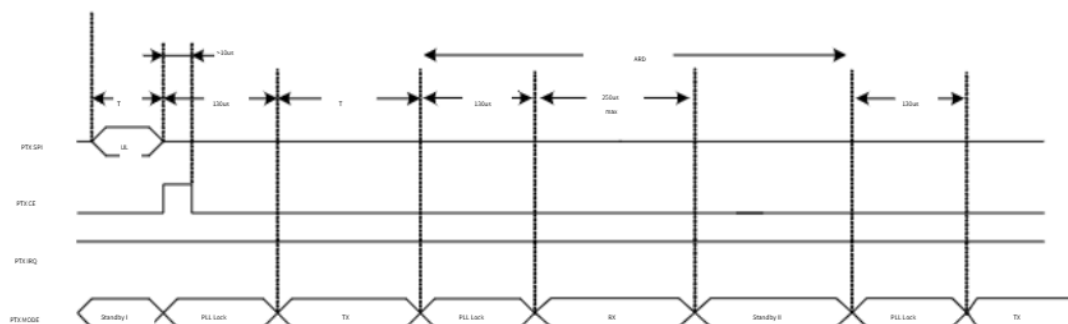


图 17.第一个 ACK 数据包丢失时增强的 ShockBurst™ 时间 (2Mbps)

7.8 Enhanced ShockBurst™ transaction diagram

This section describes several scenarios for the Enhanced ShockBurst™ automatic transaction handling. The call outs in this section's figures indicate the IRQs and other events. For MCU activity the event may be placed at a different timeframe.

Note: The figures in this section indicate the earliest possible download (DL) of the packet to the MCU and the latest possible upload (UL) of payload to the transmitter.

7.8.1 Single transaction with ACK packet and interrupts

In [Figure 18](#), the basic auto acknowledgement is shown. After the packet is transmitted by the PTX and received by the PRX the ACK packet is transmitted from the PRX to the PTX. The `RX_DR` IRQ is asserted after the packet is received by the PRX, whereas the `TX_DS` IRQ is asserted when the packet is acknowledged and the ACK packet is received by the PTX.

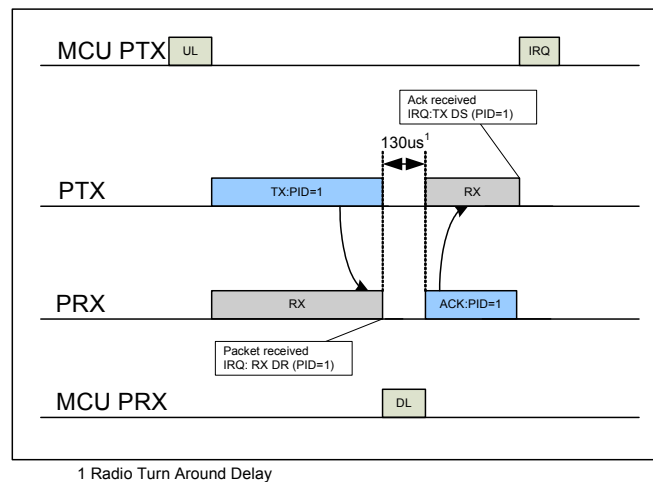


Figure 18. TX/RX cycles with ACK and the according interrupts

7.8 增强的 ShockBurst™ 事务图

本节介绍了增强型 ShockBurst™ 自动事务处理的几种场景。本节数字中的标注表示 IRQ 和其他事件。对于 MCU 活动，事件可能放置在不同的时间范围内。

注意：本节中的数字表示数据包到 MCU 的最早可能下载（DL）和负载到发射器的最晚可能上传（UL）。

7.8.1 带有 ACK 数据包和中断的单个事务

在图 18 中。将显示 Basic Auto Acknowledgement（基本自动确认）。数据包由 PTX 传输并被 PRX 接收后，ACK 数据包将从 PRX 传输到 PTX。RX_DR IRQ 在 PRX 收到数据包后断言，而 TX_DS IRQ 在数据包得到确认并且 PTX 收到 ACK 数据包时断言。

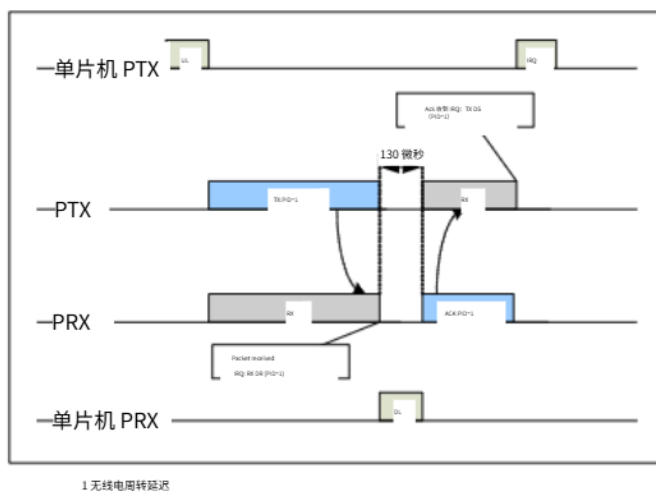


图 18.带有 ACK 和相应中断的 TX/RX 周期

7.8.2 Single transaction with a lost packet

Figure 19. is a scenario where a retransmission is needed due to loss of the first packet transmit. After the packet is transmitted, the PTX enters RX mode to receive the ACK packet. After the first transmission, the PTX waits a specified time for the ACK packet, if it is not in the specific time slot the PTX retransmits the packet as shown in Figure 19.

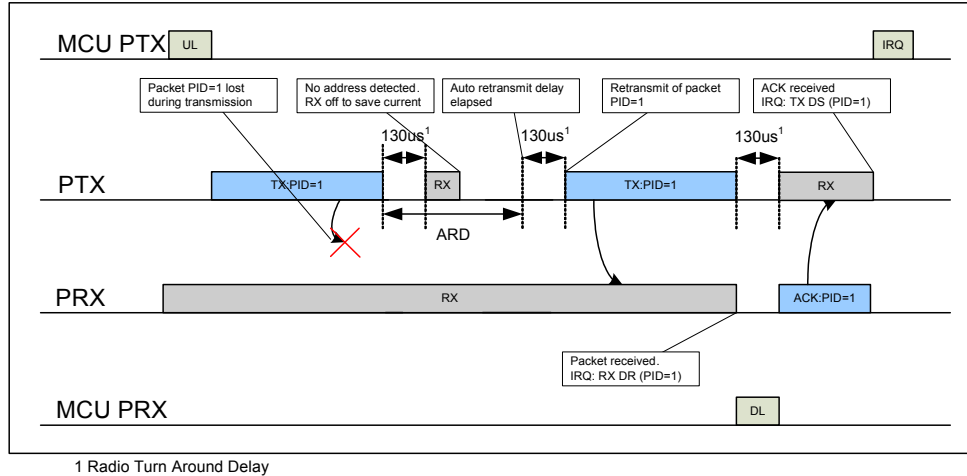


Figure 19. TX/RX cycles with ACK and the according interrupts when the first packet transmit fails

When an address is detected the PTX stays in RX mode until the packet is received. When the retransmitted packet is received by the PRX (see Figure 19.), the RX_DR IRQ is asserted and an ACK is transmitted back to the PTX. When the ACK is received by the PTX, the TX_DS IRQ is asserted.

7.8.3 Single transaction with a lost ACK packet

Figure 20. is a scenario where a retransmission is needed after a loss of the ACK packet. The corresponding interrupts are also indicated.

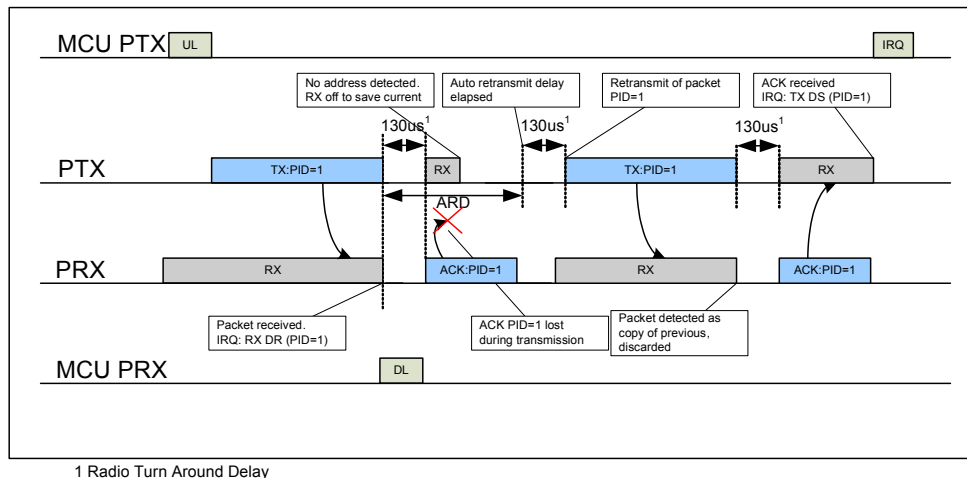


Figure 20. TX/RX cycles with ACK and the according interrupts when the ACK packet fails

7.8.2 丢失数据包的单笔交易

图 19.是由于丢失第一个数据包传输而需要重新传输的场景。数据包传输完成后，PTX 进入 RX 模式接收 ACK 数据包。第一次传输后，PTX 会等待 ACK 数据包的指定时间，如果它不在特定时隙中，PTX 会重新传输数据包，如图 19 所示。

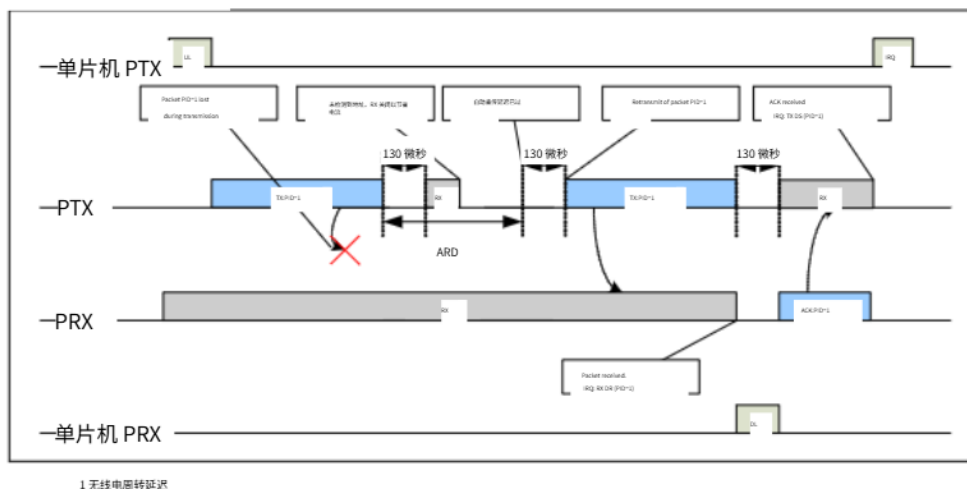


图 19.TX/RX 循环使用 ACK 和第一个数据包传输失败时的相应中断

检测到地址时，PTX 将保持 RX 模式，直到收到数据包。当 PRX 收到重新传输的数据包时（请参阅图 19.），RX_DR IRQ 将被断言，ACK 将传输回 PTX。当 PTX 收到 ACK 时，将断言 TX_DS IRQ。

7.8.3 丢失 ACK 数据包的单笔交易

图 20.是在 ACK 数据包丢失后需要重传的场景。此外，还会指示相应的中断。

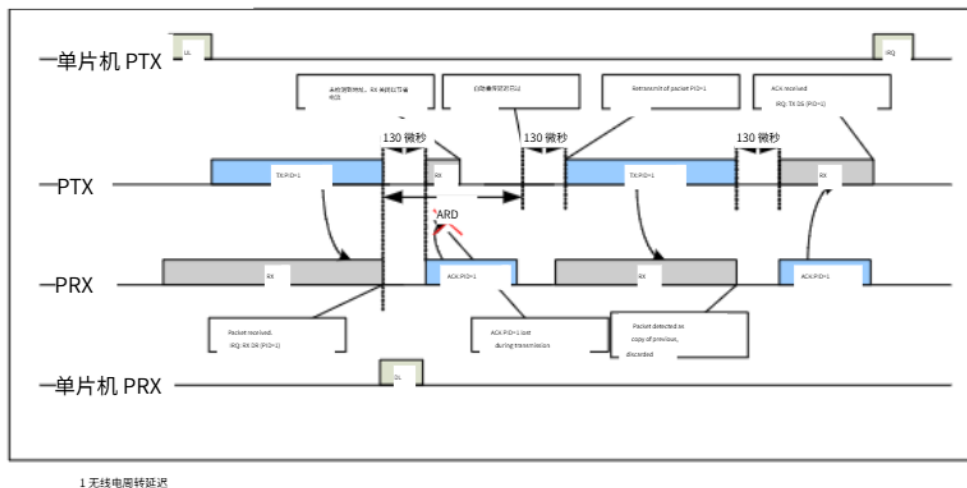


图 20.当 ACK 数据包失败时，使用 ACK 和相应的中断进行 TX/RX 循环

7.8.4 Single transaction with ACK payload packet

Figure 21. is a scenario of the basic auto acknowledgement with payload. After the packet is transmitted by the PTX and received by the PRX the ACK packet with payload is transmitted from the PRX to the PTX. The `RX_DR` IRQ is asserted after the packet is received by the PRX, whereas on the PTX side the `TX_DS` IRQ is asserted when the ACK packet is received by the PTX. On the PRX side, the `TX_DS` IRQ for the ACK packet payload is asserted after a new packet from PTX is received. The position of the IRQ in Figure 21. shows where the MCU can respond to the interrupt.

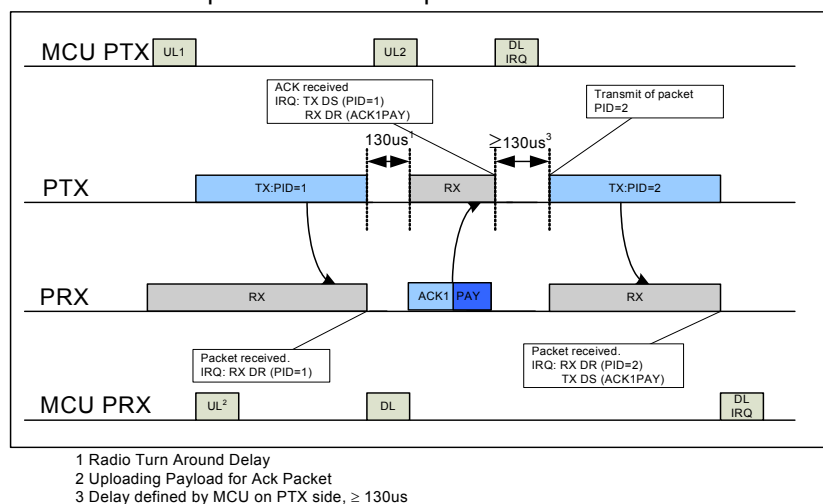


Figure 21. TX/RX cycles with ACK Payload and the according interrupts

7.8.5 Single transaction with ACK payload packet and lost packet

Figure 22. is a scenario where the first packet is lost and a retransmission is needed before the `RX_DR` IRQ on the PRX side is asserted. For the PTX both the `TX_DS` and `RX_DR` IRQ are asserted after the ACK packet is received. After the second packet (PID=2) is received on the PRX side both the `RX_DR` (PID=2) and `TX_DS` (ACK packet payload) IRQ are asserted.

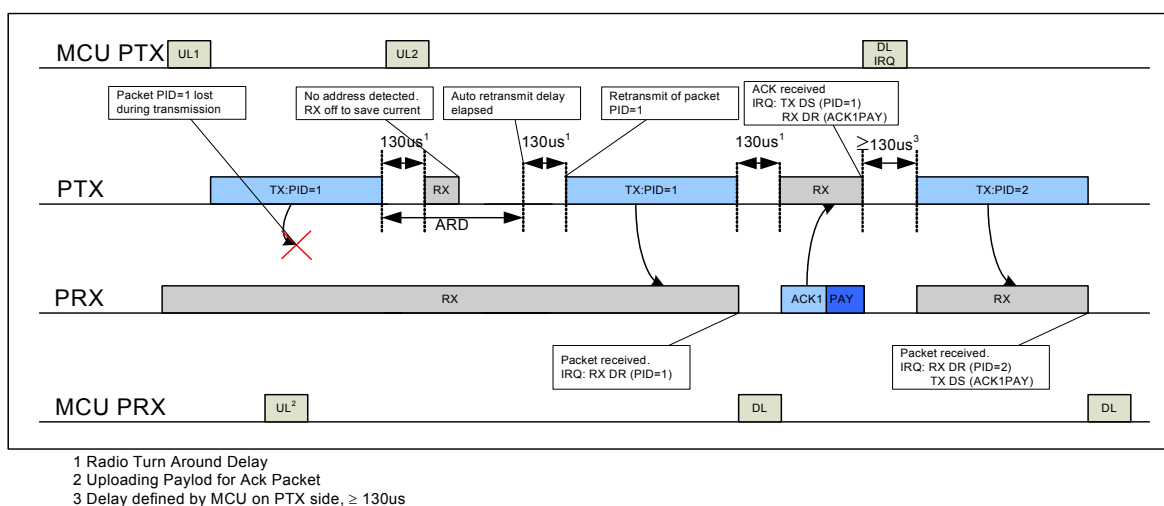
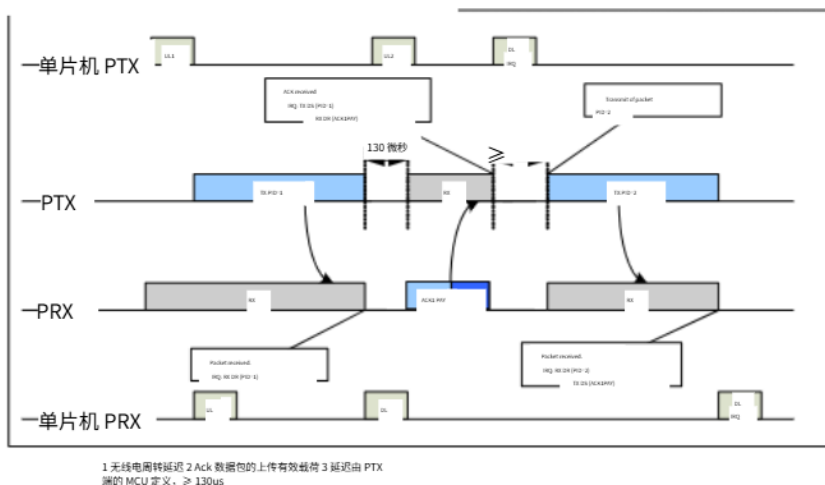


Figure 22. TX/RX cycles and the according interrupts when the packet transmission fails

7.8.4 带有 ACK payload packet 的单个事务

图 21.是具有 payload 的基本自动确认的场景。数据包由 PTX 传输并由 PRX 接收后，带有有效负载的 ACK 数据包将从 PRX 传输到 PTX。RX_DR IRQ 在 PRX 收到数据包后置位，而在 PTX 端，当 PTX 收到 ACK 数据包时，TX_DS IRQ 置位。在 PRX 端，收到来自 PTX 的新数据包后，将断言 ACK 数据包有效负载的 TX_DS IRQ。IRQ 的位置在图 21 中。显示 MCU 可以响应中断的位置。图 21.带有 ACK Payload 和相应中断的 TX/RX 周期



7.8.5 包含 ACK payload packet 和 lost packet 的单个事务

图 22.是第一个数据包丢失，并且在断言 PRX 端的 RX_DR IRQ 之前需要重新传输的场景。对于 PTX，收到 ACK 数据包后，TX_DS 和 RX_DR IRQ 都会断言。在 PRX 端收到第二个数据包 (PID=2) 后，RX_DR (PID=2) 和 TX_DS (ACK 数据包有效负载) IRQ 都将被置位。

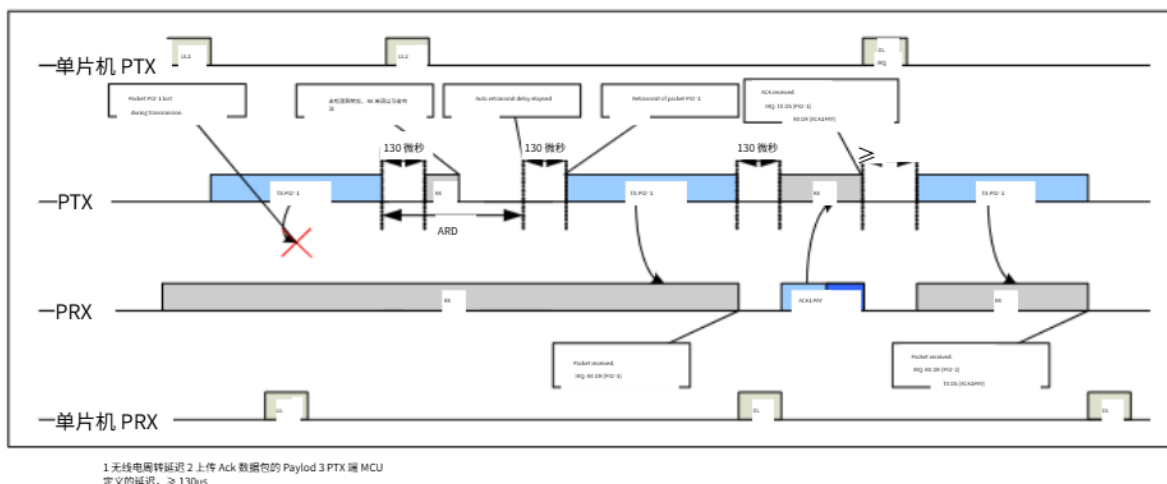


图 22.数据包传输失败时的 TX/RX 周期和相应的中断

7.8.6 Two transactions with ACK payload packet and the first ACK packet lost

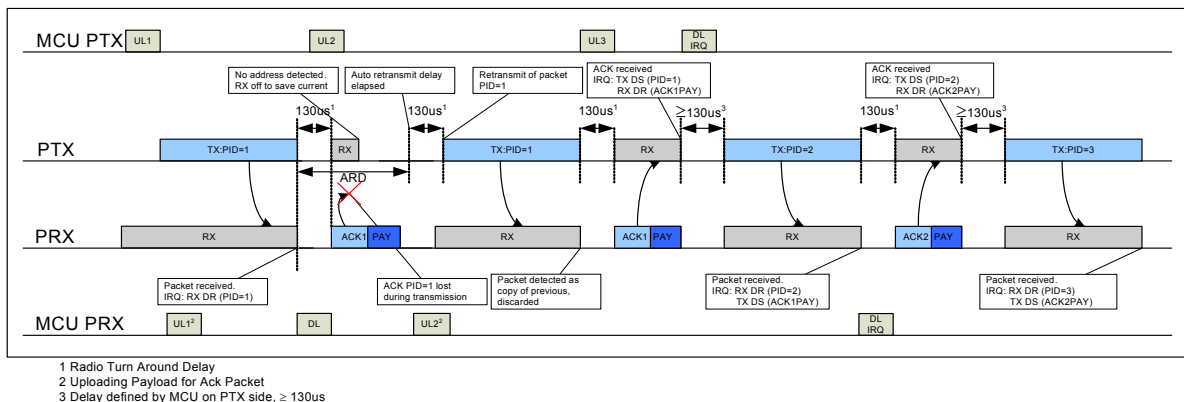


Figure 23. TX/RX cycles with ACK Payload and the according interrupts when the ACK packet fails

In Figure 23, the ACK packet is lost and a retransmission is needed before the TX_DS IRQ is asserted, but the RX_DR IRQ is asserted immediately. The retransmission of the packet (PID=1) results in a discarded packet. For the PTX both the TX_DS and RX_DR IRQ are asserted after the second transmission of ACK, which is received. After the second packet (PID=2) is received on the PRX both the RX_DR (PID=2) and TX_DS (ACK1PAY) IRQ is asserted. The callouts explains the different events and interrupts.

7.8.7 Two transactions where max retransmissions is reached

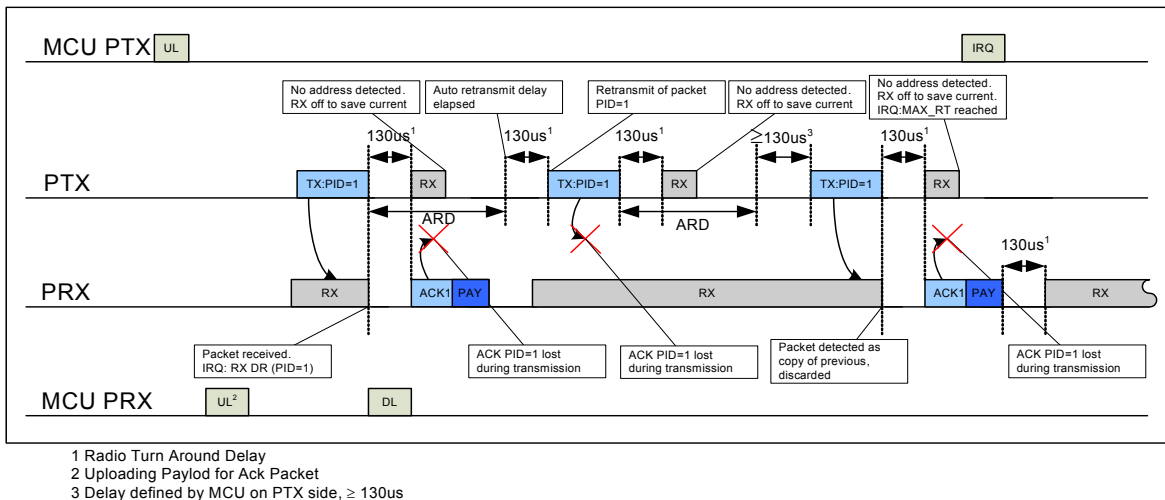


Figure 24. TX/RX cycles with ACK Payload and the according interrupts when the transmission fails. ARC is set to 2.

MAX_RT IRQ is asserted if the auto retransmit counter (ARC_CNT) exceeds the programmed maximum limit (ARC). In Figure 24, the packet transmission ends with a MAX_RT IRQ. The payload in TX FIFO is NOT removed and the MCU decides the next step in the protocol. A toggle of the CE starts a new transmitting sequence of the same packet. The payload can be removed from the TX FIFO using the FLUSH_TX command.

7.8.6 两个 ACK 负载数据包的事务和第一个 ACK 数据包丢失

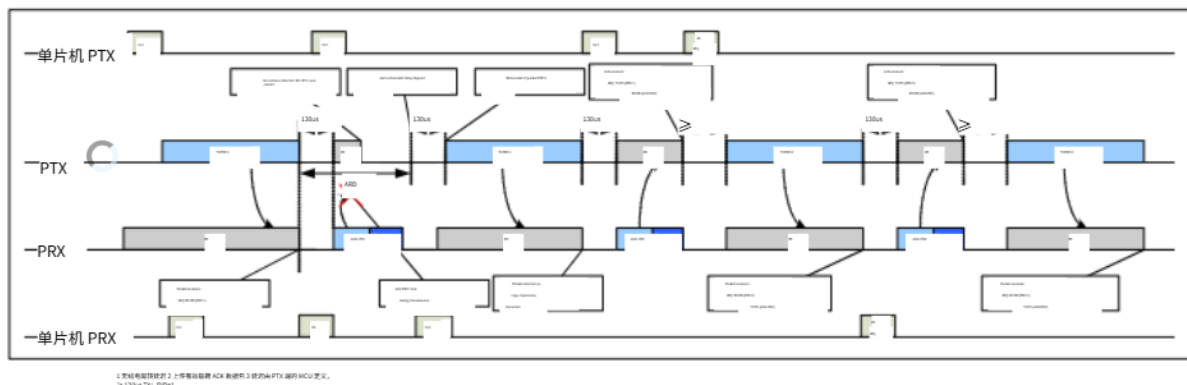


图 23.ACK Payload 的 TX/RX 循环，当 ACK 数据包失败时，会有相应的中断

在图 23.ACK 数据包丢失，并且在断言 TX_DS IRQ 之前需要重新传输，但 RX_DR IRQ 会立即断言。数据包的重新传输（PID=1）会导致数据包被丢弃。对于 PTX，TX_DS 和 RX_DR IRQ 在第二次传输 ACK 后（接收）均被断言。在 PRX 上收到第二个数据包（PID=2）后，RX_DR（PID=2）和 TX_DS（ACK1PAY）IRQ 都被置位。标注解释了不同的事件和中断。

7.8.7 达到最大重新传输数的两个事务

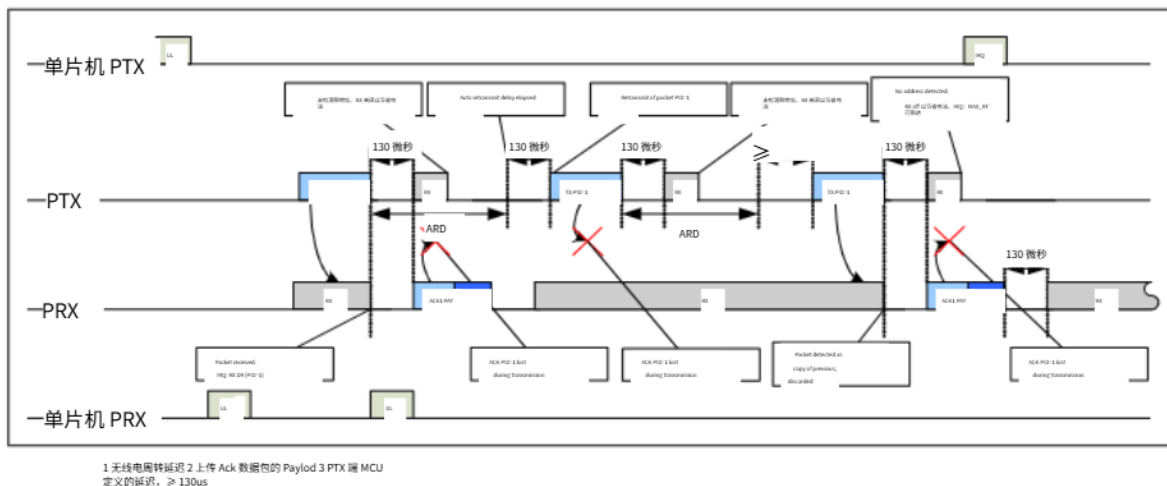


图 24.TX/RX 循环与 ACK Payload 和传输失败时的相应中断。ARC 设置为 2。

如果自动重传计数器（ARC_CNT）超过编程的最大限制（ARC），则 MAX_RT IRQ 被置位。在图 24.数据包传输以 MAX_RT IRQ 结束。TX FIFO 中的有效负载不会被删除，MCU 决定协议的下一步。CE 的切换将启动同一数据包的新传输序列。可以使用 FLUSH_TX 命令从 TX FIFO 中删除有效负载。

7.9 Compatibility with ShockBurst™

You must disable Enhanced ShockBurst™ for backward compatibility with the nRF2401A, nRF2402, nRF24E1 and nRF24E2. Set the register `EN_AA` = 0x00 and `ARC` = 0 to disable Enhanced ShockBurst™. In addition, the nRF24L01+ air data rate must be set to 1Mbps or 250kbps.

7.9.1 ShockBurst™ packet format

[Figure 25](#) shows the packet format with MSB to the left.

Preamble 1 byte	Address 3-5 byte	Payload 1 - 32 byte	CRC 1-2 byte
-----------------	------------------	---------------------	--------------

Figure 25. A ShockBurst™ packet compatible with nRF2401/nRF2402/nRF24E1/nRF24E2 devices.

The ShockBurst™ packet format has a preamble, address, payload and CRC field that are the same as the Enhanced ShockBurst™ packet format described in [section 7.3 on page 28](#).

The differences between the ShockBurst™ packet and the Enhanced ShockBurst™ packet are:

- The 9 bit Packet Control Field is not present in the ShockBurst™ packet format.
- The CRC is optional in the ShockBurst™ packet format and is controlled by the `EN_CRC` bit in the `CONFIG` register.

7.9 与 ShockBurst™ 的兼容性

您必须禁用增强型 ShockBurst™ 才能向后兼容 nRF2401A、nRF2402、nRF24E1 和 nRF24E2。设置寄存器 EN_AA = 0x00 且 ARC = 0 以禁用增强型 ShockBurst™。
此外，nRF24L01+ 空中数据速率必须设置为 1Mbps 或 250kbps。

7.9.1 ShockBurst™ 数据包格式

图 25.显示 MSB 位于左侧的数据包格式。



图 25.与 nRF2401/nRF2402/nRF24E1/nRF24E2 设备兼容的 ShockBurst™ 数据包。

ShockBurst™ 数据包格式具有前导码、地址、有效负载和 CRC 字段，它们与第 7.3 页第 28 节中描述的增强型 ShockBurst™ 数据包格式相同。

ShockBurst™ 数据包和增强型 ShockBurst™ 数据包的区别在于：

- 9 位数据包控制字段在 ShockBurst™ 数据包格式中不存在。
- CRC 在 ShockBurst™ 数据包格式中是可选的，由 CONFIG 寄存器中的 EN_CRC 位控制。

8 Data and Control Interface

The data and control interface gives you access to all the features in the nRF24L01+. The data and control interface consists of the following six 5Volt tolerant digital signals:

- **IRQ** (this signal is active low and controlled by three maskable interrupt sources)
- **CE** (this signal is active high and used to activate the chip in RX or TX mode)
- **CSN** (SPI signal)
- **SCK** (SPI signal)
- **MOSI** (SPI signal)
- **MISO** (SPI signal)

Using 1 byte SPI commands, you can activate the nRF24L01+ data FIFOs or the register map during all modes of operation.

8.1 Features

- Special SPI commands for quick access to the most frequently used features
- 0-10Mbps 4-wire SPI
- 8 bit command set
- Easily configurable register map
- Full three level FIFO for both TX and RX direction

8.2 Functional description

The SPI is a standard SPI with a maximum data rate of 10Mbps.

8.3 SPI operation

This section describes the SPI commands and timing.

8.3.1 SPI commands

The SPI commands are shown in [Table 20](#). Every new command must be started by a high to low transition on **CSN**.

The **STATUS** register is serially shifted out on the **MISO** pin simultaneously to the SPI command word shifting to the **MOSI** pin.

The serial shifting SPI commands is in the following format:

<**Command word**: MSBit to LSBit (one byte)>

<**Data bytes**: LSByte to MSByte, MSBit in each byte first>

See [Figure 26. on page 52](#) and [Figure 27. on page 52](#) for timing information.

8 数据和控制接口

数据和控制界面可让您访问 nRF24L01+ 中的所有功能。数据和控制接口由以下 6 个 5V 容差数字信号组成：

- IRQ (此信号为低电平有效，由三个可屏蔽中断源控制)
- CE (此信号为高电平有效，用于在 RX 或 TX 模式下激活芯片)
- CSN (SPI 信号)
- SCK (SPI 信号)
- MOSI (SPI 信号)
- MISO (SPI 信号)

使用 1 字节 SPI 命令，您可以在所有作模式下激活 nRF24L01+ 数据 FIFO 或寄存器映射。

8.1 特征

Mode 0

- 特殊的 SPI 命令，用于快速访问最常用的功能
- 0-10Mbps 4 线 SPI
- 8 位命令集
- 易于配置的寄存器映射
- 用于 TX 和 RX 方向的全三电平 FIFO

8.2 功能描述

SPI 是标准 SPI，最大数据速率为 10Mbps。

8.3 SPI 作

本节介绍 SPI 命令和计时。

8.3.1 SPI 命令

SPI 命令如表 20 所示。每个新命令都必须由 CSN 上的高到低转换来启动。

STATUS 寄存器在 MISO 引脚上串行移出，同时 SPI 命令字移位到 MOSI 引脚。

串行移位 SPI 命令的格式如下：

<命令字：MSBit 到 LSBit (一个字节) >

<数据字节：LSByte 到 MSByte，每个字节首先输入 MSBit>

参见图 26。在第 52 页和图 27 中。在第 52 页了解时间信息。

Command name	Command word (binary)	# Data bytes	Operation
R_REGISTER	000A AAAA	1 to 5 LSByte first	Read command and <i>status</i> registers. AAAAA = 5 bit Register Map Address
W_REGISTER	001A AAAA	1 to 5 LSByte first	Write command and status registers. AAAAA = 5 bit Register Map Address Executable in power down or standby modes only.
R_RX_PAYLOAD	0110 0001	1 to 32 <u>LSByte first</u>	Read RX-payload: 1 – 32 bytes. A read operation always starts at byte 0. Payload is deleted from FIFO after it is read. Used in RX mode.
W_TX_PAYLOAD	1010 0000	1 to 32 LSByte first	Write TX-payload: 1 – 32 bytes. A write operation always starts at byte 0 <u>used in TX payload</u> .
FLUSH_TX	1110 0001	0	Flush TX FIFO, <u>used in TX mode</u>
FLUSH_RX	1110 0010	0	Flush RX FIFO, <u>used in RX mode</u> Should not be executed during transmission of acknowledge, that is, acknowledge package will not be completed.
REUSE_TX_PL	1110 0011	0	Used for a PTX device Reuse last transmitted payload. TX payload reuse is active until W_TX_PAYLOAD or FLUSH TX is executed. TX payload reuse must not be activated or deactivated during package transmission.
R_RX_PL_WID ^a	0110 0000	1	Read RX payload width for the top R_RX_PAYLOAD in the RX FIFO. Note: Flush RX FIFO if the read value is larger than 32 bytes.
W_ACK_PAYLOAD ^a	1010 <u>1PPP</u>	1 to 32 LSByte first	<u>Used in RX mode.</u> Write Payload to be transmitted together with ACK packet on PIPE PPP. (<u>PPP valid in the range from 000 to 101</u>). Maximum three ACK packet payloads can be pending. Payloads with same PPP are handled using first in - first out principle. Write payload: 1– 32 bytes. A write operation always starts at byte 0.
W_TX_PAYLOAD_NO_ACK ^a	1011 0000	1 to 32 LSByte first	<u>Used in TX mode.</u> Disables AUTOACK on this specific packet.
NOP	1111 1111	0	No Operation. Might be used to read the STATUS register

a. The bits in the FEATURE register shown in [Table 28. on page 63](#) have to be set.

Table 20. Command set for the nRF24L01+ SPI

The W_REGISTER and R_REGISTER commands operate on single or multi-byte registers. When accessing multi-byte registers read or write to the MSBit of LSByte first. You can terminate the writing before all bytes in a multi-byte register are written, leaving the unwritten MSByte(s) unchanged. For example, the LSByte of RX_ADDR_P0 can be modified by writing only one byte to the RX_ADDR_P0 register. The content of the *status* register is always read to MISO after a high to low transition on CSN.

命令名称	命令 word (二进制)	# 数据字节	操作
R_REGISTER	000A AAAA 型	1 至 5 LSByte 优先	读取命令和状态寄存器。AAAAA = 5 位寄存器映射地址
W_REGISTER	001A AAAA 级	1 至 5 LSByte 优先	写入命令和状态寄存器。AAAAA = 5 位寄存器映射地址 仅在掉电或待机模式下可执行。
R_RX_PAYLOAD	0110 0001	1 至 32 LSByte 优先	读取 RX-payload: 1 – 32 字节。读取作始终从字节 0 开始。读取 Payload 后, 将从 FIFO 中删除 Payload。在 DV 模式下使用
W_TX_PAYLOAD	1010 0000	1 至 32 LSByte 优先	写入 TX 有效载荷: 1 – 32 字节。写入作始终从 TX 有效负载中使用的字节 0 开始。
FLUSH_TX FLUSH_RX	1110 0001 1110 0010	0 0	Flush TX FIFO, 用于 TX 模式 Flush RX FIFO, 在 RX 模式下使用不应在发送确认时执行, 即确认包不会完成。
REUSE_TX_PL	1110 0011	0	用于 PTX 设备 重复使用上次传输的有效负载。TX 负载重用处于活动状态, 直到执行 W_TX_PAYLOAD 或 FLUSH TX。在包裹传输期间, 不得激活或停用 TX 负载重用。
R_RX_PL_WID	0110 0000	1	读取 RX FIFO 中前 R_RX_PAYLOAD 的 RX payload width。 注意: 如果读取值大于 32 字节, 则刷新 RX FIFO。
W_ACK_PAYLOAD	1010 1 购买力平	1 至 32 LSByte 优先	在 RX 模式下使用。 写入 Payload 以与 PIPE PPP 上的 ACK 数据包一起传输。PPP 的有效范围为 000 到 101)。最多可以有 3 个 ACK 数据包负载处于待处理状态。具有相同 PPP 的有效载荷采用先进先出原则处理。写入有效负载: 1– 32 字节。写入作始终从字节 0 开始。
W_TX_PAYLOAD_NO_ACK NOP	1011 0000 1111 1111	1 至 32 LSByte 优先 0	在 TX 模式下使用。禁用此特定数据包的 AUTOACK。 无作。可能用于读取 STATUS 寄存器

一个。FEATURE 寄存器中的位如表 28 所示。在第 63 页。

表 20.nRF24L01+ SPI 的命令集

W_REGISTER 和 R_REGISTER 命令在单字节或多字节寄存器上运行。访问多字节寄存器时, 首先读取或写入 LSByte 的 MSBit。您可以在写入多字节寄存器中的所有字节之前终止写入, 而未写入的 MSByte 保持不变。例如, RX_ADDR_P0 的 LSByte 可以通过仅向 RX_ADDR_P0 register 写入一个字节来修改。在 CSN 上从高到低转换后, 状态寄存器的内容始终读取到 MISO。

Note: The 3 bit pipe information in the `STATUS` register is updated during the `IRQ` pin high to low transition. The pipe information is unreliable if the `STATUS` register is read during an `IRQ` pin high to low transition.

8.3.2 SPI timing

SPI operation and timing is shown in [Figure 26.](#) to [Figure 28.](#) and in [Table 22.](#) to [Table 27.](#) nRF24L01+ must be in a standby or power down mode before writing to the configuration registers.

In [Figure 26.](#) to [Figure 28.](#) the following abbreviations are used:

Abbreviation	Description
Cn	SPI command bit
Sn	<code>STATUS</code> register bit
Dn	Data Bit (Note: LSByte to MSByte, MSBit in each byte first)

Table 21. Abbreviations used in Figure 26. to Figure 28.



Figure 26. SPI read operation

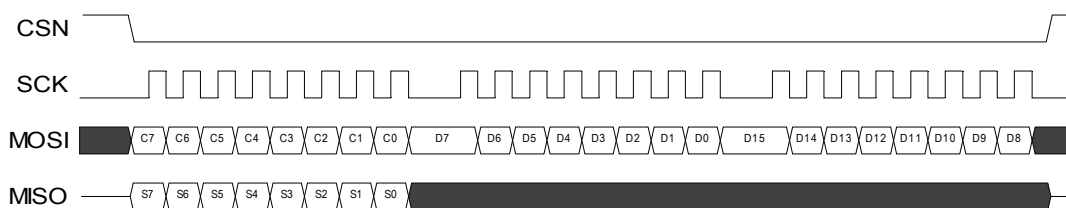


Figure 27. SPI write operation

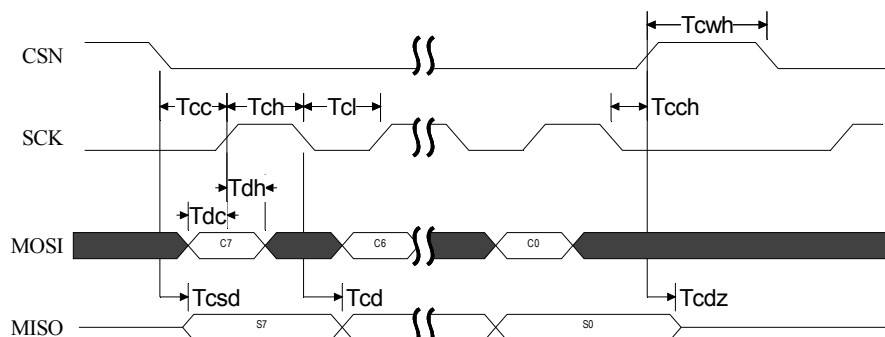


Figure 28. SPI NOP timing diagram

注意：STATUS 寄存器中的 3 位管道信息在 IRQ 引脚高到低期间更新过渡。如果在 IRQ 引脚高到低转换期间读取 STATUS 寄存器，则管道信息不可靠。

8.3.2 SPI 时序

SPI 作和时序如图 26 所示。到图 28。和表 22。到表 27..nRF24L01+ 在写入配置寄存器之前必须处于待机或掉电模式。

在图 26 中。到图 28。使用以下缩写：

缩写	描述
Cn SPI 命令位	
Sn STATUS 寄存器位 Dn	
数据位（注意：LSByte 到 MSByte，每个字节中的 MSBit 在前）	

表 21.图 26 中使用的缩写。到图 28。

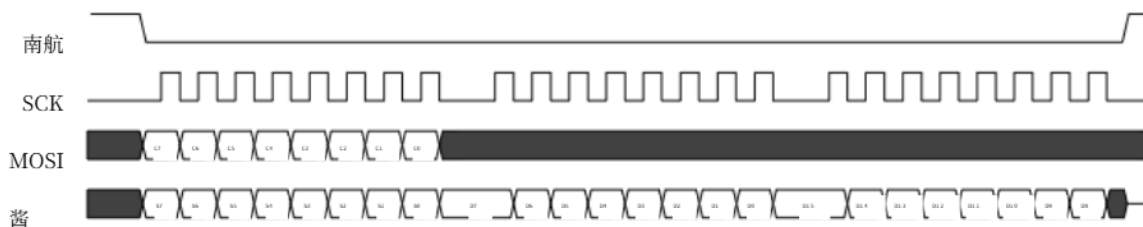


图 26.SPI 读取作

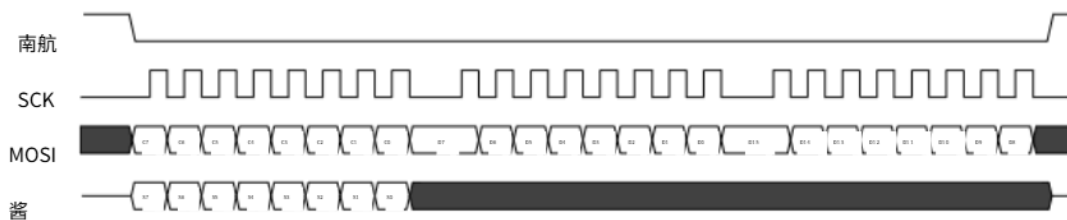


图 27.SPI 写入作

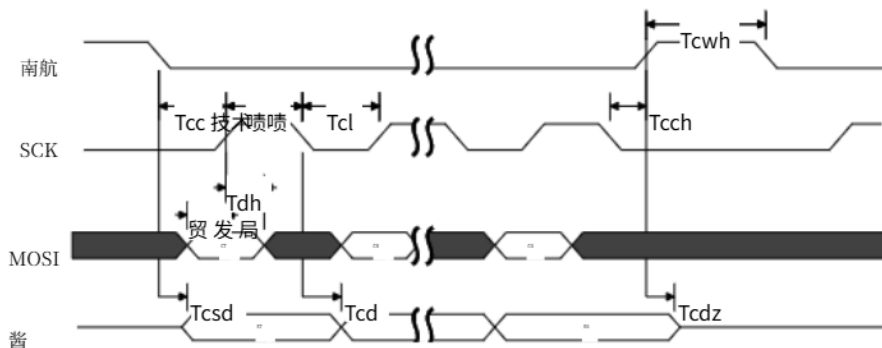


图 28.SPI NOP 时序图

Figure 29. shows the R_{pull} and C_{load} that are referenced in Table 22. to Table 27.

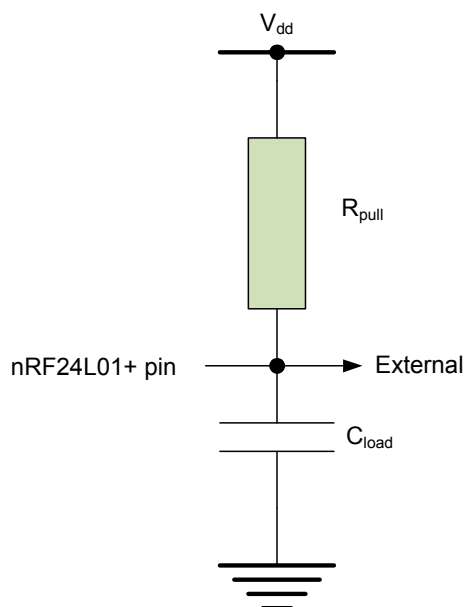


Figure 29. R_{pull} and C_{load}

Symbol	Parameters	Min.	Max	Units
Tdc	Data to sck Setup	2		ns
Tdh	sck to Data Hold	2		ns
Tcsd	csn to Data Valid		38	ns
Tcd	sck to Data Valid		55	ns
Tcl	sck Low Time	40		ns
Tch	sck High Time	40		ns
Fsck	sck Frequency	0	10	MHz
Tr,Tf	sck Rise and Fall		100	ns
Tcc	csn to sck Setup	2		ns
Tcch	sck to csn Hold	2		ns
Tcwh	csn Inactive time	50		ns
Tcdz	csn to Output High Z		38	ns

Table 22. SPI timing parameters ($C_{load} = 5pF$)

Symbol	Parameters	Min.	Max	Units
Tdc	Data to sck Setup	2		ns
Tdh	sck to Data Hold	2		ns
Tcsd	csn to Data Valid		42	ns
Tcd	sck to Data Valid		58	ns
Tcl	sck Low Time	40		ns
Tch	sck High Time	40		ns
Fsck	sck Frequency	0	8	MHz
Tr,Tf	sck Rise and Fall		100	ns
Tcc	csn to sck Setup	2		ns

图 29.显示了表 22 中引用的 R 和 C。到表 27。

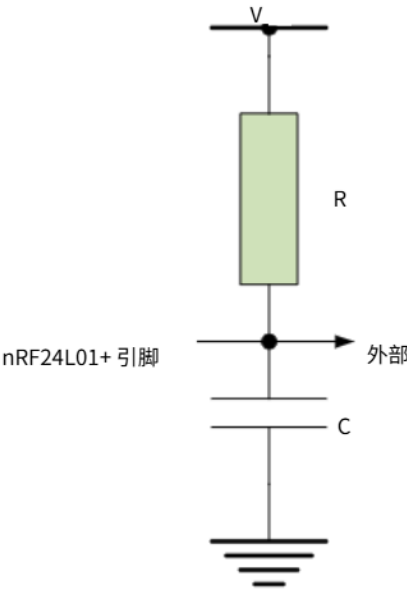


图 29.R 和 C

元件参数	最小	麦克斯	单位
Tdc 数据到 SCK 设置	2		夜行者
Tdh SCK 到数据保留	2		夜行者
tcsd csn 到 data valid		38	夜行者
tcd SCK 到数据有效		55	夜行者
Tcl SCK 低电平	40		夜行者
Tch SCK High Time	40		夜行者
Fsck SCK 频率	0	10	兆赫
Tr, Tf SCK 上升和下降		100	夜行者
Tcc CSN 到 SCK 设置	2		夜行者
TCCH SCK 到 CSN 保留	2		夜行者
Tcwh CSN 非活动时间	50		夜行者
Tcdz CSN 到输出高 Z		38	夜行者

表 22.SPI 时序参数 (C = 5pF)

元件参数	最小	麦克斯	单位
Tdc 数据到 SCK 设置	2		夜行者
Tdh SCK 到数据保留	2		夜行者
tcsd csn 到 data valid		42	夜行者
tcd SCK 到数据有效		58	夜行者
Tcl SCK 低电平	40		夜行者
Tch SCK High Time	40		夜行者
Fsck SCK 频率	0	8	兆赫
Tr, Tf SCK 上升和下降		100	夜行者
Tcc 技术 CSN 到 SCK 设置	2		夜行者

Symbol	Parameters	Min.	Max	Units
Tcch	sck to csN Hold	2		ns
Tcwh	csN Inactive time	50		ns
Tcdz	csN to Output High Z		42	ns

Table 23. SPI timing parameters ($C_{load} = 10pF$)

Symbol	Parameters	Min.	Max	Units
Tdc	Data to sck Setup	2		ns
Tdh	sck to Data Hold	2		ns
Tcsd	csN to Data Valid		75	ns
Tcd	sck to Data Valid		86	ns
Tcl	sck Low Time	40		ns
Tch	sck High Time	40		ns
Fsck	sck Frequency	0	5	MHz
Tr,Tf	sck Rise and Fall		100	ns
Tcc	csN to sck Setup	2		ns
Tcch	sck to csN Hold	2		ns
Tcwh	csN Inactive time	50		ns
Tcdz	csN to Output High Z		75	ns

Table 24. SPI timing parameters ($R_{pull} = 10k\Omega$, $C_{load} = 50pF$)

Symbol	Parameters	Min.	Max	Units
Tdc	Data to sck Setup	2		ns
Tdh	sck to Data Hold	2		ns
Tcsd	csN to Data Valid		116	ns
Tcd	sck to Data Valid		123	ns
Tcl	sck Low Time	40		ns
Tch	sck High Time	40		ns
Fsck	sck Frequency	0	4	MHz
Tr,Tf	sck Rise and Fall		100	ns
Tcc	csN to sck Setup	2		ns
Tcch	sck to csN Hold	2		ns
Tcwh	csN Inactive time	50		ns
Tcdz	csN to Output High Z		116	ns

Table 25. SPI timing parameters ($R_{pull} = 10k\Omega$, $C_{load} = 100pF$)

象征	参数	最小	麦克斯	单位
TCCH SCK 到 CSN 保留		2		夜行者
Tcwh CSN 非活动时间		50		夜行者
Tcdz CSN 到输出高 Z			42	夜行者

表 23.SPI 时序参数 (C = 10pF)

元件参数	最小	麦克斯	单位
Tdc 数据到 SCK 设置	2		夜行者
Tdh SCK 到数据保留	2		夜行者
tcsd csn 到 data valid		75	夜行者
tcd SCK 到数据有效		86	夜行者
Tcl SCK 低电平	40		夜行者
Tch SCK High Time	40		夜行者
Fsck SCK 频率	0	5	兆赫
Tr, Tf SCK 上升和下降		100	夜行者
Tcc CSN 到 SCK 设置	2		夜行者
TCCH SCK 到 CSN 保留	2		夜行者
Tcwh CSN 非活动时间	50		夜行者
Tcdz CSN 到输出高 Z		75	夜行者

表 24.SPI 时序参数 (R = 10kΩ, C = 50pF)

元件参数	最小	麦克斯	单位
Tdc 数据到 SCK 设置	2		夜行者
Tdh SCK 到数据保留	2		夜行者
tcsd csn 到 data valid		116	夜行者
tcd SCK 到数据有效		123	夜行者
Tcl SCK 低电平	40		夜行者
Tch SCK High Time	40		夜行者
Fsck SCK 频率	0	4	兆赫
Tr, Tf SCK 上升和下降		100	夜行者
Tcc CSN 到 SCK 设置	2		夜行者
TCCH SCK 到 CSN 保留	2		夜行者
Tcwh CSN 非活动时间	50		夜行者
Tcdz CSN 到输出高 Z		116	夜行者

表 25.SPI 时序参数 (R = 10kΩ, C = 100pF)

Symbol	Parameters	Min.	Max	Units
Tdc	Data to sck Setup	2		ns
Tdh	sck to Data Hold	2		ns
Tcsd	csn to Data Valid		75	ns
Tcd	sck to Data Valid		85	ns
Tcl	sck Low Time	40		ns
Tch	sck High Time	40		ns
Fsck	sck Frequency	0	5	MHz
Tr,Tf	sck Rise and Fall		100	ns
Tcc	csn to sck Setup	2		ns
Tcch	sck to csn Hold	2		ns
Tcwh	csn Inactive time	50		ns
Tcdz	csn to Output High Z		75	ns

Table 26. SPI timing parameters ($R_{pull} = 50k\Omega$, $C_{load} = 50pF$)

Symbol	Parameters	Min.	Max	Units
Tdc	Data to sck Setup	2		ns
Tdh	sck to Data Hold	2		ns
Tcsd	csn to Data Valid		116	ns
Tcd	sck to Data Valid		121	ns
Tcl	sck Low Time	40		ns
Tch	sck High Time	40		ns
Fsck	sck Frequency	0	4	MHz
Tr,Tf	sck Rise and Fall		100	ns
Tcc	csn to sck Setup	2		ns
Tcch	sck to csn Hold	2		ns
Tcwh	csn Inactive time	50		ns
Tcdz	csn to Output High Z		116	ns

Table 27. SPI timing parameters ($R_{pull} = 50k\Omega$, $C_{load} = 100pF$)

8.4 Data FIFO

The data FIFOs store transmitted payloads (TX FIFO) or received payloads that are ready to be clocked out (RX FIFO). The FIFOs are accessible in both PTX mode and PRX mode.

The following FIFOs are present in nRF24L01+:

- TX three level, 32 byte FIFO
- RX three level, 32 byte FIFO

Both FIFOs have a controller and are accessible through the SPI by using dedicated SPI commands. A TX FIFO in PRX can store payloads for ACK packets to three different PTX devices. If the TX FIFO contains more than one payload to a pipe, payloads are handled using the first in - first out principle. The TX FIFO in a PRX is blocked if all pending payloads are addressed to pipes where the link to the PTX is lost. In this case, the MCU can flush the TX FIFO using the `FLUSH_TX` command.

The RX FIFO in PRX can contain payloads from up to three different PTX devices and a TX FIFO in PTX can have up to three payloads stored.

元件参数	最小	麦克斯	单位
Tdc 数据到 SCK 设置	2		夜行者
Tdh SCK 到数据保留	2		夜行者
tcsd csn 到 data valid		75	夜行者
tcd SCK 到数据有效		85	夜行者
Tcl SCK 低电平	40		夜行者
Tch SCK High Time	40		夜行者
Fsck SCK 频率	0	5	兆赫
Tr, Tf SCK 上升和下降		100	夜行者
Tcc CSN 到 SCK 设置	2		夜行者
TCCH SCK 到 CSN 保留	2		夜行者
Tcwh CSN 非活动时间	50		夜行者
Tcdz CSN 到输出高 Z		75	夜行者

表 26.SPI 时序参数 (R = 50kΩ, C = 50pF)

元件参数	最小	麦克斯	单位
Tdc 数据到 SCK 设置	2		夜行者
Tdh SCK 到数据保留	2		夜行者
tcsd csn 到 data valid		116	夜行者
tcd SCK 到数据有效		121	夜行者
Tcl SCK 低电平	40		夜行者
Tch SCK High Time	40		夜行者
Fsck SCK 频率	0	4	兆赫
Tr, Tf SCK 上升和下降		100	夜行者
Tcc CSN 到 SCK 设置	2		夜行者
TCCH SCK 到 CSN 保留	2		夜行者
Tcwh CSN 非活动时间	50		夜行者
Tcdz CSN 到输出高 Z		116	夜行者

表 27.SPI 时序参数 (R = 50kΩ, C = 100pF)

8.4 数据 FIFO

数据 FIFO 存储已发送的 payload (TX FIFO) 或已接收的已准备好 clocked out 的 payload (RX FIFO)。FIFO 可在 PTX 模式和 PRX 模式下访问。

nRF24L01+ 中存在以下 FIFO:

- TX 三电平, 32 字节 FIFO
- RX 三电平 32 字节 FIFO

两个 FIFO 都有一个控制器, 可以使用专用的 SPI 命令通过 SPI 访问。PRX 中的 TX FIFO 可以将 ACK 数据包的有效负载存储到三个不同的 PTX 设备。如果 TX FIFO 在管道中包含多个有效负载, 则使用先进先出原则处理有效负载。如果所有待处理有效负载都寻址到与 PTX 的链路丢失的管道, 则 PRX 中的 TX FIFO 将被阻止。在这种情况下, MCU 可以使用 FLUSH_TX 命令刷新 TX FIFO。

PRX 中的 RX FIFO 可以包含来自最多三个不同 PTX 设备的有效负载, 而 PTX 中的 TX FIFO 最多可以存储三个有效负载。

You can write to the TX FIFO using these three commands; `W_TX_PAYLOAD` and `W_TX_PAYLOAD_NO_ACK` in PTX mode and `W_ACK_PAYLOAD` in PRX mode. All three commands provide access to the `TX_PLD` register (see [Table 28. on page 63](#). for details of this register).

The RX FIFO can be read by the command `R_RX_PAYLOAD` in PTX and PRX mode. This command provides access to the `RX_PLD` register.

The payload in TX FIFO in a PTX is not removed if the `MAX_RT` IRQ is asserted.

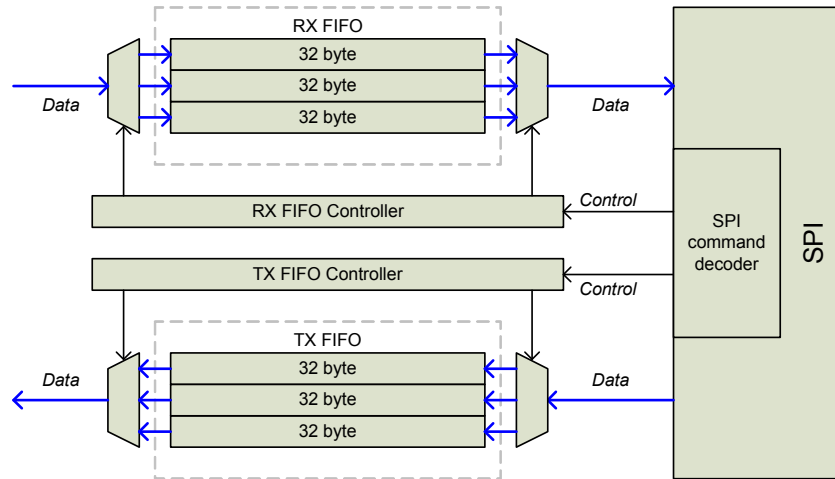


Figure 30. FIFO (RX and TX) block diagram

You can read if the TX and RX FIFO are full or empty in the `FIFO_STATUS` register.

8.5 Interrupt

The nRF24L01+ has an active low interrupt (`IRQ`) pin. The `IRQ` pin is activated when `TX_DS` IRQ, `RX_DR` IRQ or `MAX_RT` IRQ are set high by the state machine in the `STATUS` register. The `IRQ` pin resets when MCU writes '1' to the IRQ source bit in the `STATUS` register. The IRQ mask in the `CONFIG` register is used to select the IRQ sources that are allowed to assert the `IRQ` pin. By setting one of the MASK bits high, the corresponding IRQ source is disabled. By default all IRQ sources are enabled.

Note: The 3 bit pipe information in the `STATUS` register is updated during the `IRQ` pin high to low transition. The pipe information is unreliable if the `STATUS` register is read during an `IRQ` pin high to low transition.

您可以使用这三个命令写入 TX FIFO;在 PTX 模式下 W_TX_PAYLOAD 和 W_TX_PAYLOAD_NO_ACK, 在 PRX 模式下 W_ACK_PAYLOAD。所有三个命令都提供对 TX_PLD 寄存器的访问 (有关此寄存器的详细信息, 请参见第 63 页的表 20)

RX FIFO 可通过 PTX 和 PRX 模式下的命令 R_RX_PAYLOAD 读取。此命令提供对 RX_PLD 寄存器的访问。

如果断言 MAX_RT IRQ, 则不会移除 PTX 中 TX FIFO 中的有效负载。

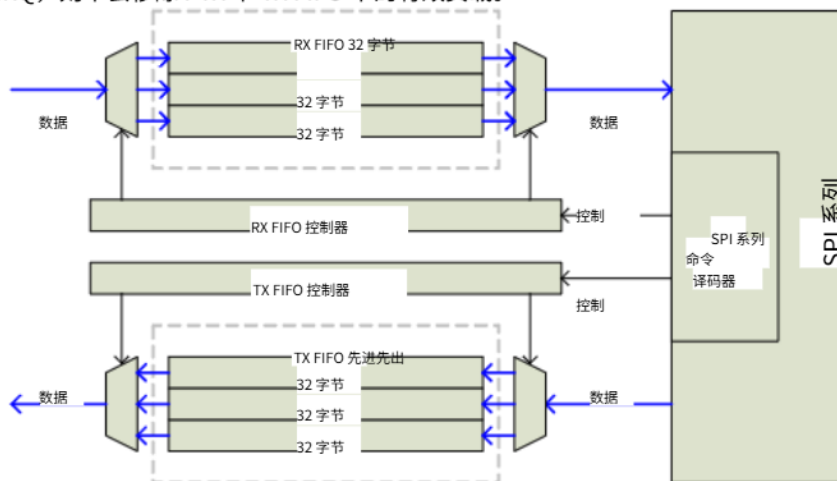


图 30.FIFO (RX 和 TX) 框图

您可以读取 FIFO_STATUS register 中的 TX 和 RX FIFO 是满还是空。

8.5 中断

nRF24L01+ 具有低电平有效中断 (IRQ) 引脚。当 IRQ、TX_DS RX_DR IRQ 或 MAX_RT IRQ 被 STATUS 寄存器中的状态机设置为高电平时, IRQ 引脚被激活。当 MCU 将 '1' 写入 STATUS 寄存器中的 IRQ 源位时, IRQ 引脚复位。CONFIG 寄存器中的 IRQ 掩码用于选择允许置位 IRQ pin 的 IRQ 源。通过将其中一个 MASK 位设置为高电平, 相应的 IRQ 源被禁用。默认情况下, 所有 IRQ 源都处于启用状态。

注意: STATUS 寄存器中的 3 位管道信息在 IRQ 引脚高到低期间更新过渡。如果在 IRQ 引脚高到低转换期间读取 STATUS 寄存器, 则管道信息不可靠。

9 Register Map

You can configure and control the radio by accessing the register map through the SPI.

9.1 Register map table Only standby state 1 and power-down state can change registers.

All undefined bits in the table below are redundant. They are read out as '0'.

Note: Addresses 18 to 1B are reserved for test purposes, altering them makes the chip malfunction.

Address (Hex)	Mnemonic	Bit	Reset Value	Type	Description
00	CONFIG				Configuration Register
	Reserved	7	0	R/W	Only '0' allowed
	MASK_RX_DR	6	0	R/W	Mask interrupt caused by RX_DR 1: Interrupt not reflected on the IRQ pin 0: Reflect RX_DR as active low interrupt on the IRQ pin
	MASK_TX_DS	5	0	R/W	Mask interrupt caused by TX_DS 1: Interrupt not reflected on the IRQ pin 0: Reflect TX_DS as active low interrupt on the IRQ pin
	MASK_MAX_RT	4	0	R/W	Mask interrupt caused by MAX_RT 1: Interrupt not reflected on the IRQ pin 0: Reflect MAX_RT as active low interrupt on the IRQ pin
	EN_CRC	3	1	R/W	Enable CRC. Forced high if one of the bits in the EN_AA is high
	CRCO	2	0	R/W	CRC encoding scheme '0' - 1 byte '1' - 2 bytes
	PWR_UP	1	0	R/W	1: POWER UP, 0: POWER DOWN
	PRIM_RX	0	0	R/W	RX/TX control 1: PRX, 0: PTX
01	EN_AA Enhanced ShockBurst™				Enable 'Auto Acknowledgment' Function Disable this functionality to be compatible with nRF2401, see page 75
	Reserved	7:6	00	R/W	Only '00' allowed
	ENAA_P5	5	1	R/W	Enable auto acknowledgement data pipe 5
	ENAA_P4	4	1	R/W	Enable auto acknowledgement data pipe 4
	ENAA_P3	3	1	R/W	Enable auto acknowledgement data pipe 3
	ENAA_P2	2	1	R/W	Enable auto acknowledgement data pipe 2
	ENAA_P1	1	1	R/W	Enable auto acknowledgement data pipe 1
	ENAA_P0	0	1	R/W	Enable auto acknowledgement data pipe 0
02	EN_RXADDR				Enabled RX Addresses
	Reserved	7:6	00	R/W	Only '00' allowed
	ERX_P5	5	0	R/W	Enable data pipe 5.
	ERX_P4	4	0	R/W	Enable data pipe 4.
	ERX_P3	3	0	R/W	Enable data pipe 3.
	ERX_P2	2	0	R/W	Enable data pipe 2.

9 注册地图

您可以通过 SPI 访问 register map 来配置和控制无线电。

9.1 注册地图表

下表中的所有未定义位都是冗余的。它们被读出为 '0'。

注意：地址 18 到 1B 保留用于测试目的，更改它们会导致芯片故障。

地址 (十六进制)	记忆	位	重置 值	类型	描述
00	配置				配置寄存器
	保留 7: 0 R/W 只允许 '0'				
	MASK_RX_DR	6	0	RX_DR	导致的 R/W 掩码中断 1: 中断未反映在 IRQ 引脚上 0: RX_DR 反映为 IRQ 引脚上的低电平有效中断
	MASK_TX_DS	5	0	TX_DS	导致的 R/W 掩码中断 1: 中断未反映在 IRQ 引脚上 0: TX_DS 反映为 IRQ 引脚上的低电平有效中断
	MASK_MAX_RT	4	0	MAX_RT	导致的 R/W 掩码中断 1: 中断未反映在 IRQ 引脚上 0: MAX_RT IRQ 引脚上反映为低电平有效中断
	EN_CRC	3	1	R/W	使能 CRC。如果 EN_AA 很高
	CRCO	2	0	R/W	CRC 编码方案 '0' - 1 字节 '1' - 2 字节
	PWR_UP 1 0 R/W	1	1		1: 开机, 0: 关机
	PRIM_RX	0	0	R/W	(英语) RX 控制 1: PRX, 0: PTX
01	EN_AA				启用“自动确认”功能 禁用此功能以与 nRF2401 兼容, 请参见第 75 页
	增强 冲击爆发™				
	预留 7: 6 00 R/W 只允许 '00'				
	ENAA_P5 5 1 R/W	5	1	R/W	启用自动确认数据管道 5
	ENAA_P4 4 1 R/W	4	1	R/W	启用自动确认数据管道 4
	ENAA_P3 3 1 R/W	3	1	R/W	启用自动确认数据管道 3
	ENAA_P2 2 1 R/W	2	1	R/W	启用自动确认数据管道 2
	ENAA_P0	0	1	R/W	(英语) 启用自动确认数据管道 0
02	EN_RXADDR				已启用的 RX 地址
	预留 7: 6 00 R/W 只允许 '00'				
	ERX_P5 5 0 R/W	5	0	R/W	启用数据管道 5
	ERX_P4 4 0 R/W	4	0	R/W	启用数据管道 4
	ERX_P2	2	0	R/W	(英语) 启用数据管道 2

Address (Hex)	Mnemonic	Bit	Reset Value	Type	Description
	ERX_P1	1	1	R/W	Enable data pipe 1.
	ERX_P0	0	1	R/W	Enable data pipe 0.
03	SETUP_AW				Setup of Address Widths (common for all data pipes)
	Reserved	7:2	000000	R/W	Only '000000' allowed
	AW	1:0	11	R/W	RX/TX Address field width '00' - Illegal '01' - 3 bytes '10' - 4 bytes '11' - 5 bytes LSByte is used if address width is below 5 bytes
04	SETUP_RETR				Setup of Automatic Retransmission
	ARD ^a	7:4	0000	R/W	Auto Retransmit Delay '0000' – Wait 250µS '0001' – Wait 500µS '0010' – Wait 750µS '1111' – Wait 4000µS (Delay defined from end of transmission to start of next transmission) ^b
	ARC	3:0	0011	R/W	Auto Retransmit Count '0000' – Re-Transmit disabled '0001' – Up to 1 Re-Transmit on fail of AA '1111' – Up to 15 Re-Transmit on fail of AA
05	RF_CH				RF Channel
	Reserved	7	0	R/W	Only '0' allowed
	RF_CH	6:0	0000010	R/W	Sets the frequency channel nRF24L01+ operates on
06	RF_SETUP				RF Setup Register
	CONT_WAVE	7	0	R/W	Enables continuous carrier transmit when high.
	Reserved	6	0	R/W	Only '0' allowed
	RF_DR_LOW	5	0	R/W	Set RF Data Rate to 250kbps. See RF_DR_HIGH for encoding.
	PLL_LOCK	4	0	R/W	Force PLL lock signal. Only used in test
	RF_DR_HIGH	3	1	R/W	Select between the high speed data rates. This bit is don't care if RF_DR_LOW is set. Encoding: [RF_DR_LOW, RF_DR_HIGH]: '00' – 1Mbps '01' – 2Mbps '10' – 250kbps '11' – Reserved

only power-down status can changed

地址 (十六进制)	记忆	位	重置 值	类型	描述
	ERX_P1 1 1	R/W	启用数据管道 1.		
	ERX_P0	0	1	R/W	禁用数据管道 0。
03	SETUP_AW				地址宽度设置（所有数据管道通用）
	预留 7: 2 000000	R/W	只允许 '000000'		
	AW	1:0	11	R/W	RX/TX 地址字段宽度 '00' - 非法 '01' - 3 字节 '10' - 4 字节 '11' - 如果地址宽度低于 5 字节，则使用 5 字节 LSByte
04	SETUP_RETR ARD	7:4	0000	R/W	自动重新传输的设置 自动重传延迟 '0000' - 等待 250μS '0001' - 等待 500μS '0010' - 等待 750μS '1111' - 等待 4000μS（定义为从传输结束到下一次传输开始的延迟）
	弧	3:0	0011	R/W	自动重传计数 '0000' - 禁用重新传输 '0001' - AA失败时最多 1 次重新传输 "1111" - AA 失败时最多 15 次重新传输
05	RF_CH				RF 通道
	保留 7 0	R/W	仅允许 '0'		
	RF_CH	6:0	0000010	R/W	设置 nRF24L01+ 的频道 上
06	RF_SETUP				RF 设置寄存器
	CONT_WAVE 7 0	R/W	在高电平时启用连续载波传输。		
	保留 6 0	R/W	仅允许 '0'		
	RF_DR_LOW	5	0	R/W	将 RF 数据速率设置为 250kbps。请参阅 RF_DR_HIGH 进行编码。
	PLL_LOCK 4 0	R/W	强制 PLL 锁定信号。仅用于测试		
	RF_DR_HIGH	3	1	R/W	在高速数据速率之间进行选择。这个位是 don't care if RF_DR_LOW set。 编码: [RF_DR_LOW, RF_DR_HIGH]: '00' - 1Mbps '01' - 2Mbps '10' - 250kbps '11' - 预留

Address (Hex)	Mnemonic	Bit	Reset Value	Type	Description
	RF_PWR	2:1	11	R/W	Set RF output power in TX mode '00' – -18dBm '01' – -12dBm '10' – -6dBm '11' – 0dBm
	Obsolete	0			Don't care
07	STATUS				Status Register (In parallel to the SPI command word applied on the MOSI pin, the STATUS register is shifted serially out on the MISO pin)
	Reserved	7	0	R/W	Only '0' allowed
	RX_DR	6	0	R/W	Data Ready RX FIFO interrupt. Asserted when new data arrives RX FIFO ^c . Write 1 to clear bit.
	TX_DS	5	0	R/W	Data Sent TX FIFO interrupt. Asserted when packet transmitted on TX. If AUTO_ACK is activated, this bit is set high only when ACK is received. Write 1 to clear bit.
	MAX_RT	4	0	R/W	Maximum number of TX retransmits interrupt. Write 1 to clear bit. If MAX_RT is asserted it must be cleared to enable further communication.
	RX_P_NO	3:1	111	R	Data pipe number for the payload available for reading from RX_FIFO 000-101: Data Pipe Number 110: Not Used 111: RX FIFO Empty
	TX_FULL	0	0	R	TX FIFO full flag. 1: TX FIFO full. 0: Available locations in TX FIFO.
08	OBSERVE_TX				Transmit observe register
	PLOS_CNT	7:4	0	R	Count lost packets. The counter is overflow protected to 15, and discontinues at max until reset. The counter is reset by writing to RF_CH. See page 75 .
	ARC_CNT	3:0	0	R	Count retransmitted packets. The counter is reset when transmission of a new packet starts. See page 75 .
09	RPD				
	Reserved	7:1	000000	R	
	RPD	0	0	R	Received Power Detector. This register is called CD (Carrier Detect) in the nRF24L01. The name is different in nRF24L01+ due to the different input power level threshold for this bit. See section 6.4 on page 25 .
0A	RX_ADDR_P0	39:0	0xE7E7E7E7E7	R/W	Receive address data pipe 0. 5 Bytes maximum length. (LSByte is written first. Write the number of bytes defined by SETUP_AW)

地址 (十六进制)	记忆	位	重置 值	类型	描述
	RF_PWR	2:1	11	R/W	在 TX 模式下设置 RF 输出功率 '00' - -18dBm '01' - -12dBm '10' - -6 分贝 '11' - -0 分贝
	过时	0			不在乎
07	地位				状态寄存器（与 MOSI 引脚上的 SPI 命令字并行，STATUS 寄存器在 MISO 引脚上串行移出）
	保留 7 0 R/W 仅允许 '0'				
	RX_DR	6	0	R/W	Data Ready RX FIFO 中断。当新数据到达 RX FIFO 时断言。 写入 1 以清除位。
	TX_DS	5	0	R/W	数据发送 TX FIFO 中断。Asserted when 数据包在 TX 上传输。如果 AUTO_ACK 被激活，则仅当收到 ACK 时，此位才设置为高电平。 写入 1 以清除位。
	MAX_RT	4	0	R/W	最大 TX 重传中断数 写入 1 以清除位。如果 MAX_RT 被置位，则必须清除它才能启用进一步的通信。 写入 1 以清除位。
	RX_P_NO	3:1	111	R	有效负载可用于的数据管道编号 从 RX_FIFO 000-101 读取：数据管道编号 110：未使用 111：RX FIFO 为空
	TX_FULL	0	0	R	TX FIFO 全标志。 1：TX FIFO 已满。 0：TX FIFO 中的可用位置。
08	OBSERVE_TX PLOS_CNT 7: 4 0 R				发送观察寄存器 计算丢失的数据包。计数器是 overflow protected 设置为 15，并在 Max 时停止，直到重置。通过写入 RF_CH 来重置计数器。见第 75 页。
	ARC_CNT	3:0	0	R	对重新传输的数据包进行计数。计数器已重置当开始传输新数据包时。见第 75 页。
09	RPD 保留 7: 1 000000 R				
	RPD	0	0	r	接收功率检测器。这个寄存器称为 CD（载波检测）在 nRF24L01 中。由于该位的输入功率电平阈值不同，因此 nRF24L01+ 中的名称不同。请参阅第 25 页的第 6.4 节。
0 安培	RX_ADDR_P0	39:0	0xE7E7E7E7	R/W	接收地址数据管道 0。最大 5 字节长度。LSByte 先写入。写入 SETUP_AW 定义的字数

Address (Hex)	Mnemonic	Bit	Reset Value	Type	Description
0B	RX_ADDR_P1	39:0	0xC2C2C2C2	R/W	Receive address data pipe 1. 5 Bytes maximum length. (LSByte is written first. Write the number of bytes defined by SETUP_AW)
0C	RX_ADDR_P2	7:0	0xC3	R/W	Receive address data pipe 2. Only LSB. MSBytes are equal to RX_ADDR_P1[39:8]
0D	RX_ADDR_P3	7:0	0xC4	R/W	Receive address data pipe 3. Only LSB. MSBytes are equal to RX_ADDR_P1[39:8]
0E	RX_ADDR_P4	7:0	0xC5	R/W	Receive address data pipe 4. Only LSB. MSBytes are equal to RX_ADDR_P1[39:8]
0F	RX_ADDR_P5	7:0	0xC6	R/W	Receive address data pipe 5. Only LSB. MSBytes are equal to RX_ADDR_P1[39:8]
10	TX_ADDR	39:0	0xE7E7E7E7	R/W	Transmit address. Used for a PTX device only. (LSByte is written first) Set RX_ADDR_P0 equal to this address to handle automatic acknowledge if this is a PTX device with Enhanced ShockBurst™ enabled. See page 75 .
11	RX_PW_P0				
	Reserved	7:6	00	R/W	Only '00' allowed
	RX_PW_P0	5:0	0	R/W	Number of bytes in RX payload in data pipe 0 (1 to 32 bytes). 0 Pipe not used 1 = 1 byte ... 32 = 32 bytes
12	RX_PW_P1				
	Reserved	7:6	00	R/W	Only '00' allowed
	RX_PW_P1	5:0	0	R/W	Number of bytes in RX payload in data pipe 1 (1 to 32 bytes). 0 Pipe not used 1 = 1 byte ... 32 = 32 bytes
13	RX_PW_P2				
	Reserved	7:6	00	R/W	Only '00' allowed
	RX_PW_P2	5:0	0	R/W	Number of bytes in RX payload in data pipe 2 (1 to 32 bytes). 0 Pipe not used 1 = 1 byte ... 32 = 32 bytes
14	RX_PW_P3				
	Reserved	7:6	00	R/W	Only '00' allowed

地址 (十六进制)	记忆	位	重置 价值	类型	描述
0B	RX_ADDR_P1	39:0	0xC2C2C2C2	R/W	接收地址数据管道 1.最大 5 字节长度。LSByte 先写入。写入 SETUP_AW) 定义的字节数
0C	RX_ADDR_P2	7:0	0xC3	R/W	接收地址数据管道 2.只有 LSB。MSBytes 等于 RX_ADDR_P1[39: 8]
0D	RX_ADDR_P3	7:0	0xC4	R/W	接收地址数据管道 3.只有 LSB。MSBytes 等于 RX_ADDR_P1[39: 8]
0E	RX_ADDR_P4	7:0	0xC5	R/W	接收地址数据管道 4.只有 LSB。MSBytes 等于 RX_ADDR_P1[39: 8]
0F	RX_ADDR_P5	7:0	0xC6	R/W	接收地址数据管道 5.只有 LSB。MSBytes 等于 RX_ADDR_P1[39: 8]
10	TX_ADDR	39:0	0xE7E7E7E7	R/W	传输地址。仅用于 PTX 设备。 (LSByte 先写) 将 RX_ADDR_P0 设置为等于此地址, 以处理自动确认 (如果这是启用了增强型 ShockBurst™ 的 PTX 设备)。见第 75 页。
11	RX_PW_P0 预留 7: 6 00 R/W 只允许 '00'	5:0	0	R/W	数据管道 0 中 RX 负载的字节数 (1 到 32 字节)。 0 未使用管道 1 = 1 字节 ... 32 = 32 字节
12	RX_PW_P1 预留 7: 6 00 R/W 只允许 '00'	5:0	0	R/W	数据管道 1 中 RX 负载的字节数 (1 到 32 字节)。 0 未使用管道 1 = 1 字节 ... 32 = 32 字节
13	RX_PW_P2 预留 7: 6 00 R/W 只允许 '00'	5:0	0	R/W	数据管道 2 中 RX 负载中的字节数 (1 到 32 字节)。 0 未使用管道 1 = 1 字节 ... 32 = 32 字节
14	RX_PW_P3 保留	7:6	00	R/W	(只读) 允许 '00'

Address (Hex)	Mnemonic	Bit	Reset Value	Type	Description
	RX_PW_P3	5:0	0	R/W	Number of bytes in RX payload in data pipe 3 (1 to 32 bytes). 0 Pipe not used 1 = 1 byte ... 32 = 32 bytes
15	RX_PW_P4				
	Reserved	7:6	00	R/W	Only '00' allowed
	RX_PW_P4	5:0	0	R/W	Number of bytes in RX payload in data pipe 4 (1 to 32 bytes). 0 Pipe not used 1 = 1 byte ... 32 = 32 bytes
16	RX_PW_P5				
	Reserved	7:6	00	R/W	Only '00' allowed
	RX_PW_P5	5:0	0	R/W	Number of bytes in RX payload in data pipe 5 (1 to 32 bytes). 0 Pipe not used 1 = 1 byte ... 32 = 32 bytes
17	FIFO_STATUS				FIFO Status Register
	Reserved	7	0	R/W	Only '0' allowed
	TX_REUSE	6	0	R	Used for a PTX device Pulse the <code>rfce</code> high for at least 10µs to Reuse last transmitted payload. TX payload reuse is active until <code>W_TX_PAYLOAD</code> or <code>FLUSH_TX</code> is executed. <code>TX_REUSE</code> is set by the SPI command <code>REUSE_TX_PL</code> , and is reset by the SPI commands <code>W_TX_PAYLOAD</code> or <code>FLUSH_TX</code>
	TX_FULL	5	0	R	TX FIFO full flag. 1: TX FIFO full. 0: Available locations in TX FIFO.
	TX_EMPTY	4	1	R	TX FIFO empty flag. 1: TX FIFO empty. 0: Data in TX FIFO.
	Reserved	3:2	00	R/W	Only '00' allowed
	RX_FULL	1	0	R	RX FIFO full flag. 1: RX FIFO full. 0: Available locations in RX FIFO.
	RX_EMPTY	0	1	R	RX FIFO empty flag. 1: RX FIFO empty. 0: Data in RX FIFO.

地址 (十六进制)	记忆	位	重置 值	类型	描述
	RX_PW_P3	5:0	0	R/W	数据管道 3 中 RX 负载的字节数 (1 到 32 字节)。 0 未使用管道 1 = 1 字节 ... 32 = 32 字节
15	RX_PW_P4				
	预留 7: 6 00 R/W 只允许 '00'				
	RX_PW_P4	5:0	0	R/W	数据管道 4 中 RX 负载的字节数 (1 到 32 字节)。 0 未使用管道 1 = 1 字节 ... 32 = 32 字节
16	RX_PW_P5				
	预留 7: 6 00 R/W 只允许 '00'				
	RX_PW_P5	5:0	0	R/W	数据管道 5 中 RX 负载的字节数 (1 到 32 字节)。 0 未使用管道 1 = 1 字节 ... 32 = 32 字节
17	FIFO_STATUS				FIFO 状态寄存器
	保留 7: 0 R/W 只允许 '0'				
	TX_REUSE	6	0	R	用于 PTX 设备 将 rfce 脉冲为高电平至少 10μs 以重用上次传输的有效载荷。TX 负载重用处于活动状态，直到执行 W_TX_PAYLOAD 或 FLUSH TX。TX_REUSE 由 SPI 命令 REUSE_TX_PL 设置，并通过 SPI 命令 W_TX_PAYLOAD 或 FLUSH TX 复位
	TX_FULL 5 0 R	TX FIFO	全标志。1:	TX FIFO	已满。0: TX FIFO 中的可用位置。
	TX_EMPTY	4	1	R	TX FIFO 空标志。 1: TX FIFO 为空。 0: TX FIFO 中的数据。
	预留 3: 2 00 R/W 只允许 '00'				
	RX_FULL	1	0	R	RX FIFO 全标志。 1: RX FIFO 已满。 0: RX FIFO 中的可用位置。
	RX_EMPTY	0	1	R	RX FIFO 空标志。 1: RX FIFO 为空。 0: RX FIFO 中的数据。

Address (Hex)	Mnemonic	Bit	Reset Value	Type	Description
N/A	ACK_PLD	255:0	X	W	Written by separate SPI command ACK packet payload to data pipe number PPP given in SPI command. Used in RX mode only. Maximum three ACK packet payloads can be pending. Payloads with same PPP are handled first in first out.
N/A	TX_PLD	255:0	X	W	Written by separate SPI command TX data payload register 1 - 32 bytes. This register is implemented as a FIFO with three levels. Used in TX mode only.
N/A	RX_PLD	255:0	X	R	Read by separate SPI command. RX data payload register. 1 - 32 bytes. This register is implemented as a FIFO with three levels. All RX channels share the same FIFO.
1C	DYNPD				Enable dynamic payload length
	Reserved	7:6	0	R/W	Only '00' allowed
	DPL_P5	5	0	R/W	Enable dynamic payload length data pipe 5. (Requires EN_DPL and ENAA_P5)
	DPL_P4	4	0	R/W	Enable dynamic payload length data pipe 4. (Requires EN_DPL and ENAA_P4)
	DPL_P3	3	0	R/W	Enable dynamic payload length data pipe 3. (Requires EN_DPL and ENAA_P3)

地址 (十六进制)	记忆	位	重置 值	类型	描述
不适用	ACK_PLD	255:0	X	W	由单独的 SPI 命令写入 ACK 数据包有效载荷到 SPI 命令中给出的数据管道编号 PPP。仅在 RX 模式下使用。最多可以有 3 个 ACK 数据包负载处于待处理状态。具有相同 PPP 的负载将按照先进先出的顺序进行处理。
不适用	TX_PLD	255:0	X	W	由单独的 SPI 命令 TX 数据有效载荷寄存器 1 - 32 字节写入。 这个 register 被实现为具有三个级别的 FIFO。 仅在 TX 模式下使用。
不适用	RX_PLD	255:0	X	R	通过单独的 SPI 命令读取。 RX 数据有效载荷寄存器。1 - 32 字节。 这个 register 被实现为具有三个级别的 FIFO。 所有 RX 通道共享相同的 FIFO。
1C	DYNPD	7:0	0	R/W	启用动态负载长度
	DPL_P5	5	0	R/W	启用动态有效载荷长度数据管道 5。 (需要 EN_DPL 和 ENAA_P5)
	DPL_P4	4	0	R/W	启用动态有效载荷长度数据管道 4。 (需要 EN_DPL 和 ENAA_P4)
	DPL_P3	3	0	R/W	启用动态有效载荷长度数据管道 3。 (需要 EN_DPL 和 ENAA_P3)

Address (Hex)	Mnemonic	Bit	Reset Value	Type	Description
	DPL_P2	2	0	R/W	Enable dynamic payload length data pipe 2. (Requires EN_DPL and ENAA_P2)
	DPL_P1	1	0	R/W	Enable dynamic payload length data pipe 1. (Requires EN_DPL and ENAA_P1)
	DPL_P0	0	0	R/W	Enable dynamic payload length data pipe 0. (Requires EN_DPL and ENAA_P0)
1D	FEATURE			R/W	Feature Register
	Reserved	7:3	0	R/W	Only '00000' allowed
	EN_DPL	2	0	R/W	Enables Dynamic Payload Length
	EN_ACK_PAY ^d	1	0	R/W	Enables Payload with ACK
	EN_DYN_ACK	0	0	R/W	Enables the W_TX_PAYLOAD_NOACK command

- Please take care when setting this parameter. If the ACK payload is more than 15 byte in 2Mbps mode the ARD must be 500µS or more, if the ACK payload is more than 5byte in 1Mbps mode the ARD must be 500µS or more. In 250kbps mode (even when the payload is not in ACK) the ARD must be 500µS or more. Please see section [7.4.2 on page 33](#) for more information.
- This is the time the PTX is waiting for an ACK packet before a retransmit is made. The PTX is in RX mode for 250µS (500µS in 250kbps mode) to wait for address match. If the address match is detected, it stays in RX mode to the end of the packet, unless ARD elapses. Then it goes to standby-II mode for the rest of the specified ARD. After the ARD it goes to TX mode and then retransmits the packet.
- The RX_DR IRQ is asserted by a new packet arrival event. The procedure for handling this interrupt should be: 1) read payload through SPI, 2) clear RX_DR IRQ, 3) read FIFO_STATUS to check if there are more payloads available in RX FIFO, 4) if there are more data in RX FIFO, repeat from step 1).
- If ACK packet payload is activated, ACK packets have dynamic payload lengths and the Dynamic Payload Length feature should be enabled for pipe 0 on the PTX and PRX. This is to ensure that they receive the ACK packets with payloads. If the ACK payload is more than 15 byte in 2Mbps mode the ARD must be 500µS or more, and if the ACK payload is more than 5 byte in 1Mbps mode the ARD must be 500µS or more. In 250kbps mode (even when the payload is not in ACK) the ARD must be 500µS or more.

Table 28. Register map of nRF24L01+

地址 (十六进制)	记忆	位	重置 值	类型	描述
	DPL_P2	2	0	R/W	启用动态有效载荷长度数据管道 2. (需要 EN_DPL 和 ENAA_P2)
	DPL_P1	1	0	R/W	启用动态有效载荷长度数据管道 1. (需要 EN_DPL 和 ENAA_P1)
	DPL_P0	0	0	R/W	启用动态有效载荷长度数据管道 0. (需要 EN_DPL 和 ENAA_P0)
1D	特征			R/W	(禁用)寄存器
	保留 7: 3 0			R/W	仅允许 '00000'
	EN_DPL 2 0			R/W	启用动态有效载荷长度
	EN_ACK_PAY 1 0			R/W	启用 ACK 的 Payload
	EN_DYN_ACK	0	0	R/W	(禁用)W_TX_PAYLOAD_NOACK 命令

一个。设置此参数时请小心。如果 ACK 有效载荷在 2Mbps 模式下超过 15 字节，则 ARD 必须为 500μS 或更高，如果 ACK 有效载荷在 1Mbps 模式下超过 5 字节，则 ARD 必须为 500μS 或更高。在 250kbps 模式下（即使有效载荷不在 ACK 中），ARD 必须为 500μS 或更高。

有关更多信息，请参阅第 33 页的第 7.4.2 节。

b.这是 PTX 在进行重新传输之前等待 ACK 数据包的时间。PTX 处于 RX 模式 250μS（在 250kbps 模式下为 500μS）以等待地址匹配。如果检测到地址匹配，则除非 ARD 过期，否则它将保持 RX 模式直到数据包末尾。然后，对于指定 ARD 的其余部分，它将进入 standby-II 模式。在 ARD 之后，它进入 TX 模式，然后重新传输数据包。c. RX_DR IRQ 由新的数据包到达事件断言。处理此中断的过程应该是：1) 通过 SPI 读取有效载荷，2) 清除 RX_DR IRQ，3) 读取 FIFO_STATUS 以检查 RX FIFO 中是否有更多有效载荷可用，4) 如果 RX FIFO 中有更多数据，则从步骤 1 开始重复)。d. 如果激活了 ACK 数据包负载，则 ACK 数据包具有动态负载长度，并且应为 PTX 和 PRX 上的管道 0 启用动态负载长度功能。这是为了确保他们收到带有有效负载的 ACK 数据包。如果 ACK 有效载荷在 2Mbps 模式下超过 15 字节，则 ARD 必须为 500μS 或更高，如果 ACK 有效载荷在 1Mbps 模式下超过 5 字节，则 ARD 必须为 500μS 或更高。在 250kbps 模式下（即使有效载荷不在 ACK 中），ARD 必须为 500μS 或更高。

表 28.nRF24L01+ 的寄存器映射

10 Peripheral RF Information

This chapter describes peripheral circuitry and PCB layout requirements that are important for achieving optimum RF performance from the nRF24L01+.

10.1 Antenna output

The **ANT1** and **ANT2** output pins provide a balanced RF output to the antenna. The pins must have a DC path to **VDD_PA**, either through a RF choke or through the center point in a balanced dipole antenna. A load of $15\Omega + j88\Omega$ is recommended for maximum output power (0dBm). Lower load impedance (for instance, 50Ω) can be obtained by fitting a simple matching network between the load and **ANT1** and **ANT2**. A recommended matching network for 50Ω load impedance is described in [chapter 11 on page 66](#).

10.2 Crystal oscillator

A crystal used with the nRF24L01+ must fulfil the specifications in [Table 11. on page 19](#).

To achieve a crystal oscillator solution with low power consumption and fast start up time use a crystal with a low load capacitance specification. A lower C_0 also gives lower current consumption and faster start up time, but can increase the cost of the crystal. Typically $C_0 = 1.5\text{pF}$ at a crystal specified for $C_{0\text{max}} = 7.0\text{pF}$.

The crystal load capacitance, C_L , is given by:

$$C_L = \frac{C_1' \cdot C_2'}{C_1' + C_2'}, \text{ where } C_1' = C_1 + C_{\text{PCB1}} + C_{\text{I1}} \text{ and } C_2' = C_2 + C_{\text{PCB2}} + C_{\text{I2}}$$

C_1 and C_2 are SMD capacitors, see the application schematics in [Figure 32. on page 66](#). C_{PCB1} and C_{PCB2} are the layout parasitic on the circuit board. C_{I1} and C_{I2} are the internal capacitance load of the **XC1** and **XC2** pins respectively; the value is typically 1pF for both these pins.

10.3 nRF24L01+ crystal sharing with an MCU

Follow the rules described in sections [10.3.1](#) and [10.3.2](#) when using an MCU to drive the crystal reference input **XC1** of the nRF24L01+ transceiver.

10.3.1 Crystal parameters

The MCU sets the requirement of load capacitance C_L when it is driving the nRF24L01+ clock input. A frequency accuracy of $\pm 60\text{ppm}$ is required to get a functional radio link. The nRF24L01+ loads the crystal by 1pF in addition to the PCB routing.

10.3.2 Input crystal amplitude and current consumption

The input signal should not have amplitudes exceeding any rail voltage. Exceeding rail voltage excites the ESD structure and consequently, the radio performance degrades below specification. You must use an external DC block if you are testing the nRF24L01+ with a reference source that has no DC offset (which is usual with a RF source).

10 外设 RF 信息

本章介绍了外围电路和 PCB 布局要求，这些要求对于从 nRF24L01+ 实现最佳射频性能非常重要。

10.1 天线输出

ANT1 和 ANT2 输出引脚为天线提供平衡的射频输出。引脚必须有一条直流路径 VDD_PA，要么通过 RF 扼流圈，要么通过平衡偶极子天线中的中心点。建议负载为 $15\Omega + j88\Omega$ ，以获得最大输出功率（0dBm）。通过在负载与 ANT1 和 ANT2 之间安装一个简单的匹配网络，可以获得较低的负载阻抗（例如，50Ω）。50Ω 负载阻抗的推荐匹配网络在第 66 页的第 11 章中介绍。

10.2 晶体振荡器

与 nRF24L01+ 一起使用的晶体必须满足表 11 中的规格。在第 19 页。

为了实现低功耗和快速启动时间的晶体振荡器解决方案，请使用具有低负载电容规格的晶体。较低的 C 还可以提供更低的电流消耗和更快的启动时间，但会增加晶体的成本。通常 $C=1.5\text{pF}$ 在指定 $C=7.0\text{pF}$ 的晶体上。

晶体负载电容 C 由下式给出：

$$C_L = \frac{C_1 + C_2}{C_1 + C_2 + C'} \quad , \text{其中 } C' = C + C_1 + C_2 \text{ 和 } C' = C + C_1 + C_2$$

Cand Care SMD 电容器，参见图 32 中的应用原理图。在第 66 页。C 和 C 是 PCB 上的寄生布局。C 和 C 分别是 XC1 和 XC2 引脚的内部电容负载;这两个 pins 的值通常为 1pF。

10.3 nRF24L01+ 晶体与 MCU 共享

使用 MCU 驱动 nRF24L01+ 收发器的晶体参考输入 XC1 时，请遵循第 10.3.1 节和第 10.3.2 节中描述的规则。

10.3.1 晶体参数

MCU 在驱动 nRF24L01+ 时钟输入时设置负载电容 C 的要求。需要 $\pm 60\text{ppm}$ 的频率精度才能获得功能性无线电路。除了 PCB 布线外，nRF24L01+ 还通过 1pF 加载晶体。

10.3.2 输入晶体振幅和电流消耗

输入信号的幅度不应超过任何电源轨电压。超过轨电压会激发 ESD 结构，因此，无线电性能会下降到低于规格。如果您使用没有直流偏移的参考源（射频源通常如此）测试 nRF24L01+，则必须使用外部直流块。

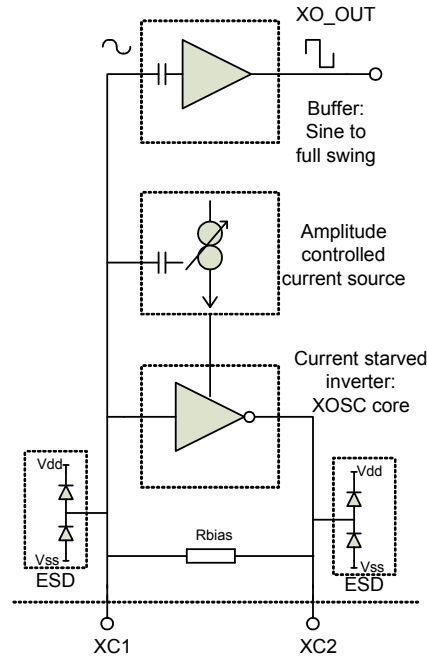


Figure 31. Principle of crystal oscillator

The nRF24L01+ crystal oscillator is amplitude regulated. It is recommended to use an input signal larger than 0.4V-peak to achieve low current consumption and good signal-to-noise ratio when using an external clock. **xc2** is not used and can be left as an open pin when clocked externally.

10.4 PCB layout and decoupling guidelines

A well designed PCB is necessary to achieve good RF performance. A poor layout can lead to loss of performance or functionality. You can download a fully qualified RF layout for the nRF24L01+ and its surrounding components, including matching networks, from www.nordicsemi.no.

A PCB with a minimum of two layers including a ground plane is recommended for optimum performance. The nRF24L01+ DC supply voltage should be decoupled as close as possible to the **VDD** pins with high performance RF capacitors, see [Table 29. on page 67](#). Mounting a large surface mount capacitor (for example, 4.7μF ceramic) in parallel with the smaller value capacitors is recommended. The nRF24L01+ supply voltage should be filtered and routed separately from the supply voltages of any digital circuitry.

Avoid long power supply lines on the PCB. All device grounds, **VDD** connections and **VDD** bypass capacitors must be connected as close as possible to the nRF24L01+ IC. The **VSS** pins should be connected directly to the ground plane for a PCB with a top-side RF ground plane. We recommend having via holes as close as possible to the **VSS** pads for a PCB with a bottom ground plane. A minimum of one via hole should be used for each **VSS** pin.

Full swing digital data or control signals should not be routed close to the crystal or the power supply lines. The exposed die attach pad is a ground pad connected to the IC substrate die ground and is intentionally not used in our layouts. We recommend to keep it unconnected.

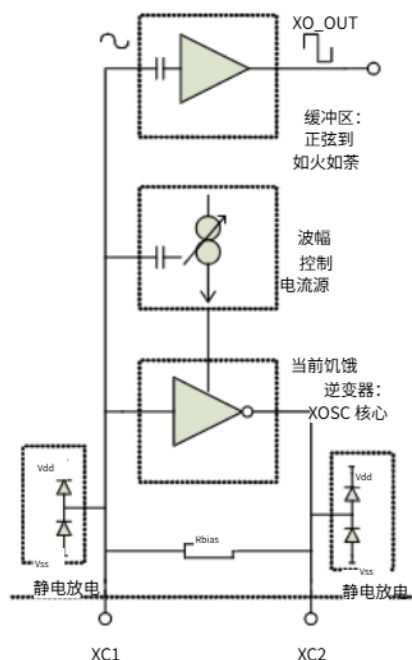


图 31.晶振原理

nRF24L01+ 晶体振荡器是幅度调节的。使用外部时钟时，建议使用大于 0.4V-peak 的输入信号，以实现低电流消耗和良好的信噪比。XC2 未使用，当外部计时时可以保留为开路引脚。

10.4 PCB 布局和去耦指南

要实现良好的射频性能，必须采用设计良好的 PCB。糟糕的布局可能会导致性能或功能损失。您可以从 www.nordicsemi.no 下载 nRF24L01+ 及其周围组件（包括匹配网络）的完全合格的射频布局。

建议使用至少具有两层（包括一个接地层）的 PCB 以获得最佳性能。nRF24L01+ 直流电源电压应尽可能靠近使用高性能射频电容器的 VDD 引脚去耦，见表 29。第 67 页。建议将大型表面贴装电容器（例如，4.7μF 陶瓷电容器）与较小值的电容器并联安装。nRF24L01+ 电源电压应与任何数字电路的电源电压分开过滤和布线。

避免在 PCB 上铺设长电源线。所有设备接地、VDD 连接和 VDD 旁路电容器必须尽可能靠近 nRF24L01+ IC 连接。VSS 引脚应直接连接到具有顶部 RF 接地层的 PCB 的接地层。对于具有底部接地层的 PCB，我们建议将通孔尽可能靠近 VSS 焊盘。每个 VSS 引脚至少应使用一个通孔。

全摆幅数字数据或控制信号不应靠近晶体或电源线布线。裸露的芯片连接焊盘是连接到 IC 载板芯片接地的接地焊盘，在我们的布局中故意不使用。我们建议保持未连接状态。

11 Application example

nRF24L01+ with single ended matching network crystal, bias resistor, and decoupling capacitors.

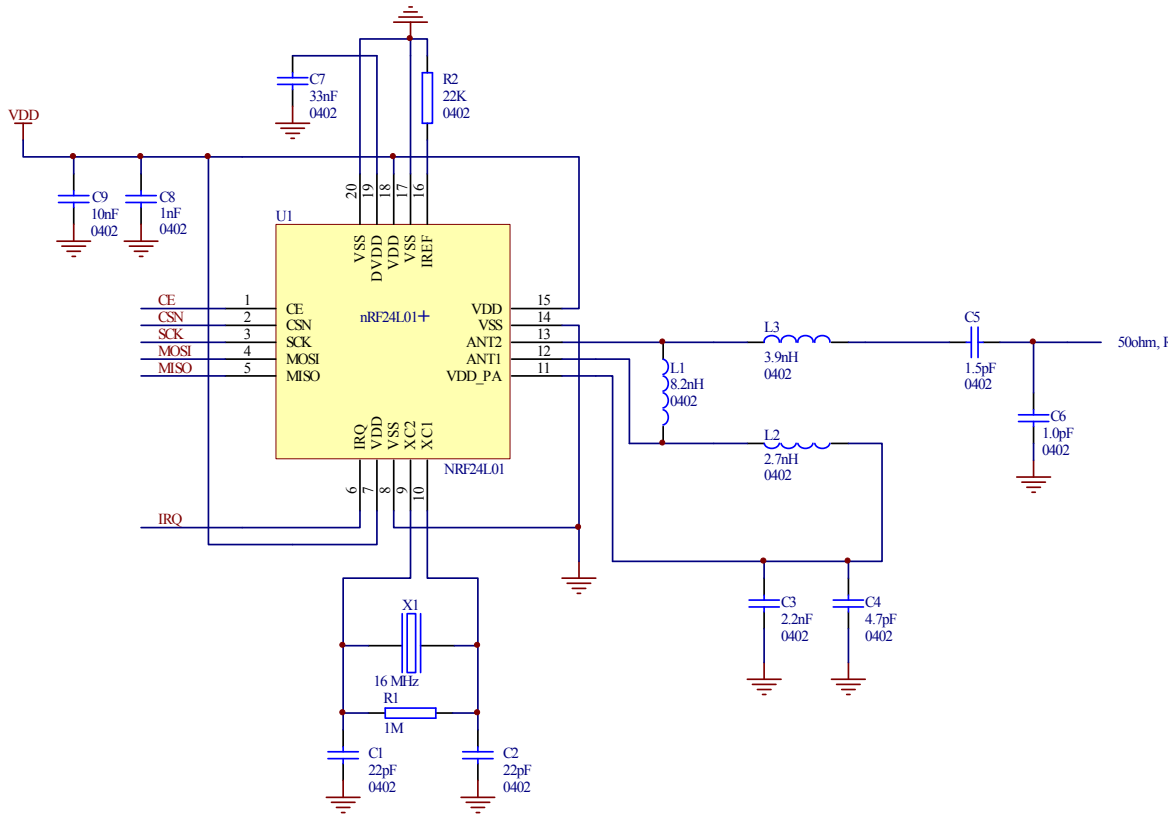


Figure 32. nRF24L01+ schematic for RF layouts with single ended 50Ω RF output

11 应用示例

nRF24L01+ 具有单端匹配网络晶体、偏置电阻器和去耦电容器。

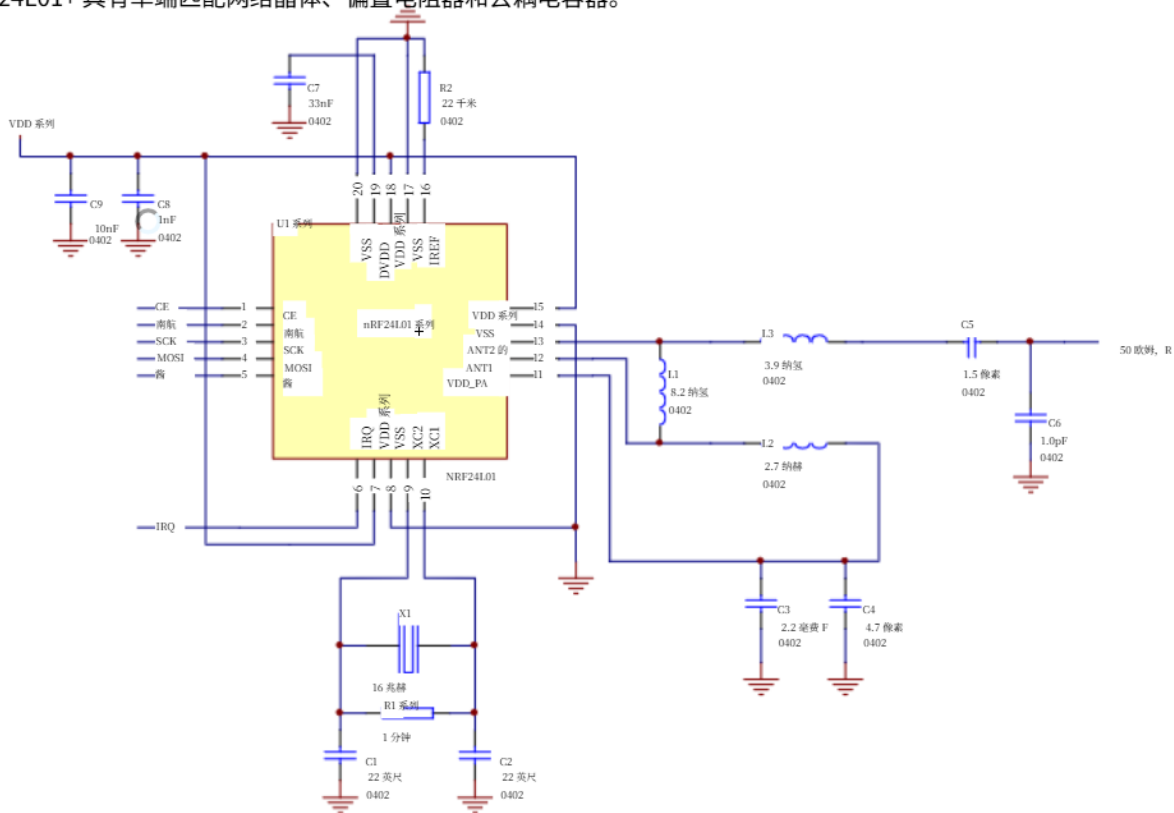


图 32.nRF24L01+ 原理图，用于具有单端 50Ω RF 输出的 RF 布局

Part	Designator	Footprint	Description
22pF ^a	C1	0402	NPO, +/- 2%
22pF ^a	C2	0402	NPO, +/- 2%
2.2nF	C3	0402	X7R, +/- 10%
4.7pF	C4	0402	NPO, +/- 0.25pF
1.5pF	C5	0402	NPO, +/- 0.1pF
1.0pF	C6	0402	NPO, +/- 0.1pF
33nF	C7	0402	X7R, +/- 10%
1nF	C8	0402	X7R, +/- 10%
10nF	C9	0402	X7R, +/- 10%
8.2nH	L1	0402	chip inductor +/- 5%
2.7nH	L2	0402	chip inductor +/- 5%
3.9nH	L3	0402	chip inductor +/- 5%
Not mounted ^b	R1	0402	
22kΩ	R2	0402	+/-1%
nRF24L01+	U1	QFN20 4x4	
16MHz	X1		+/-60ppm, C _L =12pF

- a. C1 and C2 must have values that match the crystals load capacitance, C_L.
- b. The nRF24L01+ and nRF24L01 application example and BOM are the same with the exception of R1. R1 can be mounted for backward compatibility with nRF24L01. The use of a 1Mohm resistor externally does not have any impact on crystal performance.

Table 29. Recommended components (BOM) in nRF24L01+ with antenna matching network

11.1 PCB layout examples

[Figure 33.](#), [Figure 34.](#) and [Figure 35.](#) show a PCB layout example for the application schematic in [Figure 32.](#)

A double-sided FR-4 board of 1.6mm thickness is used. This PCB has a ground plane on the bottom layer. Additionally, there are ground areas on the component side of the board to ensure sufficient grounding of critical components. A large number of via holes connect the top layer ground areas to the bottom layer ground plane.

部分	指示器	脚印	描述
22 英尺	C1	0402	非营利组织, +/- 2%
22 英尺	C2 型	0402	非营利组织, +/- 2%
2.2 毫费 F	C3 系列	0402	X7R, +/- 10%
4.7 像素	C4 系列	0402	NPO, +/- 0.25pF
1.5 像素	C5 系列	0402	NPO, +/- 0.1pF
1,0pF	C6 系列	0402	NPO, +/- 0.1pF
33nF	C7 系列	0402	X7R, +/- 10%
1nF	C8 系列	0402	X7R, +/- 10%
10nF	C9 系列	0402	X7R, +/- 10%
8.2 毫安	L1 系列	0402	片式电感器 +/- 5%
2.7 纳赫	L2 (二层)	0402	片式电感器 +/- 5%
3,9 纳赫	L3 系列	0402	片式电感器 +/- 5%
未挂载	R1 系列	0402	
22kΩ	R2	0402	+/-1%
nRF24L01+	U1 系列	接线端子 QFN20 4x4	
16 兆赫	X1		+/-60 页/分钟, C=12pF

一个。C1 和 C2 的值必须与晶体负载电容 C b 匹配。nRF24L01+ 和 nRF24L01 应用示例和 BOM 相同，但 R1 除外。可以安装 R1 以向后兼容 nRF24L01。在外部使用 1Mohm 电阻器对晶体性能没有任何影响。

表 29.具有天线匹配网络的 nRF24L01+ 中的推荐组件 （BOM）

11.1 PCB 布局示例

图 33., 图 34.和图 35.显示了图中应用程序原理图的 PCB 布局示例 32..

使用 1.6mm 厚的双面 FR-4 板。该 PCB 在底层有一个接地层。此外，电路板的元件侧有接地区域，以确保关键元件的充分接地。大量的过孔将顶层接地区域连接到底层接地层。

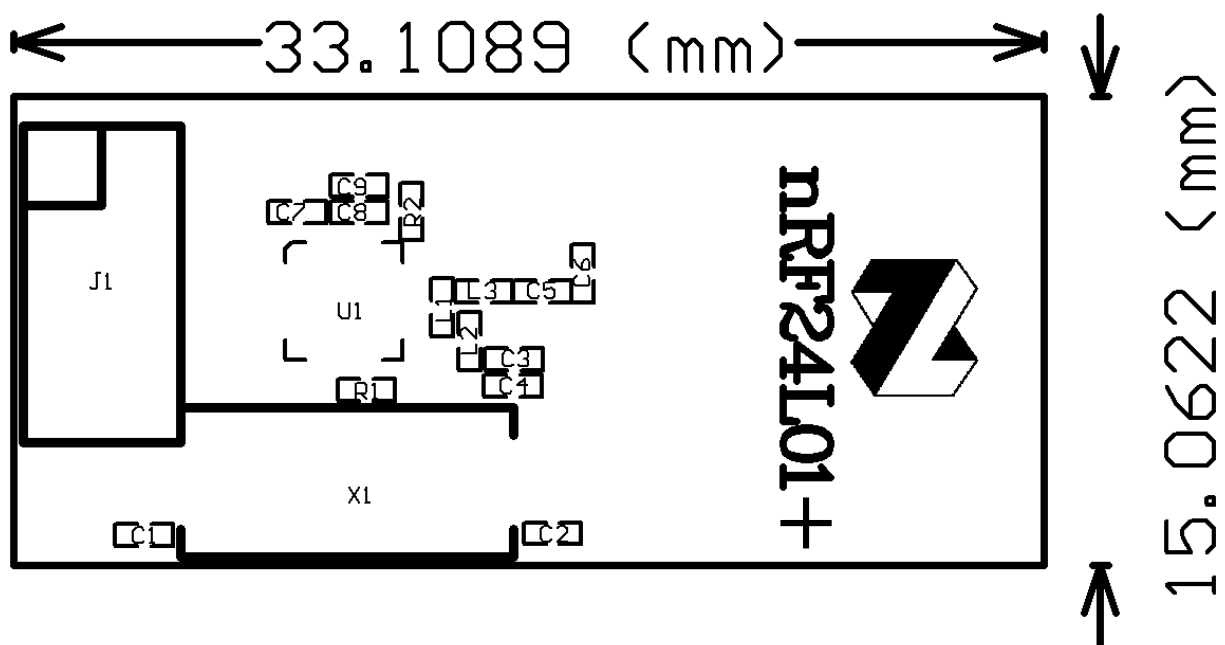


Figure 33. Top overlay (nRF24L01+ RF layout with single ended connection to PCB antenna and 0402 size passive components)

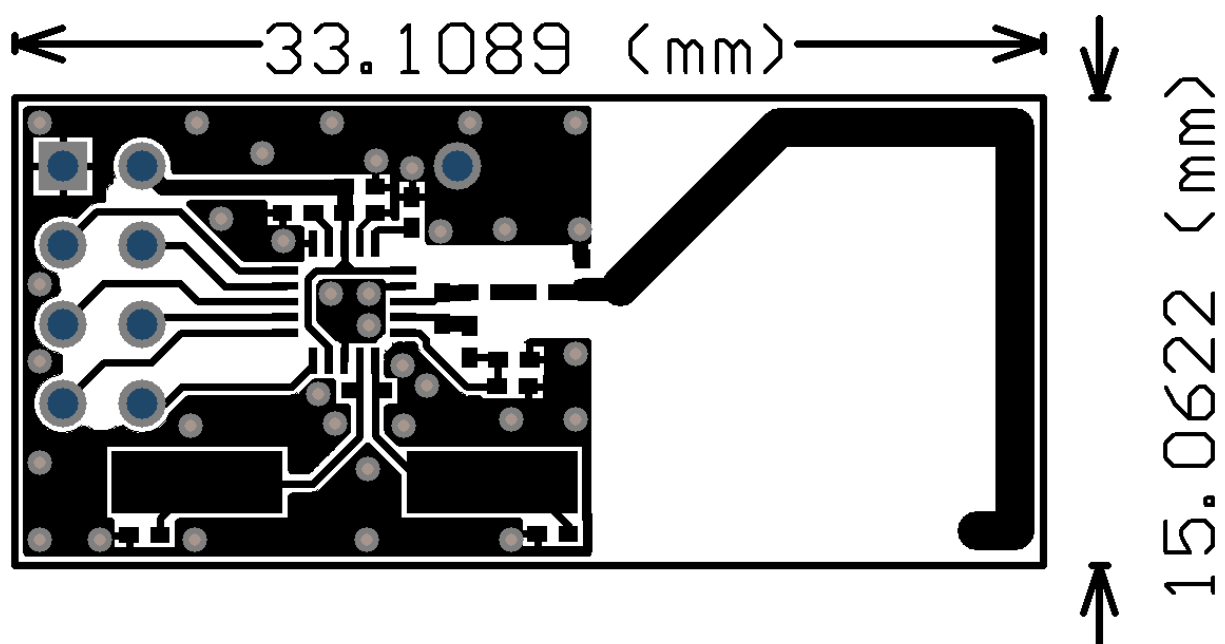


Figure 34. Top layer (nRF24L01+ RF layout with single ended connection to PCB antenna and 0402 size passive components)

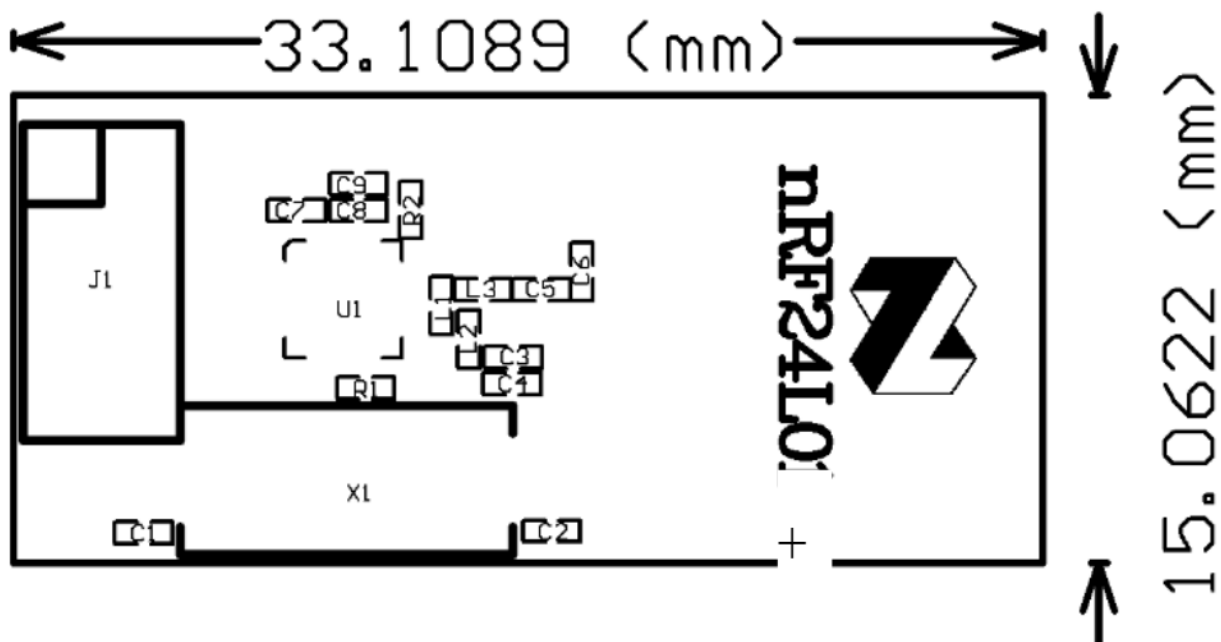


图 33.顶部覆盖层 (nRF24L01+ 射频布局, 单端连接到 PCB 天线和 0402 尺寸无源元件)

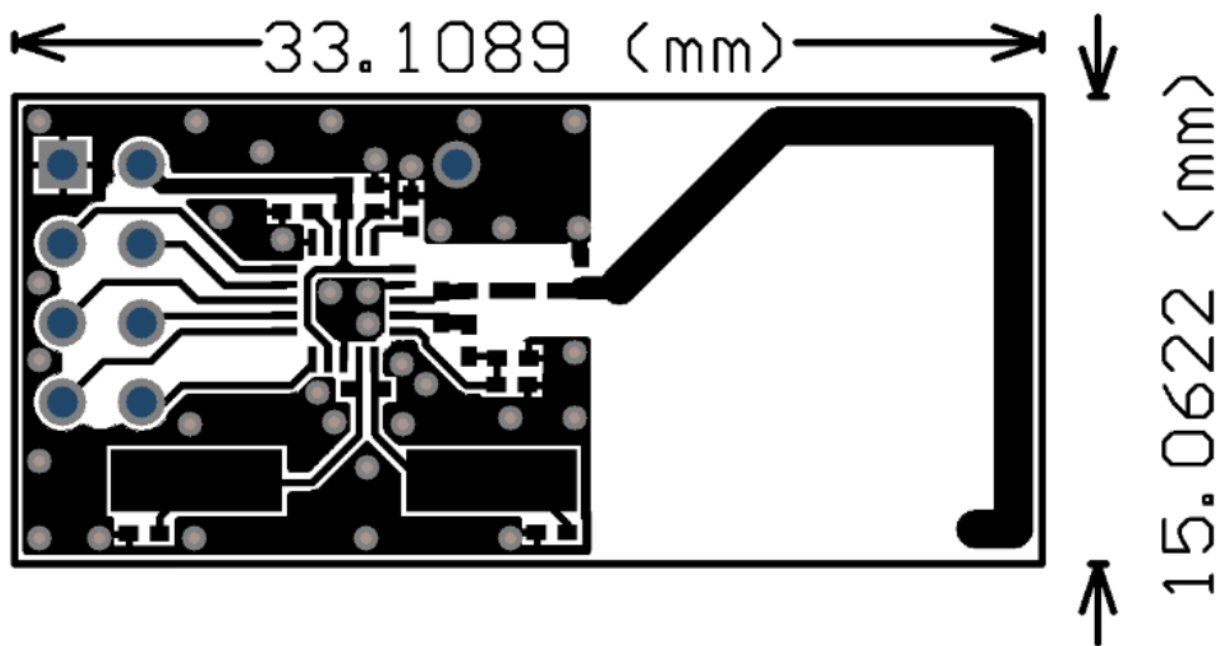


图 34.顶层 (nRF24L01+ 射频布局, 单端连接到 PCB 天线和 0402 尺寸的无源元件)

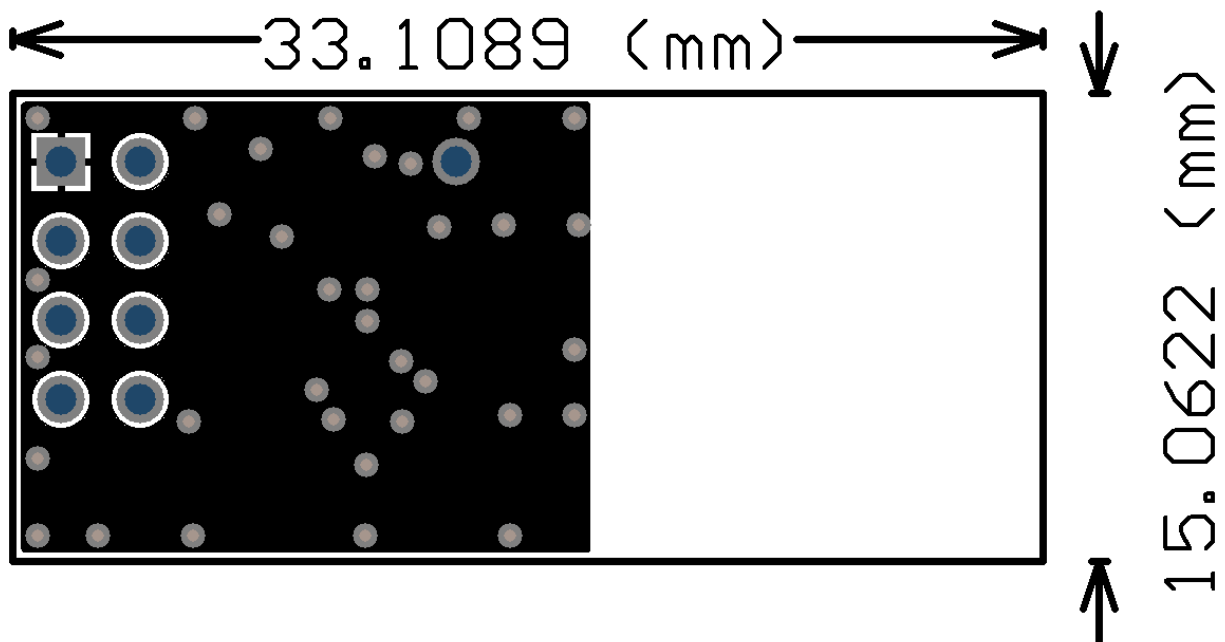


Figure 35. Bottom layer (nRF24L01+ RF layout with single ended connection to PCB antenna and 0402 size passive components)

The next figure ([Figure 36](#), [Figure 37](#), and [Figure 38](#).) is for the SMA output to have a board for direct measurements at a 50Ω SMA connector.

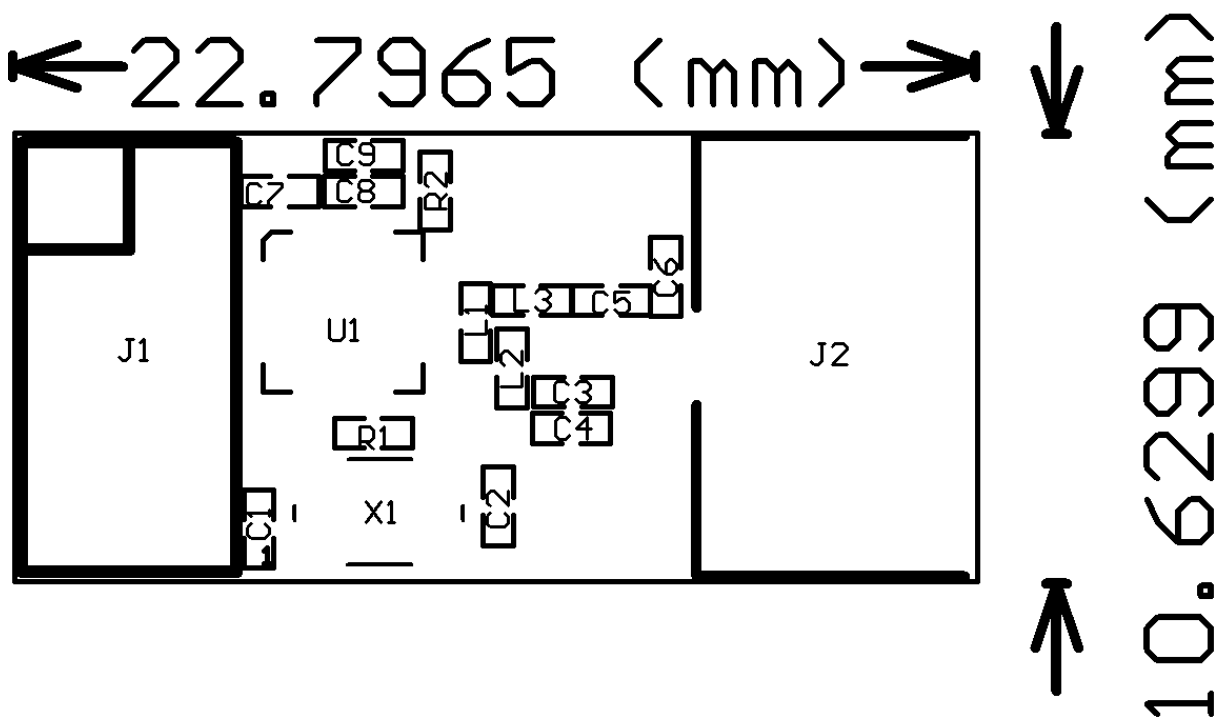


Figure 36. Top Overlay (Module with OFM crystal and SMA connector)

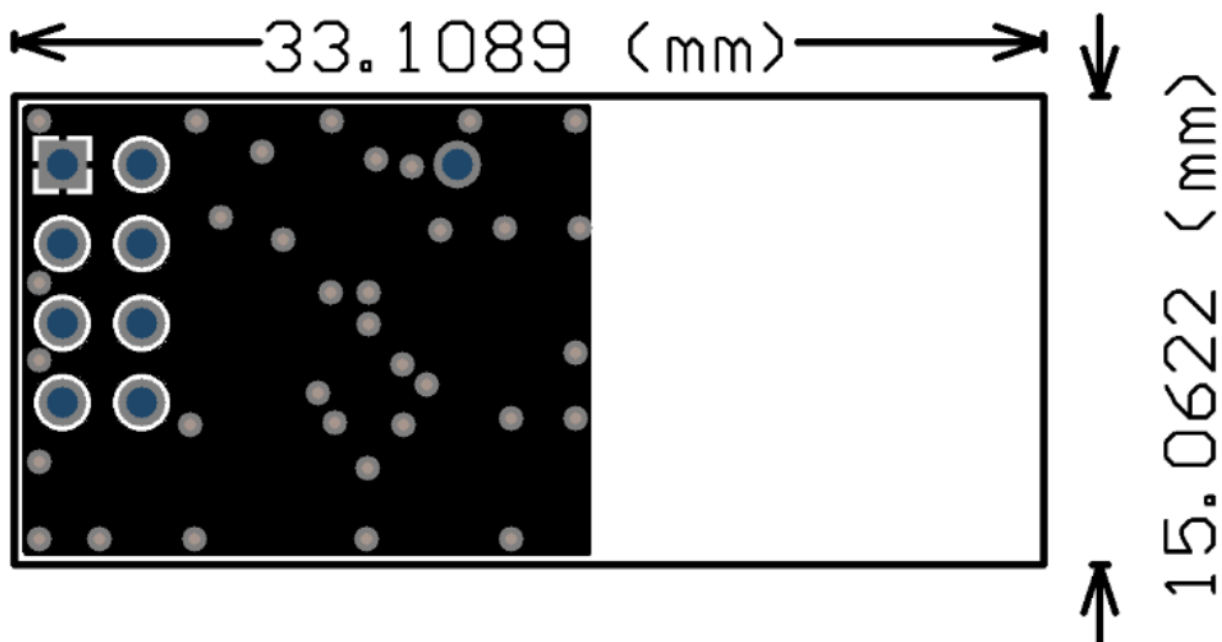


图 35.底层 (nRF24L01+ 射频布局, 单端连接到 PCB 天线和 0402 尺寸的无源元件)

嵌套图 (图 36.、图 37. 和图 38.) 是 SMA 输出在 50Ω SMA 连接器上有一个用于直接测量的板。

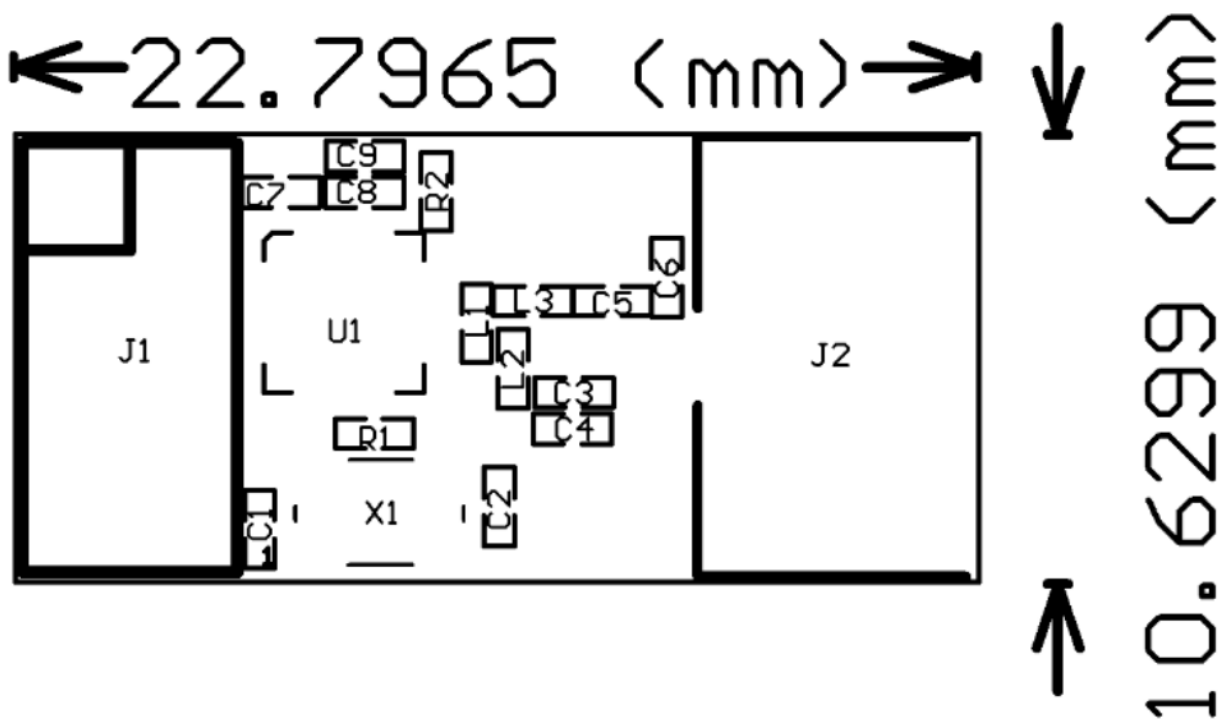


图 36.顶部覆盖层 (带有 OFM 晶体和 SMA 连接器的模块)

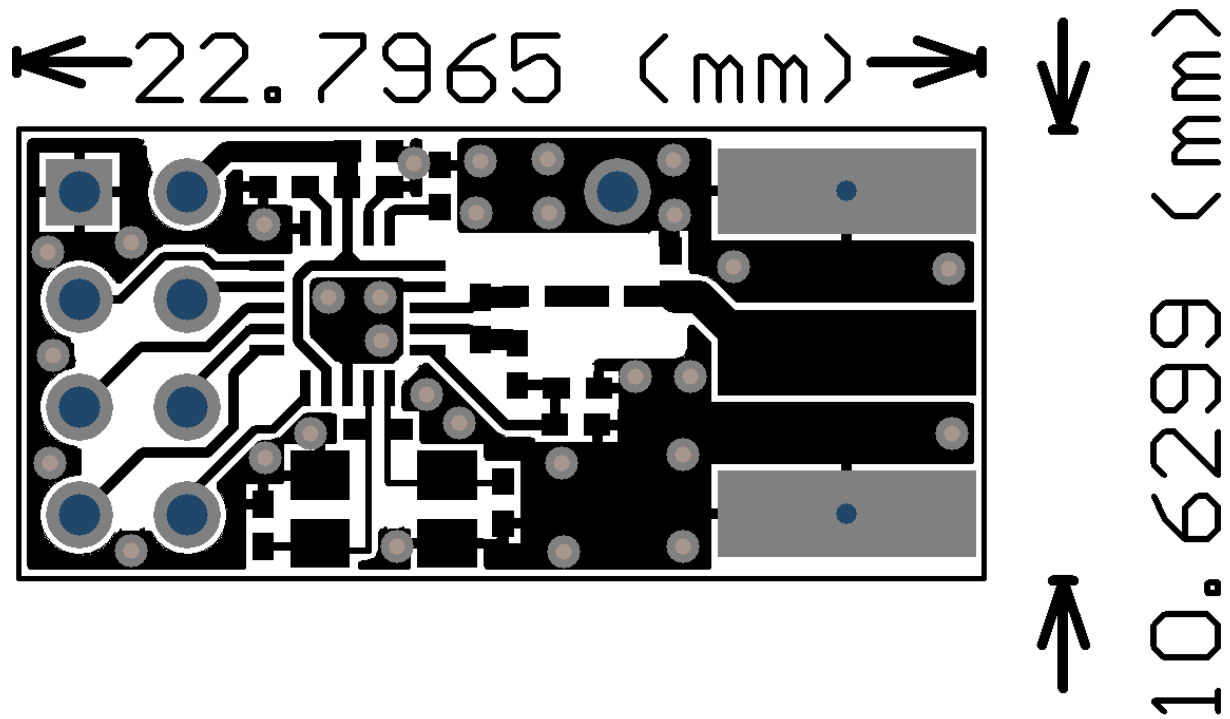


Figure 37. Top Layer (Module with OFM crystal and SMA connector)

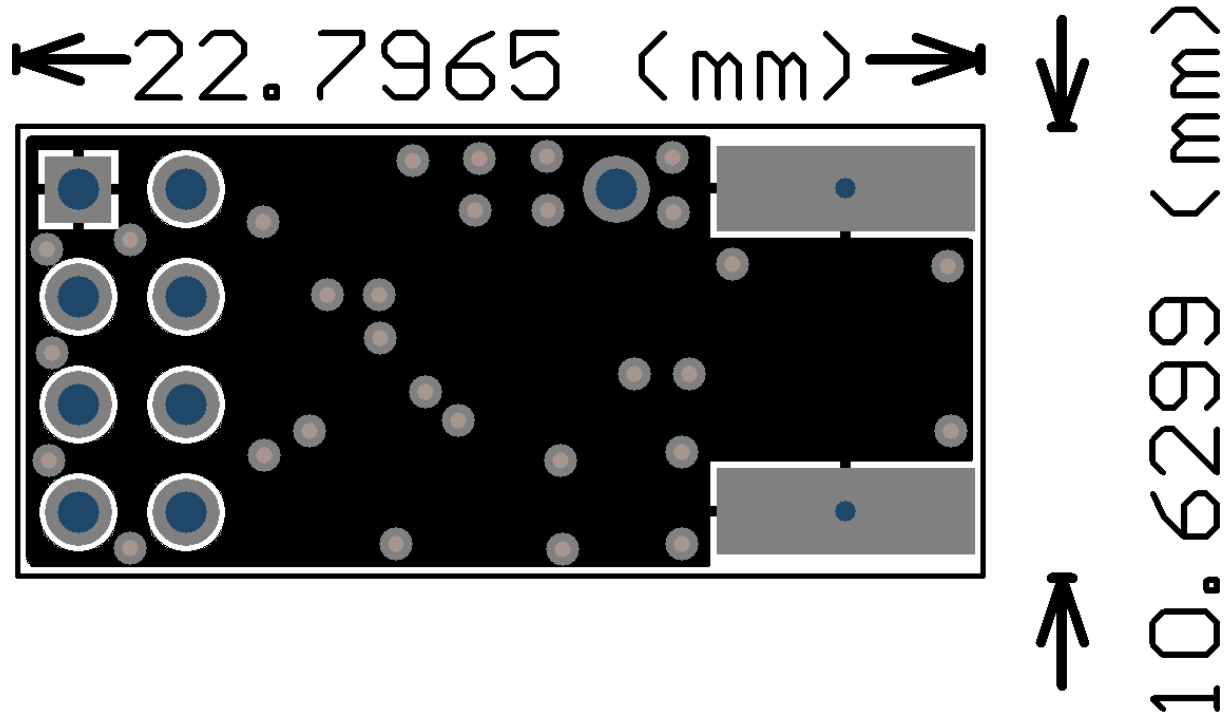


Figure 38. Bottom Layer (Module with OFM crystal and SMA connector)

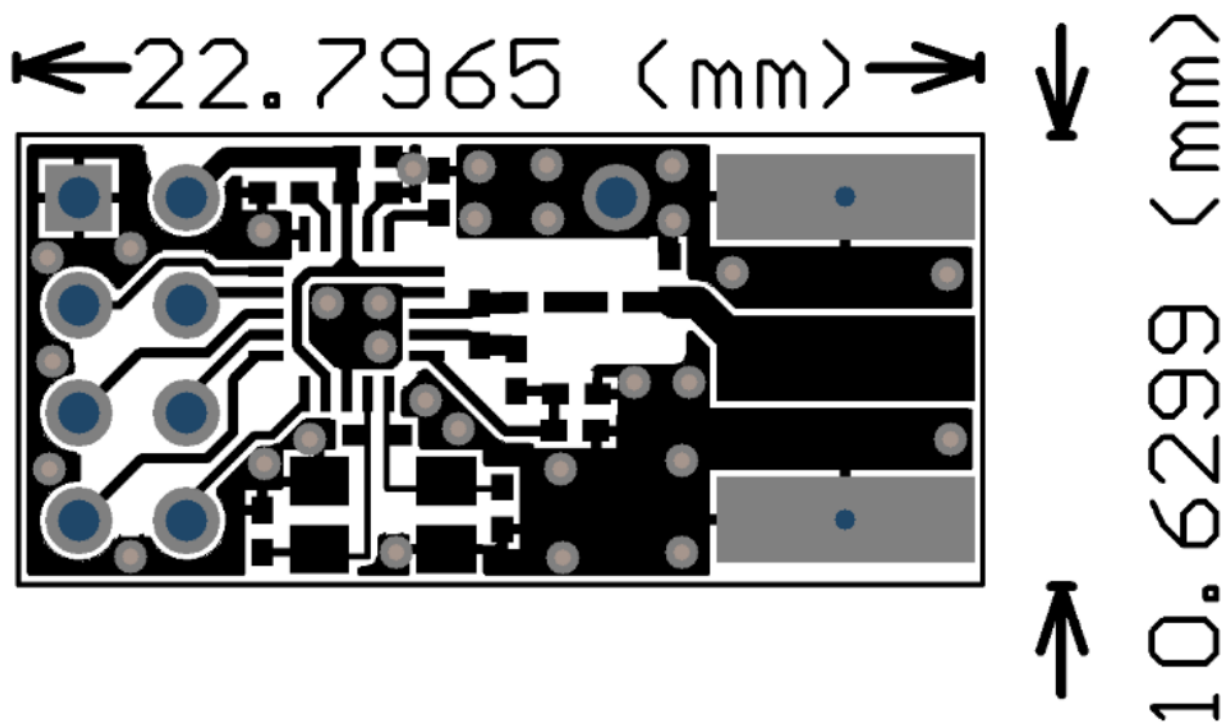


图 37.顶层（带有 OFM 晶体和 SMA 连接器的模块）

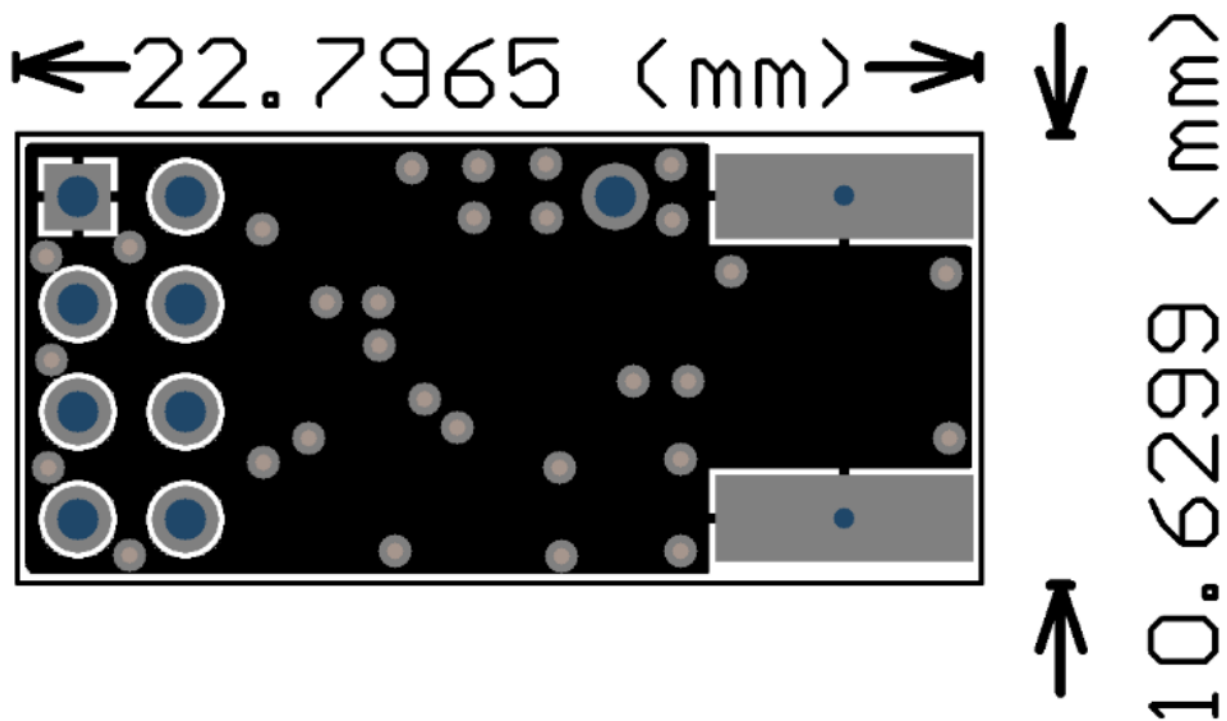
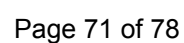


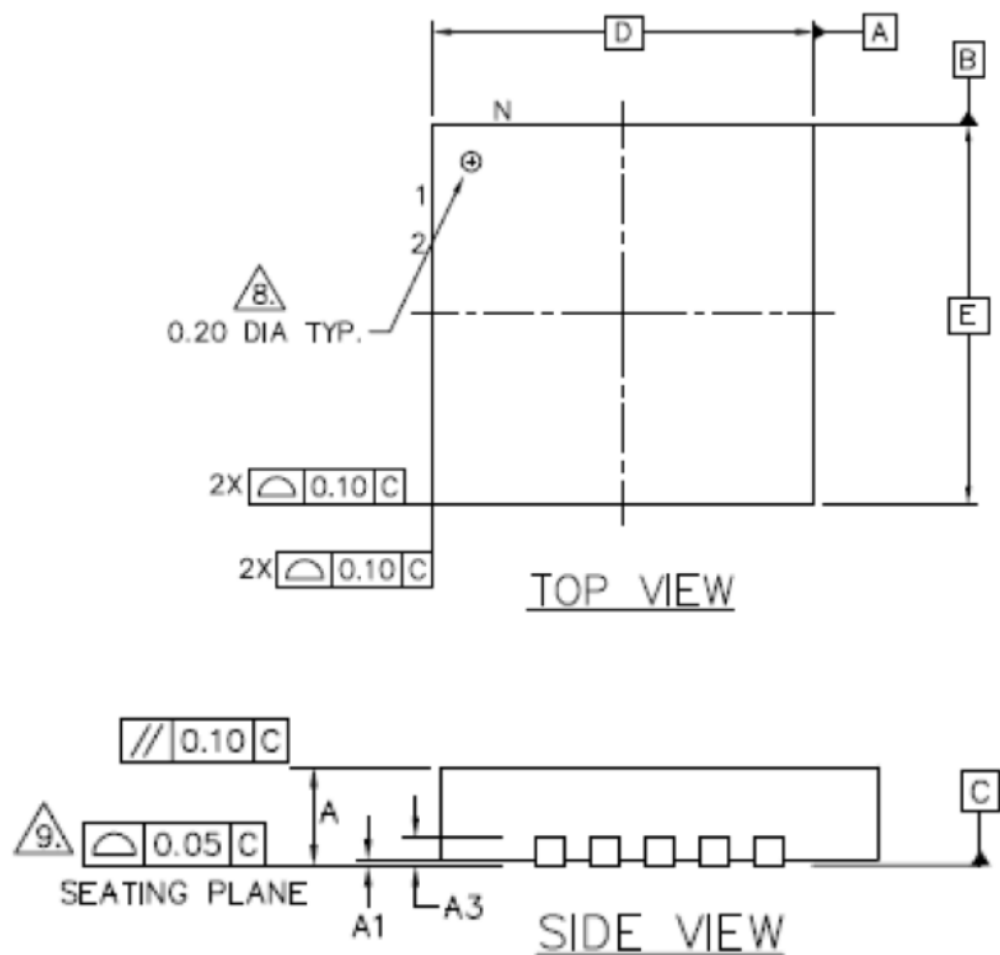
图 38.底层（带有 OFM 晶体和 SMA 连接器的模块）

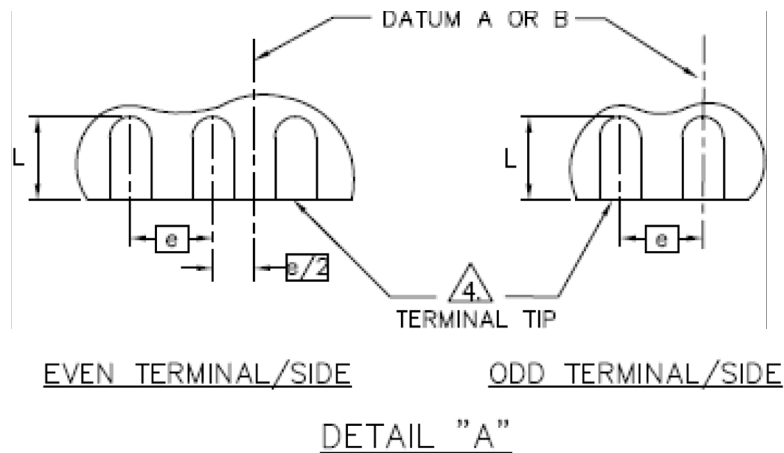
nRF24L01+ uses the QFN20 4x4 package, with matt tin plating.



12 机械规格

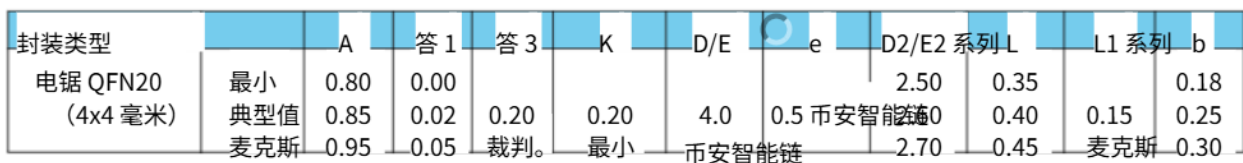
nRF24L01+ 使用 QFN20 4x4 封装, 采用亚光镀锡。





Package Type		A	A1	A3	K	D/E	e	D2/E2	L	L1	b
Saw QFN20 (4x4 mm)	Min.	0.80	0.00					2.50	0.35		0.18
	Typ.	0.85	0.02	0.20	0.20	4.0	0.5 BSC	2.60	0.40	0.15	0.25
	Max	0.95	0.05	REF.	min.	BSC ^a		2.70	0.45	max	0.30

Figure 39. nRF24L01+ Package Outline



a. BSC: 中心之间的基本间距, 参考 JEDEC 标准 95, 第 4.17-11/A 页

图 39.nRF24L01+ 封装外形

13 Ordering information

13.1 Package marking

n	R	F		A	X
2	4	L	0	1	+
Y	Y	W	W	L	L

13.2 Abbreviations

Abbreviation	Definition
nRF	Fixed text
A	Variable Build Code, that is, unique code for production sites, package type and test platform
X	"X" grade, that is, Engineering Samples (optional)
YY	2 digit Year number
WW	2 digit Week number
LL	2 letter wafer lot number code

13.3 Product options

13.3.1 RF silicon

Ordering code	Package	Container	MOQ ^a
nRF24L01P-SAMPLE	4x4mm 20-pin QFN, lead free (green)	Sample box	5
nRF24L01P-T	4x4mm 20-pin QFN, lead free (green)	Tray	490
nRF24L01P-R	4x4mm 20-pin QFN, lead free (green)	Tape and reel	4000
nRF24L01P-R7	4x4mm 20-pin QFN, lead free (green)	Tape and reel	1500

a. Minimum Order Quantity

Table 30. nRF24L01+ RF silicon options

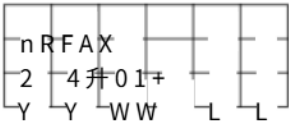
13.3.2 Development tools

Type Number	Description	Version
nRF24L01P-EVKIT	nRF24L01+ Evaluation kit	
nRF24L01P-UPGRADE	nRF24L01+ Upgrade kit for owners of nRF24L01-EVKIT	
nRF24L01P-MODULE-SMA	nRF24L01+ Evaluation kit module with SMA antenna	
nRF24L01P-MODULE-PCB	nRF24L01+ Evaluation kit module with PCB antenna	

Table 31. nRF24L01+ solution options

13 订购信息

13.1 包装标记



13.2 缩写

缩写	定义
nRF 固定文本	可变构建代码，即生产站点的唯一代码，打包-
	年龄类型和测试平台
X	“X” 级，即工程样品（可选）
YY	2 位数字年份
世界	2 位数字周数
将	2 个字母的晶圆批号代码

13.3 产品选项

13.3.1 射频硅

订购代码	包	容器	最小起订量
nRF24L01P-样品	4x4mm 20 引脚 QFN，无铅（绿色）	样品箱	5
nRF24L01P-T 系列	4x4mm 20 引脚 QFN，无铅（绿色）	托盘	490
nRF24L01P-R	4x4mm 20 引脚 QFN，无铅（绿色）	卷带	4000
nRF24L01P-R7 系列	4x4mm 20 引脚 QFN，无铅（绿色）	卷带	1500

a. 最小起订量

表 30.nRF24L01+ 射频硅选项

13.3.2 开发工具

类型编号	描述	版本
nRF24L01P-评估套件	nRF24L01+ 评估套件	
nRF24L01P-升级 nRF24L01+ 升级套件	评估套件，适用于 nRF24L01- 的所有者	
nRF24L01P-模组-SMA	带 SMA 天线的 nRF24L01+ 评估套件模块	
nRF24L01P-模块-PCB	nRF24L01+ 带 PCB 天线的评估套件模块	

表 31.nRF24L01+ 解决方案选项

14 Glossary of Terms

Term	Description
ACK	Acknowledgement
ACS	Adjacent Channel Selectivity
AGC	Automatic Gain Control
ART	Auto Re-Transmit
CD	Carrier Detect
CE	Chip Enable
CLK	Clock
CRC	Cyclic Redundancy Check
CSN	Chip Select NOT
ESB	Enhanced ShockBurst™
GFSK	Gaussian Frequency Shift Keying
IM	Intermodulation
IRQ	Interrupt Request
ISM	Industrial-Scientific-Medical
LNA	Low Noise Amplifier
LSB	Least Significant Bit
LSByte	Least Significant Byte
Mbps	Megabit per second
MCU	Microcontroller Unit
MISO	Master In Slave Out
MOSI	Master Out Slave In
MSB	Most Significant Bit
MSByte	Most Significant Byte
PCB	Printed Circuit Board
PID	Packet Identity Bits
PLD	Payload
PRX	Primary RX
PTX	Primary TX
PWR_DWN	Power Down
PWR_UP	Power Up
RoHS	Restriction of use of Certain Hazardous Substances
RPD	Received Power Detector
RX	Receive
RX_DR	Receive Data Ready
SPI	Serial Peripheral Interface
TX	Transmit
TX_DS	Transmit Data Sent

Table 32. Glossary

14 术语表

术语	描述
ACK	确认
ACS 公司	邻道选择性
AGC	自动增益控制
艺术	自动重传
光盘	载波检测
CE	芯片使能
时钟	时钟
CRC	循环冗余校验
南航	Chip Select NOT
ESB	增强的 ShockBurst™
GFSK 公司	高斯频移键控
我	互调
IRQ	中断请求
主义	工业-科学-医疗
LNA 系列	低噪声放大器
LSB	最低有效位
LSByte	最低有效字节
Mbps	每秒兆比特
MCU	微控制器单元
MISO	主控输入
从机输出	MOSI 主控输出
从机输入	MSB 最高有效位
MSB	最高有效字节
PCB	印刷电路板
PID	数据包
身份位	PLD 有效载荷
PRX	主 RX
PTX	主 TX
PWR_DWN	掉电
PWR_UP	上电
RoHS	限制使用某些有害物质
RPD	接收功率检测器
RX	接收
RX_DR	接收数据就绪
SPI	串行外设接口
TX 传输	
TX_DS	传输发送的数据

表 32.词汇表

Appendix A - Enhanced ShockBurst™ - Configuration and communication example

Enhanced ShockBurst™ transmitting payload

1. Set the configuration bit `PRIM_RX` low.
2. When the application MCU has data to transmit, clock the address for the receiving node (`TX_ADDR`) and payload data (`TX_PLD`) into nRF24L01+ through the SPI. The width of TX-payload is counted from the number of bytes written into the TX FIFO from the MCU. `TX_PLD` must be written continuously while holding `CSN` low. `TX_ADDR` does not have to be rewritten if it is unchanged from last transmit. If the PTX device shall receive acknowledge, configure data pipe 0 to receive the ACK packet. The RX address for data pipe 0 (`RX_ADDR_P0`) must be equal to the TX address (`TX_ADDR`) in the PTX device. For the example in [Figure 14. on page 41](#) perform the following address settings for the TX5 device and the RX device:
 TX5 device: `TX_ADDR = 0xB3B4B5B605`
 TX5 device: `RX_ADDR_P0 = 0xB3B4B5B605`
 RX device: `RX_ADDR_P5 = 0xB3B4B5B605`
3. A high pulse on `CE` starts the transmission. The minimum pulse width on `CE` is 10µs.
4. nRF24L01+ ShockBurst™:
 - ▶ Radio is powered up.
 - ▶ 16MHz internal clock is started.
 - ▶ RF packet is completed (see the packet description).
 - ▶ Data is transmitted at high speed (1Mbps or 2Mbps configured by MCU).
5. If auto acknowledgement is activated (`ENAA_P0=1`) the radio goes into RX mode immediately, unless the `NO_ACK` bit is set in the received packet. If a valid packet is received in the valid acknowledgement time window, the transmission is considered a success. The `TX_DS` bit in the `STATUS` register is set high and the payload is removed from TX FIFO. If a valid ACK packet is not received in the specified time window, the payload is retransmitted (if auto retransmit is enabled). If the auto retransmit counter (`ARC_CNT`) exceeds the programmed maximum limit (`ARC`), the `MAX_RT` bit in the `STATUS` register is set high. The payload in TX FIFO is NOT removed. The `IRQ` pin is active when `MAX_RT` or `TX_DS` is high. To turn off the `IRQ` pin, reset the interrupt source by writing to the `STATUS` register (see Interrupt chapter). If no ACK packet is received for a packet after the maximum number of retransmits, no further packets can be transmitted before the `MAX_RT` interrupt is cleared. The packet loss counter (`PLOS_CNT`) is incremented at each `MAX_RT` interrupt. That is, `ARC_CNT` counts the number of retransmits that were required to get a single packet through. `PLOS_CNT` counts the number of packets that did not get through after the maximum number of retransmits.
6. nRF24L01+ goes into standby-I mode if `CE` is low. Otherwise, next payload in TX FIFO is transmitted. If TX FIFO is empty and `CE` is still high, nRF24L01+ enters standby-II mode.
7. If nRF24L01+ is in standby-II mode, it goes to standby-I mode immediately if `CE` is set low.

附录 A - 增强型 ShockBurst™ - 配置和通信示例

增强的 ShockBurst™ 传输负载

1. 将 configuration bit PRIM_RX 设置为低电平。
2. 当应用 MCU 有要传输的数据时，通过 SPI 将接收节点 (TX_ADDR) 和有效载荷数据 (TX_PLD) 的地址时钟到 nRF24L01+ 中。TX-payload 的宽度是从 MCU 写入 TX FIFO 的字节数计算的。TX_PLD 必须在保持 CSN 为低电平的情况下连续写入。如果 TX_ADDR 与上次传输相比没有变化，则不必重写。如果 PTX 设备应收到确认，请配置数据管道 0 以接收 ACK 数据包。数据管道 0 (RX_ADDR_P0) 的 RX 地址必须等于 PTX 设备中的 TX 地址 (TX_ADDR)。对于图 14 中的示例。在第 41 页上为 TX5 设备和 RX 设备执行以下地址设置：TX5 设备：TX_ADDR = 0xB3B4B5B605 TX5 设备：RX_ADDR_P0 = 0xB3B4B5B605 RX 设备：RX_ADDR_P5 = 0xB3B4B5B605
3. CE 上的高脉冲将启动传输。CE 上的最小脉冲宽度为 10μs。
4. nRF24L01+ 冲击爆发™：
 - × 无线电已通电。
 - × 16MHz 内部时钟启动。
 - × RF 数据包已完成 (请参阅数据包说明)。
 - × 数据高速传输 (由 MCU 配置的 1Mbps 或 2Mbps)。
5. 如果激活了自动确认 (ENAA_P0=1)，无线电将立即进入 RX 模式，除非在接收到的数据包中设置了 NO_ACK 位。如果在有效确认时间窗口内收到有效数据包，则认为传输成功。STATUS 寄存器中的 TX_DS 位设置为高电平，并且有效载荷从 TX FIFO 中删除。如果在指定的时间窗口内未收到有效的 ACK 数据包，则重新传输负载 (如果启用了自动重新传输)。如果自动重传计数器 (ARC_CNT) 超过编程的最大限值 (ARC)，则 STATUS 寄存器中的 MAX_RT 位设置为高电平。TX FIFO 中的有效负载不会被删除。当 MAX_RT 或 TX_DS 为高电平时，IRQ 引脚处于活动状态。要关闭 IRQ 引脚，请通过写入 STATUS 寄存器来重置中断源 (参见中断章节)。如果超过最大重传次数后仍未收到 ACK 报文，则在清除 MAX_RT 中断之前，无法再发送任何报文。数据包丢失计数器 (PLOS_CNT) 在每次 MAX_RT 中断时递增。也就是说，ARC_CNT 计算单个数据包通过所需的重传次数。PLOS_CNT 计算在最大重新传输次数后未通过的数据包数。
6. 如果 CE 较低，nRF24L01+ 将进入待机 I 模式。否则，将传输 TX FIFO 中的下一个 payload。如果 TX FIFO 为空且 CE 仍然为高电平，则 nRF24L01+ 进入待机 II 模式。
7. 如果 nRF24L01+ 处于待机 II 模式，如果 CE 设置为低电平，它会立即进入待机 I 模式。

Enhanced ShockBurst™ receive payload

1. Select RX by setting the `PRIM_RX` bit in the `CONFIG` register to high. All data pipes that receive data must be enabled (`EN_RXADDR` register), enable auto acknowledgement for all pipes running Enhanced ShockBurst™ (`EN_AA` register), and set the correct payload widths (`RX_PW_Px` registers). Set up addresses as described in item 2 in the Enhanced ShockBurst™ transmitting payload example above.
2. Start Active RX mode by setting `CE` high.
3. After 130µs nRF24L01+ monitors the air for incoming communication.
4. When a valid packet is received (matching address and correct CRC), the payload is stored in the RX-FIFO, and the `RX_DR` bit in `STATUS` register is set high. The `IRQ` pin is active when `RX_DR` is high. `RX_P_NO` in `STATUS` register indicates what data pipe the payload has been received in.
5. If auto acknowledgement is enabled, an ACK packet is transmitted back, unless the `NO_ACK` bit is set in the received packet. If there is a payload in the `TX_PLD` FIFO, this payload is added to the ACK packet.
6. MCU sets the `CE` pin low to enter standby-I mode (low current mode).
7. MCU can clock out the payload data at a suitable rate through the SPI.
8. nRF24L01+ is now ready for entering TX or RX mode or power down mode.

增强的 ShockBurst™ 接收负载

1. 通过将 CONFIG 寄存器中的 PRIM_RX 位设置为高电平来选择 RX。必须启用接收数据的所有数据管道 (EN_RXADDR 寄存器)，为运行 Enhanced ShockBurst™ 的所有管道启用自动确认 (EN_AA 寄存器)，并设置正确的有效载荷宽度 (RX_PW_Px 寄存器)。按照上面增强型 ShockBurst™ 传输负载示例中的第 2 项所述设置地址。
2. 通过将 CE 设置为高电平来启动 Active RX 模式。
3. 130μs 后，nRF24L01+ 监测空气中的传入通信。
4. 当接收到有效数据包 (匹配地址和正确的 CRC) 时，有效载荷存储在 RX-FIFO 中，并且 STATUS 寄存器中的 RX_DR 位设置为高电平。当 IRQ 引脚为高电平时 RX_DR 有效。STATUS register 中的 RX_P_NO 表示接收到有效负载的数据管道。
有效负载的数据管道
5. 如果启用自动确认，则 ACK 数据包将传回，除非在接收到的数据包中设置了 NO_ACK 位。如果 TX_PLD FIFO 中有 payload，则此 payload 将添加到 ACK 数据包中。
6. MCU 将 CE 引脚设置为低电平，以进入待机 I 模式 (低电流模式)。
7. MCU 可以通过 SPI 以合适的速率输出 payload 数据。
8. nRF24L01+ 现在可以进入 TX 或 RX 模式或断电模式。

Appendix B - Configuration for compatibility with nRF24XX

How to setup nRF24L01+ to receive from an nRF2401/nRF2402/nRF24E1/nRF24E2:

1. Use the same CRC configuration as the nRF2401/nRF2402/nRF24E1/nRF24E2.
2. Set the `PWR_UP` and `PRIM_RX` bit to 1.
3. Disable auto acknowledgement on the data pipe that is addressed.
4. Use the same address width as the PTX device.
5. Use the same frequency channel as the PTX device.
6. Select data rate 1Mbps or 250kbps on both nRF24L01+ and nRF2401/nRF2402/nRF24E1/nRF24E2.
7. Set correct payload width on the data pipe that is addressed.
8. Set `CE` high.

How to setup nRF24L01+ to transmit to an nRF2401/nRF24E1:

1. Use the same CRC configuration as the nRF2401/nRF2402/nRF24E1/nRF24E2.
2. Set the `PRIM_RX` bit to 0.
3. Set the Auto Retransmit Count to 0 to disable the auto retransmit functionality.
4. Use the same address width as the nRF2401/nRF2402/nRF24E1/nRF24E2.
5. Use the same frequency channel as the nRF2401/nRF2402/nRF24E1/nRF24E2.
6. Select data rate 1Mbps or 250kbps on both nRF24L01+ and nRF2401/nRF2402/nRF24E1/nRF24E2.
7. Set `PWR_UP` high.
8. Clock in a payload that has the same length as the nRF2401/nRF2402/nRF24E1/nRF24E2 is configured to receive.
9. Pulse `CE` to transmit the packet.

附录 B - 与 nRF24XX 兼容的配置

如何设置 nRF24L01+ 以从 nRF2401/nRF2402/nRF24E1/nRF24E2 接收：

1. 使用与 nRF2401/nRF2402/nRF24E1/nRF24E2 相同的 CRC 配置。
2. 将 PWR_UP 位和 PRIM_RX 位设置为 1。
3. 在寻址的数据管道上禁用自动确认。
4. 使用与 PTX 设备相同的地址宽度。
5. 使用与 PTX 设备相同的频道。
6. 在 nRF1L24+ 和 nRF2401/nRF2402/nRF24E1/nRF24E24E2 上选择数据速率 24Mbps 或 24kbps。
7. 在寻址的数据管道上设置正确的负载宽度。
8. 将 CE 设置为高。

如何设置 nRF24L01+ 以传输到 nRF2401/nRF24E1：

1. 使用与 nRF2401/nRF2402/nRF24E1/nRF24E2 相同的 CRC 配置。
2. 将 PRIM_RX 位设置为 0。
3. 将 Auto Retransmit Count （自动重传计数） 设置为 0 以禁用自动重传功能。
4. 使用与 nRF2401/nRF2402/nRF24E1/nRF24E2 相同的地址宽度。
5. 使用与 nRF2401/nRF2402/nRF24E1/nRF24E2 相同的频道。
6. 在 nRF1L24+ 和 nRF2401/nRF2402/nRF24E1/nRF24E24E2 上选择数据速率 24Mbps 或 24kbps。
7. 将 PWR_UP 设置为高。
8. 与 nRF2401/nRF2402/nRF24E1/nRF24E2 长度相同的有效载荷中的时钟配置为接收。
9. 脉冲 CE 以传输数据包。

Appendix C - Constant carrier wave output for testing

The output power of a radio is a critical factor for achieving wanted range. Output power is also the first test criteria needed to qualify for all telecommunication regulations.

Configuration

1. Set `PWR_UP = 1` and `PRIM_RX = 0` in the `CONFIG` register.
2. Wait 1.5ms `PWR_UP`->standby.
3. In the RF register set:
 - ▶ `CONT_WAVE = 1`.
 - ▶ `PLL_LOCK = 1`.
 - ▶ `RF_PWR`.
4. Set the wanted RF channel.
5. Set `CE` high.
6. Keep `CE` high as long as the carrier is needed.

Note: Do not use `REUSE_TX_PL` together with `CONT_WAVE=1`. When both these registers are set the chip does not react when setting `CE` low. If however, both registers are set `PWR_UP = 0` will turn TX mode off.

The nRF24L01+ should now output an unmodulated centered carrier.

附录 C - 用于测试的恒定载波输出

无线电的输出功率是实现所需范围的关键因素。输出功率也是符合所有电信法规所需的首要测试标准。

配置

1. 在 CONFIG 寄存器中设置 PWR_UP = 1 和 PRIM_RX = 0。
2. 等待 1.5 毫秒 PWR_UP->standby。
3. 在 RF register set 中：
 - X CONT_WAVE = 1。
 - X PLL_LOCK = 1。
 - X RF_PWR。
4. 设置所需的 RF 信道。
5. 将 CE 设置为高。
6. 只要需要载体，就保持 CE 高。

注意：请勿将 REUSE_TX_PL 与 CONT_WAVE=1 一起使用。当这两个 registers 都设置为将 CE 设置为低电平时，芯片没有反应。但是，如果两个 registers 都设置为 PWR_UP = 0 将关闭 TX 模式。

nRF24L01+ 现在应该输出一个未调制的中心载波。